ENN581 Robot Motion, Control & Planning

Revision booklet · written exam, 2 h, any calculator, printed notes only

Print at 100 % (actual size), A4 landscape, double-sided (flip on the short edge). Every page has a grey tab on its right edge: Index, Recipes, W1–W12, Tables, Derivations, top to bottom. Slide pages are cited as [p. N]; things not on the slides are marked [added]. Every number in this booklet was checked by script.

Colour key. Boxes: **blue** formula, **green** recipe, **red dashed** trap, **purple** worked example, **amber** slide check. Tabs: **blue** W1–W6 kinematics, **orange** W7–W9 dynamics & control, **teal** W10–W11 wheeled robots, **purple** W12 planning, **slate** reference.

If the question says… go to

"transformation matrix", "indicate frames", ETS ..... p. 2, 4
"DH table", "link parameters" ..... p. 2, 4
"sketch the workspace" ..... p. 2, 5
"derive the inverse kinematics" ..... p. 2, 5
"how many solutions" ..... p. 2
"mobility", "degrees of freedom", Grübler ..... p. 2, 3, 7
"parallel", "four-bar", "loop closure" ..... p. 2, 7
"cubic", "via point", "spline", "blend" ..... p. 2, 9
"Jacobian", "singularity", "joint torques for a force" ..... p. 2, 11
"equations of motion", Lagrange ..... p. 2, 13
"inertia tensor", "Newton–Euler" ..... p. 13
"critically damped", "gains", "steady-state error" ..... p. 2, 15
"computed torque", "partitioned", "force control" ..... p. 2, 16
"wheel speeds", "omni", "differential drive", ICR ..... p. 2, 17
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"dynamics of a mobile robot" ..... p. 18
"A*", "Dijkstra", "PRM", "RRT", "C-space" ..... p. 2, 19
trig identities, rotations, inertia of shapes ..... p. 21
a derivation you have forgotten ..... p. 23

Answer layout (method marks)

1. **Draw and define**: frames, axes, positive angle directions, assumptions (no joint limits; point masses; rest to rest; $\gamma = 0$).
2. **State the general tool** before using it: ETS, cosine rule, Euler–Lagrange, G–K, boundary conditions.
3. **Substitute and simplify**, one step per line, symbols first, numbers last, with units.
4. **Box the result**; list *every* branch (IK) or mode (four-bar).
5. **Sanity check** at $\theta = 0$, $t = 0$, or a symmetric case. Write "check:".

Time (2 hours)

Read all questions first (5 min). Split time by marks. Leave 10 min to check signs and units. If stuck: write the set-up and the equation you would solve, then move on.

Ten checks before you hand in

1. Calculator in **radians**? Angles labelled ° or rad?
2. atan2, not $\tan^{-1}(y/x)$ (quadrant).
3. Both IK branches (and "no solution if $|\cos| > 1$").
4. Order of matrix products (right = about the current frame).
5. Transposes: $\mathbf{R}^{-1} = \mathbf{R}^\top$; inverse $\mathbf{T}^{-1} = [\mathbf{R}^\top \quad -\mathbf{R}^\top \mathbf{t}]$.
6. $\mathcal{L} = K - P$; gravity sign; product rule in $\frac{d}{dt}$.
7. $k_v^2 = 4k_p$ for critical damping (unit mass).
8. Ground counted as a link; $D = 3$ planar, 6 spatial.
9. Wheel row: $-\dot\theta\, {}^C y$ in the x part, $+\dot\theta\, {}^C x$ in the y part.
10. Units: mm vs m; rad/s vs rpm; N m vs N.

Notation

${}^A\mathbf{T}_B$	pose of $\{B\}$ in $\{A\}$; ${}^A\mathbf{P} = {}^A\mathbf{T}_B\, {}^B\mathbf{P}$
${}^A\mathbf{R}_B$	columns = B's axes written in A
$R_x, R_y, R_z, t_x, t_y, t_z$	elementary transforms (ETS)
$a_{i-1}, \alpha_{i-1}, d_i, \theta_i$	proximal (Craig) DH
c_i, s_i, c_{ij}	$\cos\theta_i$, $\sin\theta_i$, $\cos(\theta_i+\theta_j)$
$\mathbf{J}$	Jacobian: $\dot{\mathbf{x}} = \mathbf{J}\dot{\mathbf{q}}$; $\boldsymbol{\tau} = \mathbf{J}^\top \mathbf{F}$
$\mathbf{M}, \mathbf{C}, \mathbf{g}$	$\mathbf{Q} = \mathbf{M}\ddot{\mathbf{q}} + \mathbf{C} + \mathbf{g}$
$E = q_d - q$	servo error (desired minus actual)
M, D, N, J, F_i	mobility; 3/6; links incl. ground; joints; joint DoF
R_C, R_M	over-constraints (add); local mobility (subtract)
$u, v, \dot\theta$	chassis velocity in $\{C\}$; $\dot\varphi_i$ wheel spin
β_i, γ_i	wheel drive angle from x_C; roller angle
g, h, f	A*: cost so far, heuristic, $f = g + h$

Elementary rotations

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c & -s \\ 0 & s & c \end{bmatrix} \quad R_y(\theta) = \begin{bmatrix} c & 0 & s \\ 0 & 1 & 0 \\ -s & 0 & c \end{bmatrix} \quad R_z(\theta) = \begin{bmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Use: R_y's minus sign is bottom-left. A translation $t_x(a)$ is $\mathbf{I}$ with a in row 1, column 4.

Sample-paper map (W13 revision)

Q1 FK of RRR (ETS)	W2	p. 2
Q2 workspace sketch	W3	p. 2
Q3 IK derivation	W3	p. 2
Q4 equations of motion	W7	p. 2
Q5 four-bar IK	W4	p. 2
Q6–Q7 mobility	W1, W4	p. 2
Q8–Q9 cubic splines	W5	p. 2
Q10 omni robot	W10	p. 2

RECIPE 1 · Forward kinematics by ETS (W13 Q1)

1. Draw {S}; mark each joint axis and its *drawn* positive sense.
2. Walk base → tool. Translations: t_x, t_y, t_z. Joint: $R_{axis}(\pm\theta_i)$; minus if the drawn arrow opposes the right-hand rule about your axis.
3. End with the constant rotation that makes your last frame match {T}.
4. Group per joint and write each ${}^{i-1}\mathbf{T}_i$ (= "indicate intermediate frames").
5. Multiply; give at least the position column. Check: all $\theta = 0$ reproduces the drawing.

RECIPE 2 · Proximal DH table

1. z_i along joint axis i. Origin of {i} at the common normal to axis $i{+}1$ (or the intersection).
2. x_i along that normal towards axis $i{+}1$; $y_i = z_i \times x_i$.
3. Row i: a_{i-1} ($z_{i-1} \to z_i$ along x_{i-1}), α_{i-1} (about x_{i-1}), d_i ($x_{i-1} \to x_i$ along z_i), θ_i (about z_i).
4. ${}^{i-1}\mathbf{T}_i = R_x(\alpha_{i-1})t_x(a_{i-1})R_z(\theta_i)t_z(d_i)$; multiply rows.
5. Add a constant tool transform if {N} ≠ {tool}. Write "proximal (Craig)".

RECIPE 3 · Sketch the workspace (W13 Q2)

1. Find the joints that only *spin the plane* (base rotation) and the planar sub-arm.
2. Planar 2-link sub-arm: distance from its first joint ranges $|L_a - L_b| \ldots L_a + L_b$ (cosine rule) → annulus.
3. Spin the annulus about the base axis. If the axis passes through the annulus's centre → spherical shell; otherwise → a torus-like solid. Joint limits cut wedges; a prismatic joint extrudes.
4. Draw side and top views; label radii, centre height. Say "no joint limits, collisions ignored".

RECIPE 4 · Derive the inverse kinematics (W13 Q3)

1. Write FK first. Spherical wrist? Wrist centre $\mathbf{p}_W = \mathbf{p}_E - d_6\,\mathbf{R}\hat{z}$.
2. Top view: $\theta_1 = \operatorname{atan2}(y, x)$, second branch $\theta_1 + \pi$ (reach back, $r < 0$).
3. Side view: subtract base height; planar 2R in (r, z'):
   $\cos\theta_3 = \dfrac{r^2 + z'^2 - L_a^2 - L_b^2}{2L_aL_b}$ (relative angle; interior angle is $\pi - \theta_3$).
4. $\theta_3 = \operatorname{atan2}(\pm\sqrt{1-c_3^2}, c_3)$; $\theta_2 = \operatorname{atan2}(z', r) - \operatorname{atan2}(L_b s_3,\, L_a + L_b c_3)$.
5. Wrist: ${}^3\mathbf{R}_6 = ({}^0\mathbf{R}_3)^\top {}^0\mathbf{R}_6$ → Euler angles (two: wrist flip).
6. Count: shoulder × elbow (× flip). No solution if $|c_3| > 1$. Check one branch by FK.

RECIPE 5 · Mobility (W13 Q6, Q7)

1. $D = 3$ planar, 6 spatial (S, U joints or "still 3D" → 6).
2. Number bodies; ground = 1; a prismatic pair = 2 bodies → N.
3. List joints and F_i (R, P = 1; U, C = 2; S = 3) → J, $\sum F_i$.
4. $M = D(N - J - 1) + \sum F_i$. Implausible? Extended: $+R_C - R_M$.
5. R_M: idle spins (rod with S at both ends → 1 each). $R_C = \sum F_i - \sum u_k$ (u_k: distinct platform DoF per loop, union).

Delta: $6(11 - 15 - 1) + 39 = 9$, then $9 + 0 - 6 = 3$. *4RRR:* $-6 + 7 - 0 = 1$.

RECIPE 6 · Parallel robot: loop closure, IK, FK (W13 Q5)

1. Break the loop (at a point K, or both ends of a link) into serial chains.
2. Write the break point's pose from each chain; equate (3 equations at K; 1 length equation if broken at both ends).
3. **IK** (pose of K, ϕ given): $A = K - a(\cos\phi, \sin\phi)$, $\theta_1 = \operatorname{atan2}(A_y, A_x)$, $\varphi_2 = \phi - \theta_1$; $B = K + b(\cos\phi, \sin\phi)$, $\varphi_1 = \operatorname{atan2}(B_y, B_x - L_0)$, $\varphi_3 = \phi + \pi - \varphi_1$.
4. **FK** (θ_1 given): intersect circles (L_2 about A, L_3 about D): two assembly modes, or none.

RECIPE 7 · Trajectories (W13 Q8, Q9)

1. Rest-to-rest cubic: $a_0 = \theta_0$, $a_1 = 0$, $a_2 = \dfrac{3\Delta}{t_f^2}$, $a_3 = -\dfrac{2\Delta}{t_f^3}$; peak speed $1.5\Delta/t_f$ at $t_f/2$.
2. Given end speeds: $a_2 = \frac{3\Delta}{t_f^2} - \frac{2\dot\theta_0}{t_f} - \frac{\dot\theta_f}{t_f}$, $a_3 = -\frac{2\Delta}{t_f^3} + \frac{\dot\theta_0 + \dot\theta_f}{t_f^2}$.
3. Two segments, continuous acceleration: list 8 conditions (pos at both ends ×2, rest at start and end, $\dot\theta$ and $\ddot\theta$ match at the via) and solve.
4. Blends (LSPB): need $\ddot\theta \geq 4\Delta/t_f^2$; $t_b = \dfrac{t_f}{2} - \dfrac{\sqrt{\ddot\theta^2 t_f^2 - 4\ddot\theta\Delta}}{2\ddot\theta}$; linear speed $\ddot\theta t_b$.
5. Check: positions and speeds at both ends.

RECIPE 8 · Jacobian, singularity, statics

1. Position FK $\mathbf{x}(\mathbf{q}) \to \mathbf{J} = \partial\mathbf{x}/\partial\mathbf{q}$ (column j = how the tool moves for joint j). Geometric: revolute column $\hat{z}_j \times (\mathbf{p}_E - \mathbf{p}_j)$; prismatic $\hat{z}_j$.
2. Singular where $\det\mathbf{J} = 0$ (square) or rank drops. 2R: $\det = L_1L_2\sin q_2$.
3. $\dot{\mathbf{q}} = \mathbf{J}^{-1}\dot{\mathbf{x}}$; near-singular → huge joint rates.
4. Statics: $\boldsymbol{\tau} = \mathbf{J}^\top\mathbf{F}$ (same frame for J and F).

RECIPE 9 · Equations of motion by Lagrange (W13 Q4)

1. Coordinates $\mathbf{q}$; gravity direction drawn.
2. COM positions from FK; differentiate → velocities.
3. $K = \sum \frac{1}{2}m_i v_{C_i}^2 + \frac{1}{2}\omega_i^\top I_i \omega_i$; $P = \sum m_i g h_i$; $\mathcal{L} = K - P$.
4. Per joint: $\partial\mathcal{L}/\partial\dot{q}_i \to \frac{d}{dt}$ (product rule!) → minus $\partial\mathcal{L}/\partial q_i$.
5. Collect M ($\ddot{q}$ terms), C (velocity products), g. Check M symmetric, g has no $\dot{q}$, C = 0 at rest.

2R point masses: $M_{11} = m_1L_1^2 + m_2(L_1^2 + 2L_1L_2c_2 + L_2^2)$, $M_{12} = m_2(L_1L_2c_2 + L_2^2)$, $M_{22} = m_2L_2^2$.

RECIPE 10 · Control design (W8, W9)

1. Plant $m\ddot{x} + b\dot{x} + kx = f$ (or $\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C} + \mathbf{g} = \mathbf{Q}$).
2. Partition: $f = \alpha f' + \beta$, $\alpha = m$ (**M**), β = everything else.
3. Servo: $f' = \ddot{x}_d + k_v\dot{E} + k_pE$, $E = x_d - x \to \ddot{E} + k_v\dot{E} + k_pE = 0$.
4. Critical: $k_v = 2\sqrt{k_p}$ ($\omega_n = \sqrt{k_p}$, $\zeta = k_v/2\sqrt{k_p}$).
5. Constant disturbance: $E_{ss} = f_{dist}/(k_pm)$; add $k_i\int E$ to remove it (stable if $k_vk_p > k_i$).
6. Model error: hats on α, β; the error equation gets a right-hand side.

RECIPE 11 · Wheeled robot kinematics (W13 Q10)

1. For each wheel: $({}^Cx, {}^Cy)$, β (drive direction from x_C), γ, r.
2. Row: $r\dot{\varphi}_i = [\cos\beta \;\; \sin\beta]\,[u - \dot\theta\,{}^Cy,\; v + \dot\theta\,{}^Cx]^\top$ (+ $\tan\gamma$ term for omni).
3. Standard wheels: add no-slide $[-\sin\beta \;\; \cos\beta]\,(\cdot) = 0$.
4. Stack → invert. Inertial: $\boldsymbol{\nu} = \mathbf{J}(\theta)\boldsymbol{\zeta}$. ICR in {C}: $(-v/\dot\theta,\; u/\dot\theta)$.

RECIPE 12 · Planning by hand (W12)

1. Table: node, g, h, $f = g + h$, parent; open and closed sets.
2. Expand the lowest f (Dijkstra: lowest g); relax neighbours; record improvements.
3. Stop when the goal is *expanded*, not when discovered. Trace parents.
4. A* optimal if h admissible ($h \leq$ true cost); consistent: $h(n) \leq c(n, m) + h(m)$.
5. PRM: sample, connect, search (multi-query). RRT: grow towards random samples (single query). Both probabilistically complete, not optimal.

Words and chain types [p. 18–26]

Kinematics = geometry of motion, no forces. **Dynamics** = statics (no acceleration) + kinetics (acceleration). **DoF** = number of independent parameters fixing the configuration [p. 27]. **Mobility** M = DoF left after all joints act = minimum number of actuators [p. 39].
Closed chain: has a loop. **Open**: none. **Serial**: open, every link but first and last touches two others. **Parallel** [added]: end effector joined to base by several chains (closed loops): delta, Stewart.
Why loops [p. 26]: motors on the base so moving mass is small; larger load (legs share); interesting paths. A 6-joint arm is open and serial; a four-bar is closed and not serial.

Rigid-body DoF and joints [p. 27–36]

Rigid body: **3 DoF in 2D**, **6 DoF in 3D**. Point in 2D: 2. Pendulum in 2D: 1.

Joint	F	cuts 3D	cuts 2D
Revolute R, Prismatic P	1	5	2
Helical H	1	5	–
Cylindrical C, Universal U	2	4	–
Spherical S	3	3	–

"Cuts" = $D - F$ constraints removed. Planar mechanisms use only R and P. Two free planar links = $3 + 3 = 6$; one R pin removes $3 - 1 = 2$, leaving 4 [p. 36] [checked].

TRAP
Joint DoF ≠ mechanism mobility. Helical = 1 (pitch ties turn to slide). A 2D body is 3, not 2.

Grübler–Kutzbach [p. 39–41]

Mobility, form 2 (use this one)

$$M = D\,(N - J - 1) + \sum_{i=1}^{J} F_i$$

$D = 3$ planar, 6 spatial; N = links *including ground*; J = joints; F_i = DoF of joint i.
Use: any count. Serial chain: $N = J + 1$ so $M = \sum F_i$.

Form 1 [p. 40] (same sum, grouped)

$$M_{3D} = 6(N-1) - 5P_1 - 4P_2 - 3P_3 - 2P_4 - P_5, \quad M_{2D} = 3(N-1) - 2P_1 - P_2$$

P_i = number of joints with i DoF.
Use: cross-check; each joint removes $D - F$.

RECIPE Mobility count (W13 Q6, Q7)
1. D: all axes ⊥ page ⇒ 3. Any S, U, C or "still 3D" ⇒ 6.
2. Number links, **ground = 0**. Split every prismatic joint into two links (cylinder, rod).
3. List every joint with its F; get J and $\sum F_i$.
4. Write $M = D(N - J - 1) + \sum F_i$, then substitute.
5. Check: $M \geq 1$ for a machine; if $M \leq 0$ ask whether constraints are dependent.

Counts to know

Mechanism	D	N	J	M
Planar serial RPR [p. 42]	3	4	3	3
Four-bar [p. 44] (nb Q2)	3	4	4	1
3-RPR parallel [p. 46]	3	8	9	3
Five-bar [added]	3	5	5	2
Slider-crank [added]	3	4	4	1
Stewart 6-UPS [p. 48]	6	14	18	6
Delta, plain	6	11	15	9 → 3

WORKED four-bar, 3-RPR
Four-bar: $3(4 - 4 - 1) + 4 = 1$; form 1: $3(4 - 1) - 2 \cdot 4 = 1$. 3-RPR: $N = 1 + 1 + 3 \cdot 2 = 8$, $3(8 - 9 - 1) + 9 = 3$. [checked]

WORKED Stewart, D = 6
$N = 2 + 6 \cdot 2 = 14$, $J = 18$, $\sum F = 6(2+1+3) = 36$: $6(14 - 18 - 1) + 36 = 6$. One leg alone: $6(4 - 3 - 1) + 6 = 6$. Six SPS legs give 12 (leg spins): the U blocks them [added]. [checked]

WORKED delta, W13 Q7
$N = 2 + 3 \cdot 3 = 11$, $J = 3(1 + 4) = 15$, $\sum F = 3(1 + 4 \cdot 3) = 39$. Plain: $6(11 - 15 - 1) + 39 = \mathbf{9}$. Six rods spin freely: $R_M = 6$, $R_C = 0$:

$$M = D(N - J - 1) + \sum F_i + R_C - R_M = 9 + 0 - 6 = \mathbf{3}$$

Plain 9 if only plain G–K is quoted; 3 if the extended equation is given or the real platform DoF is asked. [checked]

TRAP
(1) Forgetting ground: 3-RPR gives 0, Stewart 0. (2) $D = 6$ on a planar drawing: four-bar −2. (3) $D = 3$ for Stewart: 21. (4) Constraints removed = F (not $D - F$): four-bar 5. (5) Prismatic counted as one link. (6) Stopping at 9 for the delta. [checked]

Where G–K fails [p. 49–51]

Three equal parallel legs (R at both ends), ground–legs–body: $N = 5$, $J = 6$, $M = 3(5 - 1) - 2 \cdot 6 = 0$, **but it moves: 1 DoF**. Rank of constraint Jacobian is 5, not 6: the third leg repeats a constraint. Delete a middle leg: $N = 4$, $J = 4$, $M = 1$ (four-bar). [checked]

RECIPE "Did you notice anything odd?"
1. Do the count, show it ($M = 0$).
2. State: the mechanism visibly moves, 1 DoF.
3. Reason (slide wording): G–K assumes *independent* constraints; here one leg is redundant.
4. Fix: remove one leg, recount ($N = 4$, $J = 4$, $M = 1$).
5. Caveat: only for equal, parallel legs; with unequal legs it locks and 0 is right.

Hoberman [p. 51]: 24 links, 36 R: $3(24 - 36 - 1) + 36 = -3$ (D=3), -42 (D=6); both negative, yet it moves [added]. [checked]

TRAP
False: "$M = 0$ means it cannot move"; "G–K always gives the true mobility". G–K is exact only for independent constraints; it can under-count (p.49) or over-count (delta, local spins).

Actuation [p. 37–38]

Controllable DoF = actuated. Compare actuators with total DoF:

equal	**fully actuated**
fewer	**under-actuated**: cheaper, simpler, lighter, but not every DoF can be commanded [p. 38]
more	**redundant** (slide wording)

6R arm, 6 motors: fully; 4 motors: under. Stewart, 6 actuators: fully. Delta, 3 motors, real $M = 3$: fully (plain 9 would wrongly say under). [checked]

SLIDE CHECK redundant
Slide 37 says "controllable > total ⇒ redundant", yet nb cell 8 calls a 7-DoF arm redundant (7 motors = 7 DoF). Standard meaning [added]: more joint DoF than the task needs (7 > 6 for a 3D pose; planar 3R positioning a point: 3 > 2). Write "redundant: more joint DoF than the 6-DoF task" and say which meaning you use.

Workspace [p. 52–57]

Reachable W_R: reached in ≥ 1 orientation. **Dexterous** W_D: reached in *all* orientations. Set by link geometry and joint limits [p. 52].

Relation [p. 57]

$$W_D \subseteq W_R \qquad \text{(i.e. } W_R \supseteq W_D\text{)}$$

Use: answer "$W_R \supseteq W_D$". $\subset$ strict and $\supset$ are traps (can be equal, e.g. both empty). $W_R \subseteq W_D$ is backwards.

Planar 2R and 3R reach [added]
2R: $|L_1 - L_2| \leq r \leq L_1 + L_2$. $L = (0.5, 0.3)$: annulus $0.2 \leq r \leq 0.8$ m. 3R, equal links L: W_R disc $r \leq 3L$; W_D disc $r \leq L$ (circle of radius L about the tip must fit in $0 \leq |\mathbf{w}| \leq 2L$). $L = 0.3$: 0.9 m and 0.3 m. [checked]
Use: fewer DoF than the pose (3 planar, 6 spatial) ⇒ W_D empty (any 2R arm).

Arm	Joints	Workspace shape [p. 53–56]
PUMA	RRR	thick spherical shell, wedge cut by joint limits
SCARA	RRP	flat ring (annulus), thickened by vertical stroke
Cylindrical	RPP	thick-walled cylinder sector
Cartesian	PPP	box, edges = the three strokes

Shapes read from the slide figures.

What the unit covers [p. 58–62]

FK (W2): $\mathbf{q} \to$ pose. IK (W3): pose $\to \mathbf{q}$. Velocity (W6), dynamics (W7–W9), trajectories (W5). Pose = 3 + 3 numbers = a 4×4 homogeneous matrix. Frames: B base, W wrist, T tool, S station, G goal, C camera [p. 61].

Practical (optional)

`test.py`: CoppeliaSim remote API, drives joint `/Joint1` to 60° for 3 s; one revolute joint. No saved outputs.

W1

Homogeneous transform [p. 12]

Pose of {B} in {A}

$$ {}^A\mathbf{T}_B = \begin{bmatrix} {}^A\mathbf{R}_B & {}^A\mathbf{t}_B \\ \mathbf{0}^\top & 1 \end{bmatrix}, \quad {}^A\mathbf{P} = {}^A\mathbf{T}_B\, {}^B\mathbf{P}, \quad {}^B\mathbf{T}_A = \begin{bmatrix} \mathbf{R}^\top & -\mathbf{R}^\top\mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix} $$

Use: columns of $\mathbf{R}$ = B's unit axes in A; $\mathbf{t}$ = B's origin in A. Inverse: transpose $\mathbf{R}$, and the translation becomes $-\mathbf{R}^\top\mathbf{t}$ (not $-\mathbf{t}$).

Compounding [p. 13]

${}^A\mathbf{T}_C = {}^A\mathbf{T}_B\, {}^B\mathbf{T}_C$ (inner indices cancel).
Right-multiply = next move about/along the *current* frame.
Left-multiply = about/along the *fixed* frame.

WORKED order matters

$R_x(30°)t_z(2)$: C's origin = $(0, -1.000, 1.732)$.
$t_z(2)R_x(30°)$: origin = $(0, 0, 2)$. [checked]

Planar RRR by trigonometry [p. 6–9]

$$ x = \textstyle\sum_k L_k \cos\phi_k, \quad y = \sum_k L_k \sin\phi_k, \quad \phi_k = q_1 + \cdots + q_k, \quad q_E = \phi_3 $$

WORKED

$L = (0.5, 0.4, 0.2)$ m, $q = (30°, 45°, -60°)$: $\phi = (30, 75, 15)°$, $x = 0.7297$, $y = 0.6881$ m, $q_E = 15°$. [checked]

TRAP

Using the joint angles themselves (cos 45°) instead of the running sums. At $q = 0$ everything agrees, so test a non-zero pose.

Elementary Transform Sequence (ETS) [p. 15–19]

Six moves: $t_x, t_y, t_z, R_x, R_y, R_z$, each with a constant or a joint variable. Walk base → tool, write every move in order, multiply.

RECIPE FK by ETS (W13 Q1)

1. Draw {S}. Mark every joint axis and its drawn positive direction.
2. Translations to the first joint; then $R_{axis}(\pm\theta_i)$ (minus if the drawn arrow is opposite to the right-hand rule).
3. Continue to the tool; finish with the constant rotation that matches the drawn {T}.
4. Group per joint: ${}^{i-1}\mathbf{T}_i$ = "indicate intermediate frames".
5. Multiply (at least the position column). Check at all $\theta = 0$.

Furuta pendulum [p. 19]

$$ {}^B\mathbf{T}_E = t_z(L_0)R_z(\theta_1)t_x(L_1)\, R_x(\theta_2)t_z(-L_2)R_y(\pi) $$

$x = L_1c_1 - L_2s_1s_2$, $y = L_1s_1 + L_2c_1s_2$, $z = L_0 - L_2c_2$ [checked]

WORKED

$L = (0.5, 0.4, 0.3)$, $\theta = (60°, 45°)$: $\mathbf{p} = (0.0163, 0.4525, 0.2879)$ m. Workspace: $\rho^2 + (z - L_0)^2 = L_1^2 + L_2^2$, a spherical band $|z - L_0| \le L_2$ (not a torus). [checked]

Proximal (modified, Craig) DH [p. 21–36]

Link transform [p. 28]

$$ {}^{i-1}\mathbf{T}_i = R_x(\alpha_{i-1})\, t_x(a_{i-1})\, R_z(\theta_i)\, t_z(d_i) $$

$$ = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Use: then ${}^0\mathbf{T}_n = {}^0\mathbf{T}_1\, {}^1\mathbf{T}_2 \cdots {}^{n-1}\mathbf{T}_n$. Revolute variable θ_i; prismatic variable d_i.

a_{i-1}	distance $z_{i-1} \to z_i$ along x_{i-1} (common normal), ≥ 0
α_{i-1}	angle $z_{i-1} \to z_i$ about x_{i-1}
d_i	distance $x_{i-1} \to x_i$ along z_i (signed)
θ_i	angle $x_{i-1} \to x_i$ about z_i

RECIPE Assign proximal frames [p. 29]

1. Joint axes. For axes i, $i{+}1$ find the common normal (or the intersection).
2. Origin of $\{i\}$ where the normal meets axis i; z_i along axis i.
3. x_i along the normal towards axis $i{+}1$ (if the axes intersect: normal to their plane).
4. y_i completes a right-handed frame.
5. $\{0\} = \{1\}$ at $q_1 = 0$; choose $\{N\}$ to zero as many parameters as possible.
6. Row i: a_{i-1}, α_{i-1} look back (axis $i{-}1 \to i$); d_i, θ_i along axis i.

WORKED Furuta DH table [p. 34]

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	L_0	$q_1{+}90°$
2	0	90°	L_1	q_2
3	0	90°	L_2	0

$\{3\} \ne \{E\}$: ${}^3\mathbf{T}_E = R_z(-90°)$ makes the product equal the ETS for all q. [checked]

Franka Panda, published proximal DH

a: 0, 0, 0, 0.0825, −0.0825, 0, 0.088; α: 0, −90, 90, 90, −90, 90, 90°
d: 0.333, 0, 0.316, 0, 0.384, 0, 0; flange $d = 0.107$. At $q = 0$: flange at $(0.088, 0, 0.926)$ m. [checked]

Distal (standard) DH [p. 37–40]

${}^{i-1}\mathbf{T}_i = R_z(\theta_i)\, t_z(d_i)\, t_x(a_i)\, R_x(\alpha_i)$; frame $\{i\}$ sits on axis $i{+}1$; joint i turns about z_{i-1}.

TRAP

Never mix: a proximal table in the distal matrix gives a different robot. Corke's toolbox `DHRobot` is distal (`RevoluteMDH` for proximal). Write "proximal (Craig) DH" on your answer.

Why 4 parameters? [p. 42–44]

6 numbers relate two bodies; DH imposes 2 rules (proximal: $x_i \perp z_{i+1}$ and x_i meets z_{i+1}), so $6 - 2 = 4$. Cost: a frame may lie off the link. A good model is **complete**, **continuous** (DH fails for nearly parallel axes: the common normal jumps) and **minimal**.

URDF [p. 45–47]

XML of `<link>`s and `<joint>`s: joint `origin` = constant ETS steps; `axis` = which R the joint is. Slide example = $R_z(q_1)t_x(1)R_z(q_2)t_x(1)$.

Notebook exercises

Two frames (Q3). Origin $(3, 5, 2)$; $x_B = -z_A$, $z_B = +y_A$, so $y_B = -x_A$:

$$ {}^A\mathbf{T}_B = \begin{bmatrix} 0 & -1 & 0 & 3 \\ 0 & 0 & 1 & 5 \\ -1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ = t(3, 5, 2)\, R_y(90°)\, R_x(-90°) \text{ [checked]} $$

PUMA by ETS (Q5).

$$ t_z(L_b)R_z(q_1)R_y(q_2)t_z(L_u)R_y(q_3)\, t_z(L_f)R_z(q_4)R_y(q_5)R_z(q_6) $$

Coincident axes: no translation between consecutive rotations (shoulder; spherical wrist). The wrist centre does not move with $q_4..q_6$. [checked]

SLIDE CHECK notebook

Cell 35 writes `tz(-30)` for −0.30 m, so its printed pose is wrong. A filled-in Q3 answer `ty(5) tx(3) Ry(90) Rx(90)` misses $t_z(2)$ and the sign of the last rotation.

W13 Q1 arm (answer shape)

$$ {}^S\mathbf{T}_T = R_z(\theta_1)t_z(L_1{+}L_2)R_y(-\theta_2)t_x(L_3)\, R_y(-\theta_3)t_x(L_4)R_x(90°) $$

$$ {}^S\mathbf{T}_T = \begin{bmatrix} c_1c_{23} & -c_1s_{23} & s_1 & c_1r \\ s_1c_{23} & -s_1s_{23} & -c_1 & s_1r \\ s_{23} & c_{23} & 0 & L_1{+}L_2{+}L_3s_2{+}L_4s_{23} \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$r = L_3c_2 + L_4c_{23}$. Check $\theta = 0$: tip $(L_3{+}L_4, 0, L_1{+}L_2)$, $y_T = z_S$. [checked]

W2

The IK problem [p. 7–20]

Joint space $\mathcal{C}$ holds $\mathbf{q}$ (two R joints: $\mathbb{S}^1 \times \mathbb{S}^1$). Task space $\mathcal{T}$ holds the pose $\xi = {}^0\mathbf{T}_E$ (or x, y, z, roll, pitch, yaw; or position + quaternion) [p. 9]. FK: $\xi = \mathcal{K}(\mathbf{q})$. IK: $\mathbf{q} = \mathcal{K}^{-1}(\xi)$ [p. 11]. $\mathcal{K}$ is many-to-one and nonlinear, so IK has 0, 1, several or infinitely many solutions.

Choosing among solutions [p. 18–19]: closest to the current configuration (smallest joint change; prefer moving small distal joints), or the one that avoids obstacles.

Methods [p. 20]: closed form (algebraic, geometric, polynomial; non-iterative; all solutions) or numerical (iterative; slower; handles long chains; returns a nearby answer when no exact one exists).

A 6-DoF arm has no continuous redundancy for a 6-number pose [p. 33] but has several discrete solutions: PUMA 560 up to **8** (shoulder L/R × elbow up/down × wrist flip) [added].

Does a solution exist? [p. 12–16]

Reachable W_R: some orientation. Dexterous W_D: all orientations. A full 3D pose needs the **dexterous** workspace.

2R reach [p. 15–16]

$$|L_1 - L_2| \le r \le L_1 + L_2$$

$L = (1.0, 0.7)$: annulus $0.3 \le r \le 1.7$ m. Joint limits $[0, \pi/2]$ on both: outer edge $q_2 = 0$ (1.7 m); inner edge $q_2 = \pi/2$ ($\sqrt{1 + 0.49} = 1.221$ m); side edges are 0.7 m circles at $q_1 = 0, \pi/2$.

Use: "sketch the workspace": give both radii and why (cosine rule). A 2R arm has no dexterous workspace (orientation is fixed by position).

Ex.: is $(0, 0.2)$ reachable? $\cos q_2 = (0.04 - 1.49)/1.4 = -1.036 < -1$: no ($0.2 < 0.3$). [checked]

Planar RP arm [p. 23–24]

$$S = \sqrt{x^2 + y^2}, \qquad q_1 = \operatorname{atan2}(y, x)$$

Slide writes $\tan^{-1}(y/x)$: use atan2. If S may be negative, $(-S,\ q_1 + \pi)$ is a second solution. *Ex.:* $(0.3, 0.4) \to S = 0.5$, $q_1 = 53.130°$. [checked]

Planar RR arm, algebraic [p. 25–30]

2R inverse kinematics

FK: $x = L_1 c_1 + L_2 c_{12}$, $y = L_1 s_1 + L_2 s_{12}$. Square and add:

$$\cos q_2 = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2 L_1 L_2}, \quad q_2 = \operatorname{atan2}\big(\pm\sqrt{1 - \cos^2 q_2},\ \cos q_2\big)$$

$$q_1 = \operatorname{atan2}(y, x) - \operatorname{atan2}\big(L_2 \sin q_2,\ L_1 + L_2 \cos q_2\big)$$

Use: two branches (the ± = elbow) [p. 28]. If $|\cos q_2| > 1$ the point is out of reach.

Expand compound angles [p. 30]: $x = (L_1 + L_2 c_2) c_1 - (L_2 s_2) s_1$, $y = (L_2 s_2) c_1 + (L_1 + L_2 c_2) s_1$: the point is the vector $(L_1 + L_2 c_2,\ L_2 s_2)$ rotated by q_1. General tool: $A \cos\varphi + B \sin\varphi = C \Rightarrow \varphi = \operatorname{atan2}(C, \pm\sqrt{A^2 + B^2 - C^2}) - \operatorname{atan2}(A, B)$.

atan2 [p. 29]: angle of (x, y) in $(-\pi, \pi]$. $\tan^{-1}(y/x)$ gives $45°$ for both $(1, 1)$ and $(-1, -1)$; the second is $-135°$. [checked]

RECIPE IK of any 2-link planar sub-problem (W13 Q3)

1. Write FK. Find the shoulder point and the target in the plane (subtract offsets and base height first).
2. $\cos q_2$ by square-and-add or cosine rule. Check $|\cos q_2| \le 1$.
3. $q_2 = \operatorname{atan2}(\pm\sqrt{1 - c_2^2},\ c_2)$: two branches.
4. $q_1 = \operatorname{atan2}(y, x) - \operatorname{atan2}(L_2 s_2,\ L_1 + L_2 c_2)$ for each branch.
5. Check one branch by FK.

WORKED 2R, both elbows

$L = (1.0, 0.7)$, target $(1.2, 0.8)$: $\cos q_2 = (1.44 + 0.64 - 1 - 0.49)/1.4 = 0.421429$, $q_2 = \pm 65.075°$; $\operatorname{atan2}(0.8, 1.2) = 33.690°$.
$q_2 = +65.075°$: $q_1 = 33.690 - 26.114 = \mathbf{7.576}°$.
$q_2 = -65.075°$: $q_1 = 33.690 + 26.114 = \mathbf{59.804}°$.
FK check: $\cos 7.576° + 0.7 \cos 72.651° = 0.9913 + 0.2087 = 1.2000$. [checked]

Planar RR arm, geometric [p. 31]

Triangle OAB: O base, A elbow, B tip, $|OB| = \sqrt{x^2 + y^2}$. Interior angle at A is $\pi - q_2$.

$$x^2 + y^2 = L_1^2 + L_2^2 - 2 L_1 L_2 \cos(\pi - q_2) = L_1^2 + L_2^2 + 2 L_1 L_2 \cos q_2$$

Sine rule for the angle μ at O: $\sin\mu / L_2 = \sin q_2 / \sqrt{x^2 + y^2}$. With $\sigma = \operatorname{atan2}(y, x)$, $q_1 = \sigma - \mu$. μ takes the sign of q_2, so for $q_2 < 0$, $q_1 = \sigma + |\mu|$. Use $\mu = \operatorname{atan2}(L_2 s_2,\ L_1 + L_2 c_2)$ and both branches are automatic. [checked]

Full marks if the triangle is drawn and labelled. Write "interior angle at elbow = $\pi - q_2$".

TRAP

Cosine rule with the joint angle: $x^2 + y^2 = L_1^2 + L_2^2 - 2 L_1 L_2 \cos q_2$ gives $\pm 114.9° = 180° - 65.075°$ for the target above (supplement). Slide 36 prints this slip. [checked]

Planar RRR with tool angle (nb Q10)

RECIPE 3R: position and tool angle φ

1. Wrist point = $(x - L_3 \cos\varphi,\ y - L_3 \sin\varphi)$.
2. Solve the 2R problem for θ_1, θ_2 to the wrist point.
3. $\theta_3 = \varphi - \theta_1 - \theta_2$.

WORKED

$L = (1.0, 0.7, 0.3)$, $(x, y) = (1.2, 0.9)$, $\varphi = 60°$: wrist $(1.0500, 0.6402)$; $\theta_2 = \pm 89.085°$; $\theta_1 = -3.319°$ or $66.061°$; $\theta_3 = -25.766°$ or $83.024°$. Both reproduce $(1.2, 0.9, 60°)$. [checked]

6-DoF arm, spherical wrist [p. 32–38]

Last three axes meet at the wrist centre W [p. 34]; joints 4–6 cannot move W. Solve q_1–q_3 from W, then q_4–q_6 from the leftover rotation.

RECIPE Decoupled IK (W13 Q3)

1. Wrist centre [added]: $\mathbf{p}_W = \mathbf{p}_E - d_6\, {}^0\mathbf{R}_E [0\ 0\ 1]^\top$.
2. **Top view** [p. 35]: $q_1 = \operatorname{atan2}(y_W, x_W)$; second branch $q_1 + \pi$.
3. **Side view** [p. 36]: shoulder height H, upper arm G, forearm F, $r_W^2 = x_W^2 + y_W^2$.

$$\cos q_3 = \frac{r_W^2 + (z_W - H)^2 - F^2 - G^2}{2FG}, \quad q_3 = \operatorname{atan2}(\pm\sqrt{1 - c_3^2}, c_3)$$

4. [p. 37]: $\sigma = \operatorname{atan2}(z_W - H, r_W)$, $\mu = \operatorname{atan2}(F \sin q_3,\ G + F \cos q_3)$, $q_2 = \sigma - \mu$ (signed, anticlockwise positive).
5. **Wrist** [p. 38]: ${}^3\mathbf{R}_6 = ({}^0\mathbf{R}_3)^\top\, {}^0\mathbf{R}_6$, read q_4–q_6 as Euler angles.
6. Count: 2 (q_1) × 2 (q_3) × 2 (wrist flip) = 8. Check by FK.

Z–Y–Z wrist [added]

${}^3\mathbf{R}_6 = R_z(q_4) R_y(q_5) R_z(q_6)$:

$$q_5 = \operatorname{atan2}\big(\sqrt{r_{13}^2 + r_{23}^2},\ r_{33}\big)$$

$$q_4 = \operatorname{atan2}(r_{23}, r_{13}), \quad q_6 = \operatorname{atan2}(r_{32}, -r_{31})$$

Use: other sign of q_5 = wrist flip (q_4, q_6 each change by π). $q_5 = 0$: singular, only $q_4 + q_6$ is fixed.

SLIDE CHECK elbow sign, pp. 36–37

Slide 36 writes $\cos(\pi - q_3)$ on the left; by the cosine rule it equals $+\cos q_3$. Slide 37's $q_2 = \sigma + \mu$ is right only if q_3 is a clockwise bend (drawn elbow-up) and $\mu > 0$. In the exam define your positive direc-

W3

tion on the sketch and keep it; signed anticlockwise angles give the formulas above. [checked]

WORKED wrist-arm position part

$H = 0.5$, $G = 0.4$, $F = 0.3$; $W = (0.4813, 0.4039, 0.5974)$: $q_1 = 40°$, $r_W = 0.6283$, $z_W - H = 0.0974$, $\cos q_3 = (0.3948 + 0.0095 - 0.09 - 0.16)/0.24 = 0.6428$, $q_3 = \pm 50°$.
$q_3 = -50°$ (elbow up): $\sigma = 8.811°$, $\mu = -21.189°$, $q_2 = \mathbf{30.0}°$.
$q_3 = +50°$ (elbow down): $q_2 = -12.378°$. With $q_1 + \pi$ (220°): 4 position solutions; ×2 wrist flips = 8. [checked]

Numerical IK [p. 40–52]

Goal potential and Newton–Raphson

$\mathbf{q}^* = \arg\min_{\mathbf{q}} \|\mathcal{K}(\mathbf{q}) \ominus \xi^*\|$ [p. 41]. Position only [p. 42]: $G = (x^* - x)^2 + (y^* - y)^2$. Scalar NR [p. 45–47]: $q^{k+1} = q^k - G'(q^k)^{-1} G(q^k)$; stop when the relative change $< \varepsilon$.

Jacobian update [p. 48–52]

$\mathbf{e} = \mathbf{x}^* - \mathcal{F}(\mathbf{q}^i)$, $\mathbf{J} = \partial\mathcal{F}/\partial\mathbf{q} \in \mathbb{R}^{m \times n}$:

$$\mathbf{q}^{i+1} = \mathbf{q}^i + \mathbf{J}^\dagger \mathbf{e}$$

$$\mathbf{J}^\dagger = \begin{cases} \mathbf{J}^{-1} & n = m \\ \mathbf{J}^\top (\mathbf{J}\mathbf{J}^\top)^{-1} & n > m \text{ “fat”: smallest } \Delta\mathbf{q} \\ (\mathbf{J}^\top \mathbf{J})^{-1} \mathbf{J}^\top & n < m \text{ “tall”: least squares} \end{cases}$$

2R: $\mathbf{J} = \begin{bmatrix} -L_1 s_1 - L_2 s_{12} & -L_2 s_{12} \\ L_1 c_1 + L_2 c_{12} & L_2 c_{12} \end{bmatrix}$, and for a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

Use: each iteration; invert the 2×2 by hand.

WORKED one Newton step

2R $(1.0, 0.7)$, target $(1.2, 0.8)$, $\mathbf{q}^0 = (0.5, 0.5)$ rad. FK $= (1.2558, 1.0685)$, $\mathbf{e} = (-0.0558, -0.2685)$.
$\mathbf{J} = \begin{bmatrix} -1.0685 & -0.5890 \\ 1.2558 & 0.3782 \end{bmatrix}$, $\det \mathbf{J} = 0.3355$.
$\mathbf{J}^{-1} = \frac{1}{0.3355} \begin{bmatrix} 0.3782 & 0.5890 \\ -1.2558 & -1.0685 \end{bmatrix}$, $\Delta\mathbf{q} = (-0.5341,\ 1.0635)$.
$\mathbf{q}^1 = (-0.0341,\ 1.5635)$ rad, $\|\mathbf{e}\|$: $0.2742 \to 0.2181$. Converges to $(7.576°, 65.075°)$ in 5 iterations. [checked]

TRAP

Using $\mathbf{J}^\top$ instead of $\mathbf{J}^{-1}$ is gradient descent: different direction. Tall $\mathbf{J}$ with $\mathbf{J}^\top(\mathbf{J}\mathbf{J}^\top)^{-1}$ is wrong ($\mathbf{J}\mathbf{J}^\top$ singular).

Why numerical IK can fail (name three): target outside the workspace ($G = 0$ impossible); local minimum, or divergence near a singular $\mathbf{J}$; the answer depends on the initial guess. A redundant arm has infinitely many solutions.

Closed form exists [p. 55–56]: any 6-DoF R/P serial chain is solvable, but analytically when three neighbouring axes meet at a point (spherical wrist; Pieper) or three consecutive axes are parallel [added]. Slide 54 “unique solution to all positions” means a *complete* set (slides 17, 20 show several).

How many solutions? (nb Q12, Q13)

Count branch by branch. Notebook 3-DoF arm (θ_1, horizontal θ_2 offset L_2, slider d_3): θ_1 faces the point or $+\pi$ (2); in the plane $|d_3| = \sqrt{\rho^2 - L_2^2}$ gives two (θ_2, d_3) pairs. Up to 4 if d_3 may be negative, 2 if $d_3 \geq 0$: state the assumption [added]. PUMA 560, full pose: up to 8.

Practical (optional)

`rrp_ik.py` is really a planar **2R** (L_1=L_2=1): analytic IK (both elbows), 2×2 Jacobian, Newton–Raphson. $\cos q_2$ is clipped to $[-1, 1]$, so an unreachable target silently returns the nearest boundary pose. W3 folder: PUMA 560 FK (proximal DH and distal ETS).

Serial vs parallel [p. 9–19, 74]

Parallel: "end effector connected to the base by multiple parallel kinematic chains" [p. 12]. More joints than DoF, so some joints are passive. Examples: Gough tyre tester, Stewart flight simulator (1965), ABB FlexPicker delta, surgical hexapod.

	Serial	Parallel
Chains	open	closed
Joints	all active	active + passive
FK	simple	**complex**
IK	complex	**simple**
Joint errors	accumulate	do not
Workspace	large	small
Payload/weight	low	very high
Moving inertia	high	low

Use: scenario "which architecture?": fast pick-and-place → delta; heavy simulator → Stewart; precise surgical tool → hexapod; justify by stiffness, low moving mass, non-accumulating errors, and note the small workspace.

Workspace of parallel robots [p. 20–23]

Four kinds [p. 21]: **reachable** (some orientation), **dexterous** (all), **constant-orientation** (points reachable with one given orientation), **orientation workspace** (all orientations at one given point). Compute by *discretisation* (step each actuator, take the boundary; slow, accuracy set by step) or *geometry* (intersect the regions each actuator allows) [p. 22–23].
Two-leg planar, prismatic legs: each leg confines the point to an annulus $r_{min} \le r \le r_{max}$ about its base pivot; **workspace = overlap of the annuli**. Sketch: draw each annulus, shade the intersection.

Mobility: G–K [p. 24–34]

Plain Grübler–Kutzbach [p. 25]

$$M = D\,(N - J - 1) + \sum_{i=1}^{J} F_i$$

$D = 3$ planar, 6 spatial; N includes ground; $F_R = F_P = 1$, $F_U = 2$, $F_S = 3$. Derivation: $D(N-1)$ free DoF, each joint removes $D - F_i$.
Use: first estimate for any mechanism.

RECIPE Mobility of a parallel robot (W13 Q6, Q7)

1. Planar or spatial? Motion out of the page means $D = 6$ ("still 3D despite this depiction", W13 p.14).
2. Number every rigid body, **ground = 1**. A prismatic pair is two bodies.
3. List every joint and its F_i. Three bodies at one pin = two joints.
4. Substitute. If M disagrees with the obvious motion, look for idle spins (R_M) or repeated constraints (R_C).

Mechanism	D	N	J	$\sum F$	M
3-RPR [p. 26]	3	8	9	9	3
Delta R–U–U legs [p. 30]	6	8	9	15	3
Delta, parallelogram [p. 33]	6	11	15	39	9 (!)
Four-bar [added]	3	4	4	4	1
Gough–Stewart 6-UPS [added]	6	14	18	36	6

Delta R–U–U: $\sum F = (1+2+2)\cdot 3 = 15$, $6(8-9-1)+15 = 3$. Common delta: $\sum F = (1+4\cdot 3)\cdot 3 = 39$, $6(11-15-1)+39 = 9$. Stewart: $6(14-18-1)+36 = 6$. 3-RPR: $3(8-9-1)+9 = 3$. [checked]
The simplest delta is not built: stiff, play-free universal joints are hard to make and play lowers the usable frequency [p. 31].

SLIDE CHECK slide 29
A handwritten note reads $D(N-J+1)$. The printed formula and every worked count use $D(N-J-1)$. With +1 the simple delta would give $6(8-9+1)+15 = 15$. [checked]

TRAP
Slider-crank (ground, crank, rod, slider; 3 R + 1 P) with $D = 6$: $6(4-4-1)+4 = -2$, "a structure". It is planar: $3(4-4-1)+4 = 1$. Spatial example: $N = 6$, $J = 7$ (5 R, 1 U, 1 S): $\sum F = 10$, $M = 6(6-7-1)+10 = -2$: report it, then look for R_C before saying "structure". [checked]

Extended G–K [p. 35–43]

Equation [p. 35]

$$M = D(N - J - 1) + \sum_{i=1}^{J} F_i + R_C - R_M$$

$$R_C = \sum_{i=1}^{J} F_i - \sum_{k=1}^{L} u_k$$

R_M = **local mobility**: internal motion that does not move the output (a rod with a ball joint at each end spins about its own axis). Plain G–K counts it, so **subtract**. R_C = **over-constraints**: joints repeating a constraint, so freedoms are removed twice: **add back**. L = number of loops; u_k = distinct platform DoF loop k allows (each distinct DoF counted once across loops).
Use: plain M implausible (delta 9; 4RRR −6). Say which of R_C, R_M applies and why.

WORKED common delta [p. 33–36]
Plain 9, but the platform only translates (3 DoF). Each of the 6 rods spins about its own axis: $R_M = 6$. No repeated constraints: $R_C = 0$. $M = 9 + 0 - 6 = \mathbf{3}$. [checked]

WORKED 4RRR [p. 37–43]
Two square plates, four RR legs: $N = 10$, $J = 12$, $\sum F = 12$. Plain: $6(10-12-1)+12 = -6$ (impossible). Loop 1 allows $\{T_y, T_z, R_x\}$, loop 2 $\{T_x, T_z, R_y\}$; union $\{T_x, T_y, T_z, R_x, R_y\}$, $\sum u_k = 5$ (T_z once). $R_C = 12 - 5 = 7$, $R_M = 0$: $M = -6 + 7 - 0 = \mathbf{1}$ (platform only moves in T_z). [checked]

"Explain" marks: plain G–K assumes every constraint is independent (R_C fixes that) and every DoF moves the output (R_M fixes that).

Active, passive joints; loop closure [p. 45–58]

Configuration $\mathbf{q} = (\boldsymbol{\theta}, \boldsymbol{\varphi})$: n active $\boldsymbol{\theta}$, m passive $\boldsymbol{\varphi}$ [p. 45]. Gough–Stewart: 6 active, 30 passive variables [p. 58].

RECIPE Loop-closure equations [p. 49]

1. Break the loop into serial chains (choose the break giving the fewest equations).
2. Write the pose of the break point from each side.
3. Equate position and orientation. Count unknowns: active θ, passive φ.

Four-bar, break link 2 at K [p. 54]
a, b = distances from K to link 2's ends ($L_2 = a + b$); D at $(L_0, 0)$:

$$L_1 c_{\theta_1} + a\cos(\theta_1+\varphi_2) = L_0 + L_3 c_{\varphi_1} + b\cos(\varphi_1+\varphi_3)$$
$$L_1 s_{\theta_1} + a\sin(\theta_1+\varphi_2) = L_3 s_{\varphi_1} + b\sin(\varphi_1+\varphi_3)$$
$$\theta_1 + \varphi_2 = \varphi_1 + \varphi_3 + \pi$$

Three equations: 1 active (θ_1), 3 passive ($\varphi_1, \varphi_2, \varphi_3$).

Freudenstein, break at both ends [p. 57]

$$\eta_1 = (L_1 c_{\theta_1} - L_0 - L_3 c_{\varphi_1})^2 + (L_1 s_{\theta_1} - L_3 s_{\varphi_1})^2 - L_2^2 = 0$$

Use: only link 2's length must be preserved: 1 equation in θ_1, φ_1. Given θ_1, two φ_1 (two assembly modes).

Forward kinematics (hard) [p. 59–67]

Given active joints, solve loop closure for the passive ones, then the pose. No general method [p. 60]; solutions can be multiple or imaginary, so numerics are common [p. 67].
Four-bar by geometry [added]: $A = L_1(\cos\theta_1, \sin\theta_1)$ and $D = (L_0, 0)$ fixed; B is on the circle L_2 about A and L_3 about D. Two intersections = two **assembly modes**; one = limit; none = cannot assemble at that θ_1.

Slide route [p. 62–66]: from $K = (x, y)$ find $\delta = \theta_1 + \varphi_2$

$A_1 = x - L_0$, $B_1 = y$, $C_1 = \dfrac{(x-L_0)^2 + y^2 + b^2 - L_3^2}{2b}$
$A_2 = x$, $B_2 = y$, $C_2 = \dfrac{L_1^2 - a^2 - x^2 - y^2}{2a}$
$A_i\cos\delta + B_i\sin\delta + C_i = 0$. Cramer:

$$\cos\delta = \frac{B_1C_2 - B_2C_1}{A_1B_2 - A_2B_1}, \quad \sin\delta = \frac{A_2C_1 - A_1C_2}{A_1B_2 - A_2B_1}$$

$\delta = \operatorname{atan2}(\sin\delta, \cos\delta)$. Consistency (slide 65): $(A_1B_2 - A_2B_1)^2 = (A_1C_2 - A_2C_1)^2 + (B_1C_2 - B_2C_1)^2$.

SLIDE CHECK slide 65
Printed:

$$\delta = \tan^{-1}\frac{A_1C_2 - A_2C_1}{(B_1C_2 - B_2C_1) + (A_1B_2 - A_2B_1)}$$

Tangent-half-angle gives

$$\delta = 2\tan^{-1}\frac{A_2C_1 - A_1C_2}{(B_1C_2 - B_2C_1) + (A_1B_2 - A_2B_1)}$$

opposite sign and a factor 2. Test: slide form 0.300°, true −0.600°. Use the atan2 of the Cramer sine and cosine. [checked]

WORKED four-bar FK, same as Sample Q5

$L_0 = 1.0$, $L_1 = 0.5$, $L_3 = 0.6$, $a = 0.2$, $b = 0.25$ ($L_2 = 0.45$), $\theta_1 = 70°$:

Mode	K (m)	φ	φ_1
B above AD	$(0.3710, 0.4678)$	$-0.600°$	$129.175°$
B below AD	$(0.2756, 0.2993)$	$-58.486°$	$171.739°$

Below mode by hand: $A = (0.1710, 0.4698)$, $|AD| = 0.9529$. $x = (L_2^2 - L_3^2 + |AD|^2)/(2|AD|) = (0.2025 - 0.36 + 0.9080)/1.9058 = 0.3938$, $h = \sqrt{0.45^2 - 0.3938^2} = 0.2178$. $B = (0.4062, 0.0862)$, $\varphi = -58.486°$, $K = A + 0.2(\cos\varphi, \sin\varphi) = (0.2756, 0.2993)$. [checked]
Non-Grashof ($0.45 + 1.0 > 0.5 + 0.6$): no link turns fully; assembles only for $|\theta_1| \le 81.52°$ ($|AD| \le L_2 + L_3$).

Inverse kinematics (easy) [p. 68–73]

Given the output pose, each leg is a serial chain with known end pose: solve each leg alone, no loop closure [p. 69]. A redundant leg gives a non-unique answer.

RECIPE Four-bar IK, from $K = (x, y)$ and φ (W13 Q5) [p. 70–72]

1. Link 2 ends: $A = K - a(\cos\varphi, \sin\varphi)$, $B = K + b(\cos\varphi, \sin\varphi)$.
2. $\theta_1 = \text{atan2}(y - a\sin\varphi,\ x - a\cos\varphi)$.
3. $\varphi_2 = \varphi - \theta_1$.
4. $\varphi_1 = \text{atan2}(y + b\sin\varphi,\ x + b\cos\varphi - L_0)$.
5. $\varphi_3 = \varphi + \pi - \varphi_1 \pmod{2\pi}$.
6. Check: with the table above $K = (0.3710, 0.4678)$, $\varphi = -0.600°$ returns $\theta_1 = 70°$, $\varphi_1 = 129.175°$.

Gough–Stewart leg (homework [p. 73])

Leg i: base point $\mathbf{b}_i$ in B, platform point $\mathbf{p}_i$ in P, ${}^B\mathbf{T}_P = [\mathbf{R} \mid \mathbf{t}]$:

$$\mathbf{l}_i = \mathbf{t} + \mathbf{R}\,\mathbf{p}_i - \mathbf{b}_i, \quad d_i = \|\mathbf{l}_i\|, \quad \hat{\mathbf{n}}_i = \mathbf{l}_i / d_i$$

d_i = active prismatic length. Base S takes the azimuth and elevation of $\hat{\mathbf{n}}_i$; platform U takes the angle between $\hat{\mathbf{n}}_i$ and the platform axes $\mathbf{R}$. One answer per leg.

WORKED Stewart leg lengths

Base points on a 1.0 m circle at $-10, 10, 110, 130, 230, 250°$; platform points on a 0.5 m circle at $-50, 50, 70, 170, 190, 290°$. Home (level, 1.0 m up), leg 1:

$\mathbf{l}_1 = (-0.6634, -0.2094, 1.0)$, $d_1 = \sqrt{0.4401 + 0.0438 + 1} = 1.2182$ m

All six legs = 1.2182 m (leg 1 azimuth $-162.48°$, elevation $55.17°$).
Pose $t(0.05, -0.02, 1.0)\,R_z(5°)R_y(3°)$: legs (m)

1.1597, 1.1974, 1.2170, 1.2812, 1.2152, 1.2504 [checked]

Delta IK (demo, base radius 0.20, platform 0.05, arm 0.20, rods 0.45 m): each arm alone, like the four-bar; at $\mathbf{p} = (0.03, -0.02, -0.38)$ the motor angles are $16.339°$, $30.871°$, $22.996°$ and every rod stays 0.45 m (that is the loop closure). Parallelograms keep the platform level: 3 translations, no rotation. [checked]

TRAP

FK/IK swapped: FK is the hard direction for parallel robots (coupled nonlinear loops), IK the easy one. For serial arms it is the opposite.

Practical: none this week.

Path, trajectory, where to build it [p. 7–13, 43–45]

Trajectory = position (and velocity, acceleration) of every joint *against time*. Path = same poses, no timing. Whatever you plan, the motors always receive **joint** positions, velocities, accelerations [p. 68].

	Joint space [p. 12]	Task space [p. 43]
You give	$q(t)$ per joint	pose of {T} in {S} vs t
Cost	cheap, no IK en route	IK at every sample
Tool path	whatever results (not a line)	exactly the line/circle asked
Use	free motion A→B	prescribed path, obstacles

Teach mode [p. 9]: jog, store, replay; no kinematics; for repeated tasks. {T} in {S} [p. 45]: independent of the robot and follows a moving station (conveyor).

Cubic polynomial [p. 15–20]

Nth order polynomial: $N+1$ coefficients, $N+1$ conditions. Start/stop at rest = 4 conditions ⇒ cubic.

Cubic, $t_0 = 0, T = t_F, \Delta q = q_F - q_0$

$$q = a_0 + a_1 t + a_2 t^2 + a_3 t^3, \quad \dot{q} = a_1 + 2a_2 t + 3a_3 t^2, \quad \ddot{q} = 2a_2 + 6a_3 t$$

$$a_0 = q_0, \quad a_1 = \dot{q}_0$$

$$a_2 = \frac{3\Delta q - (2\dot{q}_0 + \dot{q}_F)T}{T^2}, \quad a_3 = \frac{-2\Delta q + (\dot{q}_0 + \dot{q}_F)T}{T^3}$$

Rest to rest: $a_2 = \frac{3\Delta q}{T^2}$, $a_3 = -\frac{2\Delta q}{T^3}$.

Peak speed $1.5\,\Delta q/T$ at $T/2$; $\ddot{q} = \pm 6\Delta q/T^2$ at the ends (does *not* start/end at 0).

Use: any start/stop at given positions and speeds. Matrix form: rows $[1,0,0,0]$, $[1,T,T^2,T^3]$, $[0,1,0,0]$, $[0,1,2T,3T^2]$ equal $[q_0, q_F, \dot{q}_0, \dot{q}_F]$. If $t_0 \neq 0$ use $\tau = t - t_0$, $T = t_F - t_0$. Several joints: one cubic each, same T.

TRAP

a_2 divides by T^2, a_3 by T^3. Dividing a_3 by T^2 is the classic slip (20→80 in 3 s wrongly gives −13.33). Units: a_k in unit/s^k.

WORKED slides pp. 17–19

10° →50° in 2 s, rest: $a = (10, 0, 30, -10)$, $q = 10 + 30t^2 - 10t^3$; peak 30°/s at 1 s; $\ddot{q} = 60 - 60t$. [checked]

20° →80° in 3 s, $\dot{q}_F = 10$: $a_2 = 16.667$, $a_3 = -3.333$; $q(3) = 80$, $\dot{q}(3) = 10$. [checked]

Three joints, $\Delta q = (40, 90, -20)°$, $T = 2$: joint 2 $a_2 = 67.5$, $a_3 = -22.5$, peak 67.5°/s. [checked]

Via points [p. 21–28]

A cubic between each pair; each needs end velocities. If all stop at every via point [p. 25] the motion is smooth but wasteful. Velocity at via points: (1) given, (2) simple rule, (3) continuous acceleration [p. 26].

Simple rule [p. 27]

Chord slope $s_k = \Delta q_k / \Delta t_k$. At a via point: slopes change sign ⇒ $\dot{q} = 0$; same sign ⇒ average of the two slopes. Ends: $\dot{q} = 0$ (state as an assumption). Then closed-form cubic per segment, own clock $\tau \in [0, T_k]$.

Use: "apply the rule to these points". Gives continuous q and $\dot{q}$; the acceleration jumps.

WORKED notebook Q2

$q = 10, 50, 20, 40, 45°$ at $t = 0, 5, 10, 15, 20$ s. Slopes $+8, -6, +4, +1$. $\dot{q} = 0, 0, 0, 2.5, 0$ (qA, qB sign change; qC average). [checked]

seg	a_0	a_1	a_2	a_3
q_0–A	10	0	4.8	−0.64
A–B	50	0	−3.6	0.48
B–C	20	0	1.9	−0.22
C–F	40	2.5	−0.4	0.02

Last seg: $q(5) = 40 + 12.5 - 10 + 2.5 = 45$. Acceleration jumps at A, B, C: $-2.4, +3.4, -2.0$ °/s^2. [checked]

Two cubics, continuous acceleration (W13 Q8)

8 unknowns; 8 conditions: 4 positions (start, via twice, end), 2 end velocities, $\dot{q}$ continuous, $\ddot{q}$ continuous at via. With $d_1 = q_v - q_0$, $d_2 = q_F - q_v$ and end velocities $(0, w)$, $(w, 0)$:

$$\ddot{q}_1(T_1) = \frac{-6d_1 + 4wT_1}{T_1^2} = \ddot{q}_2(0) = \frac{6d_2 - 4wT_2}{T_2^2}$$

$$w = \dot{q}_{via} = \frac{3\,[\,T_1^2 d_2 + T_2^2 d_1\,]}{2T_1T_2(T_1 + T_2)}; \quad T_1 = T_2 = T: \; w = \frac{3(q_F - q_0)}{4T}$$

Use: "smooth through a via point". Equal times: w is independent of q_v; shared $\ddot{q} = 3(q_0 + q_F - 2q_v)/T^2$. "Constant acceleration" is read as continuous.

WORKED sample Q8

$q = 10, 40, 20°$, $T_1 = T_2 = 1$: $w = 7.5$°/s; seg 1 $10 + 82.5t^2 - 52.5t^3$; seg 2 $40 + 7.5t - 75t^2 + 47.5t^3$; shared $\ddot{q} = -150$. [checked]

$T_1 = 1, T_2 = 2$: $w = 3[1(-20) + 4(30)]/12 = 25$°/s; seg 1 $10 + 65t^2 - 35t^3$; seg 2 $40 + 25t - 40t^2 + 11.25t^3$; shared $\ddot{q} = -80$. [checked]

RECIPE Two cubics through a via point

1. Write the 8 conditions as a list (most marks are here).
2. Own clock and duration per segment.
3. Cubic with end velocities $(0, w)$ and $(w, 0)$; unknown w only.
4. Equate end-of-1 and start-of-2 acceleration; solve w.
5. Check $q_1(T_1) = q_v$, $q_2(T_2) = q_F$, accelerations equal.

Many via points, spline [added]

Interior w_i from $\frac{w_{i-1}}{h_{i-1}} + 2\left(\frac{1}{h_{i-1}} + \frac{1}{h_i}\right) w_i + \frac{w_{i+1}}{h_i} = 3\left(\frac{s_{i-1}}{h_{i-1}} + \frac{s_i}{h_i}\right)$. For the Q2 data: $0, 2.304, -3.214, 4.554, 0$. [checked]

Quintic and why not polynomials [p. 29–30]

Six coefficients, six conditions: $q, \dot{q}, \ddot{q}$ at both ends. Linear 2, cubic 4, quintic 6.

Quintic, rest to rest, zero end accelerations

$$a_0 = q_0, \; a_1 = a_2 = 0, \; a_3 = \frac{10\Delta q}{T^3}, \; a_4 = -\frac{15\Delta q}{T^4}, \; a_5 = \frac{6\Delta q}{T^5}$$

Peak speed $1.875\,\Delta q/T$ at $T/2$; peak $|\ddot{q}| = 10\Delta q/(\sqrt{3}\,T^2) \approx 5.77\Delta q/T^2$.

Use: in an exam write the 6×6 matrix from the conditions; these closed forms are not on the slides [added].

WORKED

0→15° in 1 s: quintic $a_3, a_4, a_5 = 150, -225, 90$, peak 28.125°/s, end $\ddot{q} = 0$. Cubic: peak 22.5, ends ±90°/s^2. Mean/peak: cubic 66.7 %, quintic 53.3 %. [checked]

Why quintic: acceleration continuous at start/stop (less jerk). Why not polynomials everywhere [p. 30]: the velocity is a hump touching its peak briefly; a trapezoid holds near-peak speed, so for the same $\Delta q, T$ it needs a lower peak speed (between $\Delta q/T$ and $2\Delta q/T$).

Linear segment with parabolic blends (LSPB) [p. 31–37]

Blend $+a$ for t_B, coast at v, blend $-a$ for t_B. v–t is a trapezoid; q–t is parabola, line, parabola.

LSPB, $t_0 = 0, a > 0$ = blend acceleration magnitude

$$v = a\,t_B, \qquad a\,t_B^2 - a\,t_F\,t_B + \Delta q = 0$$

$$t_B = \frac{t_F}{2} - \frac{\sqrt{a^2 t_F^2 - 4a\Delta q}}{2a} \quad \text{(smaller root)}, \qquad \boxed{a \geq \frac{4\Delta q}{t_F^2}}$$

Coast length $t_F - 2t_B$; $v = \frac{1}{2}\left[at_F - \sqrt{a^2 t_F^2 - 4a\Delta q}\right] \in (\Delta q/t_F,\, 2\Delta q/t_F]$.

Choose v instead (toolbox, nb. cell 40): $t_B = t_F - \Delta q/v$, $a = v/t_B$; needs $\Delta q/t_F < v \leq 2\Delta q/t_F$.

Use: trapezoidal velocity, "minimum acceleration", "is it possible?". At equality $t_B = t_F/2$, no coast, triangle with peak $2\Delta q/t_F$. Bigger a ⇒ shorter blend, lower coast speed.

RECIPE LSPB blend time (p. 37)

1. List Δq, $t_0 = 0$, t_F, a.
2. Existence: compute $4\Delta q/t_F^2$; need $a \geq$ it. Else "no solution: larger a or longer t_F".
3. t_B with the minus sign (the + root exceeds $t_F/2$).
4. $v = at_B$.
5. Check $2 \cdot \frac{1}{2}at_B^2 + v(t_F - 2t_B) = \Delta q$.

WORKED slide p. 37; slide gives no answer

120° in 2 s, $a = 300$: min $a = 120$, ok. $\sqrt{360000 - 144000} = 464.76$, $t_B = 1 - 464.76/600 = 0.2254$ s, $v = 67.62$°/s; check $15.24 + 104.76 = 120$. [checked]

$a = 120$: $t_B = 1$, $v = 120$, no coast. $a = 100$: discriminant −8000, impossible. $V = 70$ given: $t_B = 2 - 120/70 = 0.2857$ s, $a = 245$. [checked]

TRAP

Strict ">" (slide says ≥); the + root ($t_B = 1.7746$, $v = 532$); t_F instead of t_F^2 in the minimum; taking the real part of a negative discriminant; "bigger a gives faster coast" (false: v falls from 120 toward the mean 60: 82.9 at a=150, 67.6 at 300).

SLIDE CHECK slides p. 33–34, nb. cell 35

The general-t_0 quadratic $at_B^2 - a(t_F - t_0)t_B + (q_F - q_0 - at_0^2/2) = 0$ is inconsistent. 120° shifted to t_0=1, t_F=3 gives roots 2.049, −0.049, not 0.2254. Use the duration $T = t_F - t_0$: $at_B^2 - aTt_B + \Delta q = 0$. All slide examples have $t_0 = 0$.

Several segments with blends [p. 38–41]

Robot passes *near* the via points (rounded corner), not through them; hitting each exactly and stopping at the ends is over-constrained. Symbols [p. 39]: t_{djk} segment duration, t_{jk} linear part, t_j blend at q_j, v_{jk} slope, a_j blend acceleration. Given: points, t_{djk}, $|a_j|$. Find: blend times. (Craig §7.3 [added])

Craig blend times [added]

First: $\ddot{\theta}_1 = \text{sgn}(\theta_2 - \theta_1)|\ddot{\theta}_1|$, $t_1 = t_{d12} - \sqrt{t_{d12}^2 - \frac{2(\theta_2 - \theta_1)}{\ddot{\theta}_1}}$, $v_{12} = \frac{\theta_2 - \theta_1}{t_{d12} - \frac{1}{2}t_1}$.

W5

Interior: $v_{jk} = \dfrac{\theta_k - \theta_j}{t_{djk}}$, $\ddot{\theta}_j = \text{sgn}(v_{jk}-v_{ij})|\ddot{\theta}_j|$, $t_j = \dfrac{v_{jk} - v_{ij}}{\ddot{\theta}_j}$.

Last: $\ddot{\theta}_n = \text{sgn}(\theta_{n-1}-\theta_n)|\ddot{\theta}_n|$, $t_n = t_d - \sqrt{t_d^2 + \dfrac{2(\theta_n - \theta_{n-1})}{\ddot{\theta}_n}}$.

Linear part $t_{jk} = t_{djk} - \frac{1}{2}t_j - \frac{1}{2}t_k$. Corner miss = $a_j t_j^2/8$.

WORKED built from the formulas

$0, 40, 10°$; $t_d = 2,2$ s; $|a| = 50$: $t_1 = 0.4508$, $v_{12} = 22.54$; $t_3 = 0.3267$, $v_{23} = -16.33$; $t_2 = 0.7775$; linear parts 1.160, 1.285 s; misses 40° by 3.78° (reaches 36.22°). [checked]

Explain-level on the slides: say what is given/found, why via points are only approached, that each a_j sets its blend duration.

Task-space straight lines [p. 44–47]

Joint-linear ≠ Cartesian-linear [p. 46]: a straight tool path needs IK at every sample.

WORKED RR, L_1=4, L_2=3; (0,4.75)→(4.75,0)

Joint-linear from $(51.07°, 95.83°)$ to $(-38.93°, 95.83°)$: q_2 constant, so the tool stays 4.75 m from the base (an arc). Midpoint (3.359, 3.359) vs line midpoint (2.375, 2.375): 1.39 m off. [checked]

RECIPE Three problems with a Cartesian line (p. 47)

1. **Hole/limit**: ends reachable, middle not. L_1=1, L_2=0.7: inner radius 0.3. Line y=0.2, $x \in [-1, 1]$: ends $r = 1.02$, middle $r = 0.2 < 0.3$. [checked]
2. **Singularity**: $\det J = L_1 L_2 \sin q_2$; joint rates $\propto 1/\sin q_2$. L_1=L_2=1, 1 m/s along x: max rate 1.16 (x=1), 3.20 (1.9), 10.0 (1.99), 31.6 rad/s (1.999). [checked]
3. **Different IK branch** at A and B: needs a jump or a pass through a singularity.

Fixes: replan in joint space round the problem; move base or path.

Cartesian recipe [p. 48–65]

RECIPE Task-space trajectory

1. Sample the tool path at a few points.
2. IK at each sample (same branch throughout).
3. Fit a cubic (quintic if needed) to each joint's samples.
4. Resample at the controller rate; FK to check the drawn path.

Fit first: numerical IK output is not smooth.

Planar RR IK [p. 56]

$$\cos q_2 = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1 L_2}, \quad q_2 = \text{atan2}(\pm\sqrt{1 - \cos^2 q_2}, \cos q_2)$$

$$q_1 = \text{atan2}(y, x) - \text{atan2}(L_2 \sin q_2,\ L_1 + L_2 \cos q_2)$$

WORKED pp. 49–64

Path $y = -x + 4.75$; length 6.7175 m; at 3 m/s time 2.2392 s; 11 samples $t_i = i\,t_F/10$, $x = 3t/\sqrt{2}$. Start $(q_1, q_2) = (0.8914, 1.6725)$ rad = $(51.07°, 95.83°)$; end $(-38.93°, 95.83°)$. Fit error up to 0.46°; fitted tool path wanders up to 37 mm; a quintic is > 5× closer. [checked]
Other branch ($q_2 < 0$): start $(128.93°, -95.83°)$, end $(38.93°, -95.83°)$; same tool path, mirrored arm. Never switch branch mid-path. [checked]

SLIDE CHECK p. 59

Slide $q_2(t)$ cubic term 0.0004 (and −0.4043, 0.9036) is not reproduced: q_2 is equal at both ends, so a cubic fit has cubic term 0.0000 ($-0.4032t^2$+0.9029t). $q_1 = 0.1346t^3 - 0.3375t^2 - 0.6182t + 0.8858$ agrees. Method matters, not the last digits.

Poses cannot be interpolated entry by entry [p. 65] (rotation matrices); use ctraj.

Likely questions & traps

Four conditions (cubic) vs eight (two cubics); who gets t_F^2 vs t_F^3; each via segment has its own clock; "stops at every via point" is smooth but wasteful; "joint-linear gives a straight line" is false; "Cartesian commands go straight to the motors" is false. Quick: 90° in 1.5 s ⇒ $a_{min} = 160$, $t_B = 0.75$ (triangle, peak 120°/s). [checked]

Jacobian: joint rates → tool velocity [p. 6–7, 21]

Spatial velocity (twist)

$$^0\boldsymbol{\nu} = \begin{bmatrix}\boldsymbol{v}\\ \boldsymbol{\omega}\end{bmatrix} = {}^0\mathbf{J}(\mathbf{q})\,\dot{\mathbf{q}}, \qquad \underbrace{{}^0\boldsymbol{\nu}}_{6\times1} = \underbrace{{}^0\mathbf{J}}_{6\times n}\ \underbrace{\dot{\mathbf{q}}}_{n\times1}$$

$\boldsymbol{v}$ linear (m/s) on top, $\boldsymbol{\omega}$ angular (rad/s) below, both in frame {0}. n = number of joints.

Use: column j = tool spatial velocity when only joint j moves at 1 rad/s; effects add. Revolute: linear rows in m per rad, angular rows dimensionless. 6-joint arm: 6×6. Planar RR: 6×2, rows v_z, ω_x, ω_y are zero; keep x, y, ω_z.

Skew, vex, $\dot{\mathbf{R}} = [\boldsymbol{\omega}]_\times\mathbf{R}$ [p. 9–12, 18–19]

Skew matrix

$$[\mathbf{s}]_\times = \begin{bmatrix}0 & -s_3 & s_2\\ s_3 & 0 & -s_1\\ -s_2 & s_1 & 0\end{bmatrix}, \qquad \mathbf{s} = \vee_\times(\mathbf{S}) = (S_{32}, S_{13}, S_{21})^\top$$

$[\boldsymbol{\omega}]_\times\mathbf{P} = \boldsymbol{\omega}\times\mathbf{P}$. $\mathbf{R}\mathbf{R}^\top = \mathbf{I} \Rightarrow \dot{\mathbf{R}}\mathbf{R}^\top$ is skew, so $\dot{\mathbf{R}} = [\boldsymbol{\omega}]_\times\mathbf{R}$, $\boldsymbol{\omega} = \vee_\times(\dot{\mathbf{R}}\mathbf{R}^\top)$. Augmented skew (p.12): $\hat{\mathbf{S}} = \begin{bmatrix}[\mathbf{s}_{4:6}]_\times & \mathbf{s}_{1:3}\\ \mathbf{0} & 0\end{bmatrix}$, linear part first.

WORKED

$\boldsymbol{\omega} = (1,2,3)$, $\mathbf{P} = (4,5,6)$: $\boldsymbol{\omega}\times\mathbf{P} = (-3,6,-3)$. $\boldsymbol{\omega} = (0,0,2)$ rad/s, $\mathbf{r} = (0.3, 0.4, 0)$ m: $\mathbf{v} = (-0.8, 0.6, 0)$ m/s, length 2×0.5. [checked]

SLIDE CHECK cross-product note

The note prints the $\hat{\mathbf{j}}$ coefficient as $+(\omega_x P_z - \omega_z P_x)$. The determinant expansion has a minus: $\hat{\mathbf{j}}$ coefficient = $\omega_z P_x - \omega_x P_z$ (here +6, not −6). Its skew-matrix form is right, so its two forms disagree as printed. The note also drops rotations and frame labels of Craig's formula.

Jacobian from the ETS [p. 13–16, 27–39]

Differentiate the pose, column by column

$\mathbf{T} = {}^0\mathbf{T}_E = \mathbf{E}_1\cdots\mathbf{E}_M$, $\dot{\mathbf{T}} = \sum_j \frac{\partial\mathbf{T}}{\partial q_j}\dot{q}_j$, with $\mu(j)$ the position of the elementary transform holding q_j:

$$\frac{\partial\mathbf{T}}{\partial q_j} = \Big(\prod_{i<\mu(j)}\mathbf{E}_i\Big)\frac{\partial\mathbf{E}_{\mu(j)}}{\partial q_j}\Big(\prod_{i>\mu(j)}\mathbf{E}_i\Big) = \begin{bmatrix}\mathbf{J}_{R_j} & \mathbf{J}_{t_j}\\ \mathbf{0} & 0\end{bmatrix}$$

$$\mathbf{J}_{v_j} = \tau\Big(\frac{\partial\mathbf{T}}{\partial q_j}\Big), \qquad \mathbf{J}_{\omega_j} = \vee_\times\Big(\rho\Big(\frac{\partial\mathbf{T}}{\partial q_j}\Big)\rho(\mathbf{T})^\top\Big)$$

$\tau = 3\times1$ translation column, $\rho = 3\times3$ rotation block. Elementary derivatives: $\partial\mathbf{T}_{t_k}/\partial d = \hat{\mathbf{t}}_k$ (single 1 at entry $(k,4)$); $\partial\mathbf{T}_{R_k}/\partial\theta = \hat{\mathbf{R}}_k\mathbf{T}_{R_k}(\theta)$, with $\hat{\mathbf{R}}_k$ the 4×4 skew matrix of axis k, e.g. $\hat{\mathbf{R}}_z$: rows $(0,-1,0,0)$, $(1,0,0,0)$, zeros.

Use: "derive the Jacobian of this arm": show every step; the order of the product is the point.

RECIPE Jacobian by ETS (p. 38)

1. Write the ETS base → tool; mark which transform holds each joint.
2. Joint j: replace that one factor by its derivative, keep the rest.
3. Drawn positive sense opposite to the axis (e.g. $R_y(-\theta_2)$, $t_z(-d)$): flip the sign of that derivative.
4. Multiply in order; read τ (linear) and $\vee_\times(\rho\rho^\top)$ (angular); stack linear above angular.

5. Check at a pose you can see ($q = 0$).

Planar RR, two ways [p. 34–38]

$x = L_1c_1 + L_2c_{12}$, $y = L_1s_1 + L_2s_{12}$ ($s_{12} = \sin(q_1+q_2)$):

$$^0\mathbf{J}_{xy} = \begin{bmatrix}-L_1s_1 - L_2s_{12} & -L_2s_{12}\\ L_1c_1 + L_2c_{12} & L_2c_{12}\end{bmatrix},\ \omega_z \text{ row } (1,1)$$

In the tool frame ({3}, slide 45; depends on q_2 only):

$$^E\mathbf{J}_{xy} = \begin{bmatrix}L_1s_2 & 0\\ L_1c_2 + L_2 & L_2\end{bmatrix}$$

TRAP

Running sum: $\sin(q_1+q_2)$, not $\sin q_2$, in frame {0}. State frame and size (2×2 block or 6×2).

WORKED notebook Q1

$L = (3,4)$ m, $q = (30°, 15°)$, $q_1+q_2 = 45°$: $J_{11} = -3(0.5) - 4(0.7071) = -4.3284$, $J_{12} = -2.8284$, $J_{21} = 5.4265$, $J_{22} = 2.8284$; det = $3.1058 = 12\sin15°$. $\dot{\mathbf{q}} = (0.2, -0.1)$ rad/s: $\boldsymbol{v} = (-0.5828, 0.8025)$ m/s, $\omega_z = 0.1$ rad/s. [checked]

WORKED same pose, tool frame

$^E\mathbf{J}_{xy} = \begin{bmatrix}0.7765 & 0\\ 6.8978 & 4\end{bmatrix}$, det = 3.1058 (same). $\boldsymbol{v}_E = (0.1553, 0.9796)$ m/s $= \mathbf{R}_z(45°)^\top(-0.5828, 0.8025)$; speed 0.9918 in both frames. [checked]

Frames, geometric rule, Craig [p. 21, 40, 45]

Change of frame

$$^A\mathbf{J} = \begin{bmatrix}{}^A\mathbf{R}_B & \mathbf{0}\\ \mathbf{0} & {}^A\mathbf{R}_B\end{bmatrix}{}^B\mathbf{J}, \qquad {}^E\mathbf{R}_0 = ({}^0\mathbf{R}_E)^\top$$

Use: rotate both the linear and the angular half. det and singular values of the linear block are the same in every frame, so singularities do not depend on the frame. Do not rotate $\mathbf{J}$ by a transpose, nor apply a 3×3 rotation to a $6\times n$ matrix.

Geometric column rule [added]

$\hat{\mathbf{z}}_j$ = axis of joint j, $\mathbf{o}_j$ a point on it, $\mathbf{p}_E$ the tool point, all in {0}:

$$\text{revolute: } \mathbf{J}_{:,j} = \begin{bmatrix}\hat{\mathbf{z}}_j\times(\mathbf{p}_E - \mathbf{o}_j)\\ \hat{\mathbf{z}}_j\end{bmatrix}, \qquad \text{prismatic: } \begin{bmatrix}\hat{\mathbf{z}}_j\\ \mathbf{0}\end{bmatrix}$$

Use: fastest way to write a planar or simple arm by inspection. Planar RR: col 1 = $(-y, x, 0)$, col 2 = $L_2(-s_{12}, c_{12}, 0)$.

Craig velocity propagation [added]

Revolute $i{+}1$: ${}^{i+1}\boldsymbol{\omega}_{i+1} = {}^{i+1}_i\mathbf{R}\,{}^i\boldsymbol{\omega}_i + \dot{\theta}_{i+1}\,{}^{i+1}\hat{\mathbf{Z}}_{i+1}$, ${}^{i+1}\mathbf{v}_{i+1} = {}^{i+1}_i\mathbf{R}({}^i\mathbf{v}_i + {}^i\boldsymbol{\omega}_i\times{}^i\mathbf{P}_{i+1})$. Prismatic: no $\dot{\theta}$ term; add $\dot{d}_{i+1}\,{}^{i+1}\hat{\mathbf{Z}}_{i+1}$ to $\mathbf{v}$. Start ${}^0\boldsymbol{\omega}_0 = {}^0\mathbf{v}_0 = \mathbf{0}$; result is in the last frame.

Use: only if the question says "velocity propagation". Planar RR gives ${}^3\mathbf{J} = \begin{bmatrix}L_1s_2 & 0\\ L_1c_2 + L_2 & L_2\end{bmatrix}$, ω_z row $(1,1)$. [checked]

SLIDE CHECK slides 22–24, 21 vs 45

Slide 22–23 prints a 6×6 entry as j_{76} (should be j_{66}). Slide 21 stresses frame {0}; slide 45 prints the RR matrix in the tool frame {3}: both right, they differ by the rotation above.

Reading a real Jacobian: PUMA 560 [p. 24]

Slide gives no pose, frame or units. It matches the toolbox PUMA at $\mathbf{q} = (0, 45°, 180°, 0, 45°, 0)$ in {0}: say "assumed nominal pose, frame {0}". [checked]Angular rows (bottom three) are the joint axes in {0}: col 1 $(0,0,1)$ vertical waist; cols 2, 3, 5 all $(0,-1,0)$ (three parallel horizontal axes); col 4 $(0.707, 0, -0.707)$; col 6 $(1,0,0)$. Angle between axes 4 and 6 $= q_5 = 45°$. Col 1 linear = $(-p_y, p_x, 0) = (0.13, 0.664, 0)$. det = −0.0806, rank 6: not singular. [checked]

Inverse velocity [p. 41–42]

Joint rates from a wanted tool velocity

$$\dot{\mathbf{q}} = \mathbf{J}^{-1}\boldsymbol{\nu}, \qquad \mathbf{J}_{RR}^{-1} = \frac{1}{L_1L_2\sin q_2}\begin{bmatrix}L_2c_{12} & L_2s_{12}\\ -L_1c_1 - L_2c_{12} & -L_1s_1 - L_2s_{12}\end{bmatrix}$$

Slide questions: square? ($\neq$: pseudo-inverse); det $\neq$ 0?; which poses fail? Tall $\mathbf{J}$ ($m > n$): $\mathbf{J}^+ = (\mathbf{J}^\top\mathbf{J})^{-1}\mathbf{J}^\top$, least squares. Fat $\mathbf{J}$ ($n > m$, redundant): $\mathbf{J}^+ = \mathbf{J}^\top(\mathbf{J}\mathbf{J}^\top)^{-1}$, minimum norm [added].
Resolved rate: each step $\dot{\mathbf{q}}_k = \mathbf{J}^{-1}(\mathbf{q}_k)\boldsymbol{\nu}$, $\mathbf{q}_{k+1} = \mathbf{q}_k + \dot{\mathbf{q}}_k\Delta t$ (radians; recompute $\mathbf{J}$ every step).
Damped least squares [added]: $\mathbf{J}^\top(\mathbf{J}\mathbf{J}^\top + \lambda^2\mathbf{I})^{-1}$ keeps rates finite near a singularity at the cost of tracking error.

WORKED rates blow up, $L = (3,4)$, 0.1 m/s ⊥ arm

$\dot{q}_1 = 0.1/(3\sin q_2)$, $\dot{q}_2 = -\dot{q}_1(3\cos q_2 + 4)/4$.

q_2	90°	30°	15°	5°	1°
$\dot{q}_1$	0.0333	0.0667	0.1288	0.3825	1.9100
$\dot{q}_2$	−0.0333	−0.1100	−0.2221	−0.6682	−3.3422

Growth 30° →5°: $\sin30°/\sin5° = 5.7\times$. [checked]

WORKED notebook Q4: $q = (30°, 60°)$, $\boldsymbol{\nu} = (0.2, -0.3, -0.2)$

$\mathbf{J} = \begin{bmatrix}-5.5 & -4\\ 2.5981 & 0\\ 1 & 1\end{bmatrix}$, $\det_{xy} = 10.3923 = 12\sin60°$. Two joints, three components: over-determined; $\omega_z = \dot{q}_1 + \dot{q}_2$ is not free. Reading A (follow position): $\dot{\mathbf{q}} = \mathbf{J}_{xy}^{-1}(0.2, -0.3) = (-0.1155, 0.1088)$ rad/s, $\omega_z = -0.0067$; after 0.1 s $q = (29.338°, 60.623°)$. Reading B (least squares, $\mathbf{J}^+$): $(-0.1056, 0.0840)$, residual $(0.0446, 0.0258, 0.1784)$; $q = (29.395°, 60.481°)$. Say which reading you use. After 5 s (50 steps) A ends tip (3.591, 3.997), B (3.795, 4.058); ideal line ends (3.598, 4.000). [checked]

Singularities [p. 43–47]

Definition

Singular $\iff$ $\mathbf{J}(\mathbf{q})$ loses rank, $\det\mathbf{J} = 0$ (non-square: rank drop, $\det(\mathbf{J}\mathbf{J}^\top) = 0$, or $\sigma_{\min} = 0$). There is a direction $\mathbf{u}$ with $\mathbf{u}^\top\mathbf{J} = \mathbf{0}$ the tool cannot move in; rates for nearby velocities $\to\infty$ ($\mathbf{J}^{-1}$ has $1/\det$).
Boundary: stretched or folded. **Interior**: axes aligned (e.g. wrist).

$$\det\begin{bmatrix}L_1s_2 & 0\\ L_1c_2 + L_2 & L_2\end{bmatrix} = L_1L_2\sin q_2 = 0 \iff q_2 = 0°, 180°$$

Use: independent of q_1 and of the frame. Stretched: columns of $\mathbf{J}_v$ parallel, both ⊥ arm, rank 1.

W6

Other arms: factorise $\det$, then solve each factor

W13 arm $(R_z(\theta_1)t_z(1.1)R_y(-\theta_2)t_x(L_3)R_y(-\theta_3)t_x(L_4))$: $\det \mathbf{J}_v = -L_3L_4\sin\theta_3\,[L_3\cos\theta_2 + L_4\cos(\theta_2+\theta_3)]$. $\sin\theta_3 = 0$: stretched/-folded (boundary). Bracket = 0: tool on the vertical axis, θ_1 no longer moves it (interior). L_3=0.4, L_4=0.3, θ_2=60°: $\theta_3 = 71.81°$. [checked]

Spherical-wrist PUMA (zero offset): $\det \mathbf{J} = -L_u L_f \sin q_3\,[L_u \sin q_2 + L_f \sin(q_2+q_3)]\sin q_5$: elbow $\sin q_3 = 0$; shoulder (wrist centre on waist axis; L_u=0.5, L_f=0.3: q_2=22.92°, q_3=−63.39°); wrist $\sin q_5 = 0$ (axes 4, 6 align, rank 5). [checked]

WORKED RR $L = (3,4)$, $q_1 = 30°$

$q_2 = 0$: ${}^0\mathbf{J}_{xy} = \begin{bmatrix} -3.5 & -2 \\ 6.0622 & 3.4641 \end{bmatrix}$, col 1 = 1.75× col 2, along $(-0.5, 0.866) \perp$ arm $(0.866, 0.5)$. Tool frame ${}^E\mathbf{J}_{xy} = \begin{bmatrix} 0 & 0 \\ 7 & 4 \end{bmatrix}$: the row along the arm is zero. $q_2 = 180°$: $\begin{bmatrix} 0 & 0 \\ 1 & 4 \end{bmatrix}$, rank 1. [checked]

RECIPE Singularities

1. Write $\mathbf{J}$ (or the square block that matters), compute $\det$, factorise.
2. Solve each factor for the angles; list every solution in the range.
3. Name each: boundary or interior.
4. State what is lost (tool cannot move along …) and that rates $\to \infty$.

Manipulability, velocity ellipsoid [p. 49–55]

Velocity ellipsoid

Unit joint rates $\dot{\mathbf{q}}^\top\dot{\mathbf{q}} \le 1$ with $\dot{\mathbf{q}} = \mathbf{J}^{-1}\boldsymbol{\nu}$:

$$\boldsymbol{\nu}^\top(\mathbf{J}\mathbf{J}^\top)^{-1}\boldsymbol{\nu} \le 1, \quad \mathbf{A} = \mathbf{J}\mathbf{J}^\top, \quad \text{semi-axes } \sqrt{\lambda_i} = \sigma_i \text{ along } \mathbf{v}_i$$

$$m = \frac{\sqrt{\lambda_{\min}}}{\sqrt{\lambda_{\max}}} = \frac{\sigma_{\min}}{\sigma_{\max}} \in [0,1] \quad (1 = \text{isotropic},\ 0 = \text{singular})$$

Yoshikawa [added]: $w = \sqrt{\det \mathbf{J}\mathbf{J}^\top} = \sigma_1\sigma_2\cdots$; square: $w = |\det \mathbf{J}|$; RR: $L_1L_2|\sin q_2|$. m = round, w = big.

Use: some books define m as the reciprocal: state your definition. 2×2: $\lambda = \frac{1}{2}(\mathrm{tr} \pm \sqrt{\mathrm{tr}^2 - 4\det})$, $\det\mathbf{A} = (\det\mathbf{J})^2$, $\mathrm{tr}\,\mathbf{A} = L_1^2 + 2L_2^2 + 2L_1L_2\cos q_2$.

WORKED RR $L = (3,4)$, linear block

q_2	σ_1	σ_2	m	w
15°	8.002	0.3881	0.0485	3.1058
60°	7.1328	1.4570	0.2043	10.3923
90°	6.0927	1.9696	0.3233	12.0000
150°	4.2711	1.4048	0.3289	6.0000

$\sigma_1\sigma_2 = w$ each row. At 15°: tr = 64.182, det = 9.646, λ = 64.032, 0.1506. Best $m = 0.4267$ at $q_2 \approx 125.8°$ (equal links: 0.618 at 131.8°); a planar 2R never reaches $m = 1$; w peaks at 90°, m elsewhere. [checked]

TRAP

Reporting λ where the semi-axis is $\sqrt{\lambda}$; m as $\lambda_{\min}$ alone; "near a singularity the ellipsoid is large so the arm is faster": false, one axis collapses ($\sigma_{\min} \to 0$).

SLIDE CHECK slide 51

Unit-norm condition printed $\dot{\mathbf{q}}\dot{\mathbf{q}}^\top = 1$: for a column vector that is an $n \times n$ matrix. The scalar condition is $\dot{\mathbf{q}}^\top\dot{\mathbf{q}} = 1$.

Statics and force ellipsoid [p. 57–59]

Joint torques for a tool wrench

$$\boldsymbol{\tau} = \mathbf{J}^\top(\mathbf{q})\,\mathbf{F}_E, \quad \mathbf{F}_E = (F_x, F_y, F_z, n_x, n_y, n_z)^\top, \quad \boldsymbol{\tau}^\top\dot{\mathbf{q}} = \mathbf{F}^\top\boldsymbol{\nu}$$

force ellipsoid ($\boldsymbol{\tau}^\top\boldsymbol{\tau} \le 1$): $\mathbf{F}^\top\mathbf{J}\mathbf{J}^\top\mathbf{F} \le 1$ (no inverse)

Same axes as the velocity ellipsoid, semi-axes $1/\sqrt{\lambda_i}$: fast = weak. At a singularity the force semi-axis $\to \infty$: the structure carries load along the lost direction with zero torque.

Use: $\mathbf{F}$ and $\mathbf{J}$ in the same frame. Convention (not on the slide [added]): $\mathbf{F}$ is the wrench the tool applies to the environment; $\boldsymbol{\tau}$ is what the actuators supply. If $\mathbf{F}$ is the load on the tool the sign flips. $\mathbf{J}^\top$ exists even when $\mathbf{J}^{-1}$ does not.

WORKED $L = (0.5, 0.4)$ m, $q = (30°, 60°)$

$\mathbf{J}_{xy} = \begin{bmatrix} -0.65 & -0.4 \\ 0.4330 & 0 \end{bmatrix}$, $\mathbf{F} = (10, -20)$ N: $\boldsymbol{\tau} = \mathbf{J}^\top\mathbf{F} = (-6.5 - 8.660, -4.0) = (-15.160, -4.000)$ N m. Wrong $\mathbf{JF}$ gives $(1.5, 4.330)$. Moment $n_z = 2$ adds $2(1,1)$. [checked]

Stretched $q = (30°, 0)$: 50 N along the arm needs $\boldsymbol{\tau} = (0, 0)$; 50 N perpendicular needs $(45, 20)$ N m. [checked]

$\mathbf{J}\mathbf{J}^\top$ eigenvalues 0.04116, 0.72884: velocity semi-axes 0.2029, 0.8537; force semi-axes 4.929, 1.171. [checked]

SLIDE CHECK slide 58

Slide 58 gives neither the frame of $\mathbf{F}_E$ nor the sign convention for $\boldsymbol{\tau}$. Write both in your answer.

Offset RP arm (extra, W7–8 folder)

$x = l_1c_1 - d_2s_1$, $y = l_1s_1 + d_2c_1$ (d_2 slides $\perp$ link): $\mathbf{J} = \begin{bmatrix} -l_1s_1 - d_2c_1 & -s_1 \\ l_1c_1 - d_2s_1 & c_1 \end{bmatrix}$, $\det = -d_2$. Singular only at $d_2 = 0$: the two columns are parallel, the radial direction is lost; null motion $\dot{q}_1 = -1/l_1$, $\dot{d}_2 = 1$. [checked]

Traps and likely questions

Frame mix-up (**J** rotates, det does not); **JF** for $\mathbf{J}^\top\mathbf{F}$; inverse in the force ellipsoid; tall **J** needs a pseudo-inverse; degrees into a calculator set to radians; units on every answer (m/s, rad/s, N m). Forms to expect: **J** of a planar RR at given **q**; det and singular poses; joint rates for a tool velocity; torques for a tool force; sketch of the velocity ellipse near and far from a singularity. No W13 sample question is on velocity kinematics (marks not published).

W6

Equations of motion [p. 6–14, 25–33]

General form [p. 25]

$$\mathbf{Q} = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q},\dot{\mathbf{q}}) + \mathbf{g}(\mathbf{q}) + \mathbf{f}(\dot{\mathbf{q}}) + \mathbf{J}^{\top}\mathbf{w}$$

Use: sizes $n\times1 = (n\times n)(n\times1) + n\times1 + \dots$. $\mathbf{Q}$: N m (revolute) or N (prismatic). $\mathbf{f}$ friction; $\mathbf{w}$ external wrench.

Inverse motion $(\mathbf{q},\dot{\mathbf{q}},\ddot{\mathbf{q}}) \to$ torques. For *control*.

Forward torques $\to$ motion, $\ddot{\mathbf{q}} = \mathbf{M}^{-1}(\mathbf{Q}-\mathbf{C}-\mathbf{g}-\mathbf{f}-\mathbf{J}^{\top}\mathbf{w})$, integrate. For *simulation* [p. 48].

Task space $\mathcal{F} = \mathbf{M}_x\ddot{\mathbf{x}} + \mathbf{C}_x + \mathbf{g}_x$, $\mathbf{Q} = \mathbf{J}^{\top}\mathcal{F}$ [p. 32]

	Lagrange	Newton–Euler
basis	energy, robot as a whole	force/moment balance, link by link
output	closed-form equations	numbers, iterative
best for	analysis, control design	real-time inverse dynamics
cost n=6	66 394 mult., $O(n^4)$	547 mult., $O(n)$

NE: $104n-77$ mult., $92n-70$ add. (n=7: 651 mult.) [p. 47] [checked]

Meaning of each term [p. 27–33]

- $\mathbf{M}$: symmetric, positive definite (so invertible; KE $=\frac{1}{2}\dot{\mathbf{q}}^{\top}\mathbf{M}\dot{\mathbf{q}} > 0$). M_{jj} = inertia joint j feels; M_{ij} $(i\neq j)$ couples $\ddot{q}_j$ into torque i. PUMA 560: $M_{11}=3.659$, $M_{12}=-0.4044$.
- $\mathbf{C}$: centripetal (squares $\dot{q}_i^2$) + Coriolis (products $\dot{q}_i\dot{q}_j$, $i\neq j$) [p. 30].
- $\mathbf{g}$: present at rest, usually dominant. PUMA 560 at rest: [0, 31.64, 6.035, 0, 0.02825, 0] N m; stretched out [0, 46.01, 8.772, …]; pointing up [0, −0.7752, 0.2489, …].
- Held still $\Rightarrow$ only $\mathbf{g}$. Constant joint speed $\Rightarrow$ $\mathbf{C}$ and $\mathbf{g}$.
- [added] $\dot{\mathbf{M}} - 2\mathbf{C}_{mat}$ skew-symmetric; energy conserved if $\mathbf{Q}=0$, no friction.

Lagrangian method [p. 15–24]

Euler–Lagrange [p. 16–17]

$$\mathcal{L} = K - P, \qquad Q_i = \frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{q}_i}\right) - \frac{\partial\mathcal{L}}{\partial q_i}$$

Use: any arm. $K = \sum \frac{1}{2}m_i\mathbf{v}_{C_i}^{\top}\mathbf{v}_{C_i} + \frac{1}{2}\boldsymbol{\omega}_i^{\top}\mathbf{I}_i\boldsymbol{\omega}_i$ (drop 2nd term for point masses); $P = \sum m_i g h_i$.

RECIPE Equations of motion (W13 Q4)

1. Generalised coordinates $\mathbf{q}$ = joint variables. Draw gravity's direction.
2. COM position of each link from FK; differentiate for velocities.
3. Write K and P in full; $\mathcal{L} = K - P$.
4. Per joint: $\partial\mathcal{L}/\partial\dot{q}_i \to d/dt$ (product rule on $\cos q_2\cdot\dot{q}_1$) $\to -\partial\mathcal{L}/\partial q_i$.
5. Collect: $\mathbf{M}$ = coefficients of $\ddot{q}$; $\mathbf{C}$ = velocity products; $\mathbf{g}$ = the rest.
6. Check: $\mathbf{M}$ symmetric, $M_{22}>0$, $\mathbf{g}$ has no $\dot{q}$, $\mathbf{C}=0$ at rest, units.

Two-link arm (point masses) [p. 18–25]

Masses m_1, m_2 at tips of L_1, L_2; q_1 from horizontal, q_2 relative; gravity $-y$.

$$x_2 = L_1c_1 + L_2c_{12}, \quad y_2 = L_1s_1 + L_2s_{12}$$

$$K_2 = \tfrac{1}{2}m_2\left[(L_1^2+2L_1L_2c_2+L_2^2)\dot{q}_1^2 + 2(L_2^2+L_1L_2c_2)\dot{q}_1\dot{q}_2 + L_2^2\dot{q}_2^2\right]$$

$$P = m_1gL_1s_1 + m_2g(L_1s_1 + L_2s_{12})$$

Result [p. 22–25]

$$M_{11} = m_1L_1^2 + m_2(L_1^2 + 2L_1L_2c_2 + L_2^2)$$

$$M_{12} = M_{21} = m_2(L_1L_2c_2 + L_2^2), \quad M_{22} = m_2L_2^2$$

$$C_1 = -m_2L_1L_2s_2(2\dot{q}_1\dot{q}_2 + \dot{q}_2^2), \quad C_2 = m_2L_1L_2s_2\dot{q}_1^2$$

$$g_1 = (m_1+m_2)gL_1c_1 + m_2gL_2c_{12}, \quad g_2 = m_2gL_2c_{12}$$

Use: $h = m_2L_1L_2s_2$: $\mathbf{C} = \begin{bmatrix}0 & -h\\ h & 0\end{bmatrix}\begin{bmatrix}\dot{q}_1^2\\ \dot{q}_2^2\end{bmatrix} + \begin{bmatrix}-2h\\ 0\end{bmatrix}\dot{q}_1\dot{q}_2$ (centripetal + Coriolis; [added] this split, slide handwriting illegible). Elbow bends $\Rightarrow$ M_{11}, M_{12} change via c_2.

WORKED 2R torques

$m=(2,1)$ kg, $L=(1.0, 0.8)$ m, $g=9.81$, $q=(30°, 45°)$, $\dot{q}=(1,-0.5)$, $\ddot{q}=(0.5,2)$:

$\mathbf{M} = \begin{bmatrix}4.7714 & 1.2057\\ 1.2057 & 0.6400\end{bmatrix}$, $\mathbf{C}=(0.4243,\ 0.5657)$, $\mathbf{g}=(27.5183,\ 2.0312)$

$\mathbf{Q}=(32.7397,\ 4.4797)$ N m. NE gives the same. [checked]

By hand τ_2: $M_{21}\ddot{q}_1$=1.2057(0.5), $M_{22}\ddot{q}_2$=0.64(2), C_2=0.5657, g_2=0.8(9.81)cos 75°=2.0312; sum 4.4797.

TRAP

$\mathcal{L} = K + P$. Missing cross term $2L_1L_2c_2$ in v_2^2. $g_2 = \dots\sin(q_1+q_2)$ (with q from the horizontal and gravity $-y$ it is cos). Calling $\dot{q}_1\dot{q}_2$ centripetal (it is Coriolis). M_{12} without the L_2^2. "g depends on $\dot{q}$", "M diagonal", "forward dynamics is for control". [checked]

RP arm (Craig; unit script)

Revolute θ_1, prismatic d_2, link 1 COM at l_1, gravity $-y$:

$$\tau_1 = (m_1l_1^2 + I_{zz1} + I_{zz2} + m_2d_2^2)\ddot{\theta}_1 + 2m_2d_2\dot{\theta}_1\dot{d}_2 + (m_1l_1 + m_2d_2)g\cos\theta_1$$

$$f_2 = m_2\ddot{d}_2 - m_2d_2\dot{\theta}_1^2 + m_2g\sin\theta_1$$

$\mathbf{M}$ diagonal; $2m_2d_2\dot{\theta}_1\dot{d}_2$ Coriolis; $-m_2d_2\dot{\theta}_1^2$ centrifugal. Values $m=(2,1)$, $l_1=0.3$, $I_{zz}=(0.05, 0.02)$, $\theta_1=0.7$, $d_2=0.6$, $\dot{\theta}_1=1.3$, $\dot{d}_2=-0.8$, $\ddot{\theta}_1=2.0$, $\ddot{d}_2=1.5$: $\tau_1=8.9757$ N m, $f_2=6.8058$ N. [checked]

Rigid body [p. 34–39]

Newton and Euler [p. 34]

$$\mathbf{F} = m\dot{\mathbf{v}}_C, \qquad \mathbf{N} = {}^C\mathbf{I}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega}\times{}^C\mathbf{I}\boldsymbol{\omega}$$

Use: per-link balance in Newton–Euler. $\boldsymbol{\omega}\times\mathbf{I}\boldsymbol{\omega}=0$ when $\boldsymbol{\omega}$ is along a principal axis.

Inertia tensor [p. 36]

$I_{xx} = \iiint(y^2+z^2)\rho\,dv$ (cyclic), $I_{xy} = -\iiint xy\rho\,dv$ (cyclic).

Cuboid $W\times L\times H$ (x,y,z), centroid [p. 39]:

$${}^C\mathbf{I} = \tfrac{m}{12}\,\mathrm{diag}(L^2+H^2,\ W^2+H^2,\ L^2+W^2)$$

Corner frame [p. 38]: $I_{xx} = \frac{m}{3}(L^2+H^2)$, $I_{yy} = \frac{m}{3}(W^2+H^2)$, $I_{zz} = \frac{m}{3}(L^2+W^2)$,
$I_{xy} = -\frac{m}{4}WL$, $I_{xz} = -\frac{m}{4}HW$, $I_{yz} = -\frac{m}{4}HL$.

Parallel-axis theorem [added]

$${}^A\mathbf{I} = {}^C\mathbf{I} + m\left(\mathbf{p}^{\top}\mathbf{p}\,\mathbf{1}_3 - \mathbf{p}\mathbf{p}^{\top}\right)$$

Use: $\mathbf{p}$ = COM in $\{A\}$. With $\mathbf{p}=(W/2, L/2, H/2)$: centroid $\to$ corner. Same body, different frame, different tensor.

WORKED box 6 kg, $0.2\times0.3\times0.1$ m

Centroid diag$(0.050, 0.025, 0.065)$. Corner $I_{xx} = 6(0.09+0.01)/3 = 0.200$; $I_{xy} = -6(0.2)(0.3)/4 = -0.090$ kg m^2. Parallel axis: $0.05 + 6(0.15^2+0.05^2) = 0.200$. Full corner tensor: rows $(0.200, -0.090, -0.030)$, $(-0.090, 0.100, -0.045)$, $(-0.030, -0.045, 0.260)$. [checked]

SLIDE CHECK sign of I_{xy}

Slide 37 writes $I_{xy} = \iiint xy\rho\, dv = mWL/4$ without the minus; slide 38's matrix has $-mWL/4$. Slide 36 defines the off-diagonal entry with the minus. Put the minus in.

Iterative Newton–Euler [p. 40–46]

Outward pass, $i = 1 \dots n$ (revolute)

$R = {}^{i-1}_{i}\mathbf{R}$, $\mathbf{P}_i$ = origin of $\{i\}$ in $\{i{-}1\}$, $\hat{Z}$ joint axis:

$$\boldsymbol{\omega}_i = R^\top \boldsymbol{\omega}_{i-1} + \dot{q}_i \hat{Z}$$

$$\dot{\boldsymbol{\omega}}_i = R^\top \dot{\boldsymbol{\omega}}_{i-1} + R^\top \boldsymbol{\omega}_{i-1} \times \dot{q}_i \hat{Z} + \ddot{q}_i \hat{Z}$$

$$\dot{\mathbf{v}}_i = R^\top \left(\dot{\boldsymbol{\omega}}_{i-1} \times \mathbf{P}_i + \boldsymbol{\omega}_{i-1} \times (\boldsymbol{\omega}_{i-1} \times \mathbf{P}_i) + \dot{\mathbf{v}}_{i-1} \right)$$

$$\dot{\mathbf{v}}_{C_i} = \dot{\boldsymbol{\omega}}_i \times \mathbf{P}_{C_i} + \boldsymbol{\omega}_i \times (\boldsymbol{\omega}_i \times \mathbf{P}_{C_i}) + \dot{\mathbf{v}}_i$$

$\mathbf{F}_i = m_i \dot{\mathbf{v}}_{C_i}$, $\mathbf{N}_i = \mathbf{I}_i \dot{\boldsymbol{\omega}}_i + \boldsymbol{\omega}_i \times \mathbf{I}_i \boldsymbol{\omega}_i$.

Use: prismatic: add $2\boldsymbol{\omega}_i \times \dot{d}_i \hat{Z} + \ddot{d}_i \hat{Z}$ to $\dot{\mathbf{v}}_i$, $\boldsymbol{\omega}$ unchanged.

Inward pass, $i = n \dots 1$, $\mathbf{f}_{n+1} = \mathbf{n}_{n+1} = 0$

$R' = {}^{i}_{i+1}\mathbf{R}$:

$$\mathbf{f}_i = R'\mathbf{f}_{i+1} + \mathbf{F}_i$$

$$\mathbf{n}_i = \mathbf{N}_i + R'\mathbf{n}_{i+1} + \mathbf{P}_{C_i} \times \mathbf{F}_i + \mathbf{P}_{i+1} \times R'\mathbf{f}_{i+1}$$

$$\tau_i = \mathbf{n}_i \cdot \hat{Z}$$

Use: gravity trick: start with $\dot{\mathbf{v}}_0 = (0, g, 0)$ (base accelerates up) instead of adding gravity to each link. [added] *Matches Lagrange at 50 random states.* [checked]

SLIDE CHECK NE slide 45

Slide 45 writes ${}^{i}T_{i+1}$ where a rotation ${}^{i}\mathbf{R}_{i+1}$ is meant, and its moment balance omits rotating $\mathbf{n}_{i+1}$ into $\{i\}$. Use Craig's equations above.

NE is rarely derived by hand: expect "explain the two passes and why NE is $O(n)$" or "$\dot{\mathbf{v}}_C$ for one link".

Worked: single link [p. 34]

$m = 1$, ${}^{1}\mathbf{P}_C = [3, 0, 0]$, ${}^{C}\mathbf{I} = \text{diag}(1, 2, 3)$, $q = bt + ct^2$. $\ddot{q} = 2c$, $\dot{\boldsymbol{\omega}} = [0, 0, 2c]$.
$\dot{\mathbf{v}}_C = [\, -3(b{+}2ct)^2,\ 6c,\ 0\,]$ (tangential $6c$, centripetal $-3\dot{q}^2$).
$N_z = 6c$, $\tau = N_z + 3(6c) = 24c$. Check $(I_{zz}{+}m\,3^2)\ddot{q} = 12(2c)$. [checked]

3-link planar arm along a cubic (practical sheet Q3)

$L = (3, 2, 1)$ m, point masses $3, 2, 1$ kg at distal ends, vertical plane, gravity $-y$. Start tool pose $(5, 1, 90°)$, goal $(-3, 3, 180°)$, $T = 2$ s.

RECIPE Joint angles and torque history

1. Wrist = $(x - L_3\cos\varphi,\ y - L_3\sin\varphi)$; 2-link IK: $\cos\theta_2 = (w^2 - L_1^2 - L_2^2)/2L_1L_2$.
2. Start: wrist $(5, 0)$, $\cos\theta_2 = 1 \Rightarrow q_0 = (0°, 0°, 90°)$ (one solution, singular).
3. Goal: wrist $(-2, 3)$, $\cos\theta_2 = 0$; $\theta_2 = +90°$ gives $q_f = (90°, 90°, 0°)$. Other elbow: $(157.38°, -90°, 112.62°)$.
4. Rest-to-rest cubic: $a_2 = 3\Delta/T^2$, $a_3 = -2\Delta/T^3$; $\Delta = (90°, 90°, -90°)$ gives $a_2 = (67.5, 67.5, -67.5)°/\text{s}^2$, $a_3 = (-22.5, -22.5, 22.5)°/\text{s}^3$. $|\ddot{q}(0)| = 2.3562$ rad/s^2.
5. $\mathbf{Q} = \mathbf{M}\ddot{\mathbf{q}} + \mathbf{C} + \mathbf{g}$ at each t (build as for the 2R, with three tips).
6. Check $t = 0$: at rest so $\mathbf{Q} = \mathbf{M}\ddot{\mathbf{q}}{+}\mathbf{g}$; $g_1 = 9.81(3{\cdot}3{+}2{\cdot}5{+}1{\cdot}5) = 235.44$.

t (s)	q (°)	τ_1	τ_2	τ_3 (N m)
0	(0,0,90)	548.814	160.176	2.356
1	(45,45,45)	70.817	20.781	5.078
2	(90,90,0)	−268.947	−141.712	−21.591

[checked] $\tau(0)$ agrees with the W9 tab.

Quick-fire answers

- **Why is M invertible?** Symmetric positive definite: $\frac{1}{2}\dot{\mathbf{q}}^\top\mathbf{M}\dot{\mathbf{q}} > 0$ for any motion.
- **Simulator vs computed-torque controller?** Simulator = forward (torques in, motion out). Controller = inverse (motion in, torques out).
- **Multiplications for NE, $n = 7$?** $104(7) - 77 = 651$.
- **Which terms survive when held still? at constant speed?** **g** only; **C** and **g**.

TRAP student's 2R

"$\mathcal{L} = K + P$; $g_2 = m_2 g L_2 \sin(q_1{+}q_2)$; $-2m_2L_1L_2s_2\dot{q}_1\dot{q}_2$ is centripetal; $M_{12} = m_2L_1L_2c_2$." Correct: $K - P$; $\cos(q_1{+}q_2)$; Coriolis; $M_{12} = m_2(L_1L_2c_2 + L_2^2)$.

Exam shape (W13 Q4)

Marks for: COM positions from FK; velocities by differentiating; K, P in full; $\mathcal{L} = K - P$; one Euler–Lagrange line per joint; **M**, **C**, **g** identified; a physical check. Not on slides: the check list, $\dot{\mathbf{M}} - 2\mathbf{C}$ skew-symmetry. Lost: $K + P$, dropped cross term, d/dt taken before $\partial/\partial\dot{q}$, gravity sign.

Open vs closed loop, error [p.8–14]

Servo error [p. 13]

$$E = \mathbf{q}_d - \mathbf{q}, \qquad \dot{E} = \dot{\mathbf{q}}_d - \dot{\mathbf{q}} \quad \text{(desired − actual)}$$

Use: every later formula assumes this sign. Open loop = inverse dynamics $\mathbf{Q} = \mathbf{M}\ddot{\mathbf{q}} + \mathbf{C} + \mathbf{g}$ with no measurement of the output [p. 11]; closed loop feeds back $\mathbf{q}, \dot{\mathbf{q}}$ [p. 12].

Open loop fails: wrong model, outside disturbance, wrong start state. Unit mass: $\ddot{E} = 0 \Rightarrow E = E(0) + \dot{E}(0)t$, so a 0.1 rad offset stays 0.1 rad. [checked] Independent joint control [p. 14]: n single-joint loops, coupling ignored.

Spring–mass–damper [p. 16–26]

Characteristic equation [p. 16–17]

$$m\ddot{x} + b\dot{x} + kx = 0, \quad ms^2 + bs + k = 0, \quad s_{1,2} = \frac{-b \pm \sqrt{b^2 - 4mk}}{2m}$$

Use: test $b^2 - 4mk$ with strict inequalities; = 0 is critical.

b^2-4mk	case	$x(t)$
> 0	over (real, unequal) [p. 18]	$A_1e^{s_1t}+A_2e^{s_2t}$
< 0	under (complex $\lambda \pm i\mu$) [p. 21]	$e^{\lambda t}(A_1 \cos \mu t + A_2 \sin \mu t)$
$= 0$	critical $s = -\frac{b}{2m}$ [p. 24]	$(A_1+A_2t)e^{-bt/2m}$

$\lambda = -\frac{b}{2m}$, $\mu = \frac{\sqrt{4mk-b^2}}{2m}$ [added]. Constants from $x(0), \dot{x}(0)$: unequal A_1+A_2, $s_1A_1+s_2A_2$; complex A_1, $\lambda A_1+\mu A_2$; equal A_1, sA_1+A_2. [added]
$\omega_n = \sqrt{k/m}$, $\zeta = \frac{b}{2\sqrt{mk}}$ (> 1 over, $= 1$ crit., < 1 under).

WORKED deck examples, $x(0) = -1, \dot{x}(0) = 0$ [p. 19, 22, 25]

Over $m, b, k = 1, 3, 2$: disc $= 1$, $s = -1, -2$, $x = -2e^{-t} + e^{-2t}$, $x(1) = -0.6004$.
Under $1, 1, 1$: $\lambda = -\frac{1}{2}$, $\mu = \frac{\sqrt{3}}{2}$, $A_2 = -0.5774$; first peak $+0.1630$ at $t = 2\pi/\sqrt{3} = 3.628$ s (16.3%), trough -0.027 at 7.26 s.
Critical $1, 4, 4$: $s = -2$ twice, $x = (-1 - 2t)e^{-2t}$.
2% settling: critical 2.92 s < over 4.60 s < under 8.08 s. [checked]

RECIPE Classify and solve $x(t)$

1. $ms^2 + bs + k = 0$; compute $b^2 - 4mk$; name the case.
2. Roots ($\lambda \pm i\mu$ if complex); write the matching form.
3. Two initial-condition equations; solve the 2×2 for A_1, A_2.
4. Check $x(0), \dot{x}(0)$ from the final formula and $x \to 0$.

WORKED P8.1

$m, b, k = 1, 5, 6$, $x(0) = 2$: disc $= 1$, $s = -2, -3$; $A_1+A_2 = 2$, $-2A_1-3A_2 = 0 \Rightarrow A_1 = 6$, $A_2 = -4$, $x(1) = 0.6129$. [checked]

TRAP

$\lambda = -b/m$ (it is $-b/2m$). $\dot{x}(0) = 0 \Rightarrow A_2 = 0$ (only if $\lambda = 0$). Slide p.26's hand sketch shows over-damped slowest; the exact solutions say under-damped is slowest here. Trust the equations: only "critical is fastest without overshoot" is safe.

PD control [p. 27–32]

PD on the raw plant [p. 29, 32]

$$m\ddot{x} + b\dot{x} + kx = f, \; f = -k_px - k_v\dot{x} \Rightarrow m\ddot{x} + \underbrace{(b+k_v)}_{b'}\dot{x} + \underbrace{(k+k_p)}_{k'}x = 0$$

Use: critical when $b'^2 = 4mk'$. Never forget the plant's own b and k. k_p: N/m, k_v: N s/m.

WORKED

$m = 2, b = 3, k = 8$ (under: $9 - 64 < 0$), $k_p = 10$: $k' = 18$, $b' = \sqrt{4 \cdot 2 \cdot 18} = 12$, $k_v = 9$, double pole -3. Trap: $2\sqrt{k_p} = 6.32$ stays under-damped (disc -57.05); wrong answers 12, 6.32. [checked]

Control-law partitioning [p. 33–37]

Model-based + servo [p. 33–36]

$$f = \alpha f' + \beta, \; \alpha = m, \; \beta = b\dot{x} + kx \Rightarrow \ddot{x} = f' \quad \text{(unit mass)}$$

$$f' = -k_v\dot{x} - k_px \Rightarrow \ddot{x} + k_v\dot{x} + k_px = 0, \quad s^2 + k_vs + k_p = 0$$

$$\textbf{critical: } k_v^2 = 4k_p \iff k_v = 2\sqrt{k_p}$$

Use: regulation to 0. After partitioning f' is an acceleration: k_p in s^{-2}, k_v in s^{-1}. Poles p_1, p_2: $k_v = -(p_1+p_2)$, $k_p = p_1p_2$ [added].

SLIDE CHECK p.36 critical damping

Slide p.36 prints $k_p^2 = 4k_v$: subscripts swapped. Correct: $k_v^2 = 4k_p$ (agrees with p.32's $b'^2 = 4mk'$). With the slide's rule $k_p = 16 \Rightarrow k_v = 64$ gives roots $-63.75, -0.25$ (heavily over-damped); correct $k_v = 8$, double root -4. Test: $s^2 + 2s + 1$ has $k_v = 2, k_p = 1$; $k_p^2 = 4k_v$ would need $1 = 8$. [checked]

RECIPE Partitioned force at a state (P8.3)

1. Plant $m\ddot{x} + b\dot{x} + kx = f$; set $f = \alpha f' + \beta$; $\alpha = m, \beta = b\dot{x} + kx$.
2. Servo $f' = -k_v\dot{x} - k_px$ (tracking: section below).
3. Closed loop $\ddot{x} + k_v\dot{x} + k_px = 0$; critical $k_v^2 = 4k_p$.
4. Force $f = mf' + b\dot{x} + kx$. Label both portions on the block diagram.
5. Check: $\ddot{x} = (f - b\dot{x} - kx)/m$ must equal f'.

WORKED

$m, b, k = 2, 3, 8$; $k_p = 25$, $k_v = 10$ (critical, pole -5); $x = 0.5$, $\dot{x} = 1$: $f' = -10 - 12.5 = -22.5$; $f = 2(-22.5) + 3 + 4 = -38$ N; plant $\ddot{x} = (-38 - 3 - 4)/2 = -22.5$ ✓. [checked]

Wrong model [added]

$\hat{b} \neq b$: $\ddot{x} + \left(k_v + \frac{b-\hat{b}}{m}\right)\dot{x} + k_px = 0$. $\hat{m} \neq m$: both gains scaled by $\hat{m}/m$.
$\hat{b} = 1$ (true 3), $m = 2$: $k_v^{eff} = 11, 121 > 100$: slightly over-damped.
$\hat{m} = 1.5$: $s^2 + 7.5s + 18.75$, $56.25 - 75 < 0$: under-damped. [checked]

Choosing gains [added]

ω_n, ζ [added]

$$\omega_n = \sqrt{k_p}, \quad \zeta = \frac{k_v}{2\sqrt{k_p}}, \quad s = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

$$\text{overshoot } (0 < \zeta < 1) = \exp\left(-\pi\zeta/\sqrt{1-\zeta^2}\right)$$

Use: critical, speed ω_n: $k_p = \omega_n^2$, $k_v = 2\omega_n$. Given ζ: $k_v = 2\zeta\omega_n$. Doubling k_p (fixed k_v): $\zeta \to \zeta/\sqrt{2}$, $\omega_n \to \sqrt{2}\,\omega_n$.

(a) $\omega_n = 10$ crit.: $k_p = 100$, $k_v = 20$. (b) $\zeta = 0.7, \omega_n = 8$: 64, 11.2. (c) $k_p = 25$, $k_v = 6$: $\zeta = 0.6$, poles $-3 \pm 4i$, overshoot 9.48%. (d) poles $-2, -8$: $k_p = 16$, $k_v = 10$, $\zeta = 1.25$. [checked]
$k_p = 25$: $k_v = 4$: $\zeta = 0.4$, poles $-2 \pm 4.58i$, 25.4%; $k_v = 10$: -5 double; $k_v = 20$: $-1.34, -18.66$ (slow pole dominates, slower than critical). [checked]

Trajectory following [p. 38–40]

Servo law with feed-forward [p. 39]

$$f' = \ddot{x}_d + k_v\dot{E} + k_pE \Rightarrow \ddot{E} + k_v\dot{E} + k_pE = 0, \quad f = mf' + b\dot{x} + kx$$

Use: needs $x_d, \dot{x}_d, \ddot{x}_d$ (smooth, twice differentiable). Without $\ddot{x}_d$: constant a leaves $E_{ss} = a/k_p$.

$k_p = 100, k_v = 20$, $E(0) = 0.5$, $\dot{E}(0) = 0$: $E = E_0(1 + 10t)e^{-10t}$, $E(0.5) = 0.0202$ m. No FF on $x_d = t^2$: $E_{ss} = 2/100 = 0.02$ m. [checked]

Disturbance and PID [p. 41–45]

Steady-state error under a constant disturbance [p. 42–43]

$$\ddot{E} + k_v\dot{E} + k_pE = \frac{f_{dist}}{m} \Rightarrow E_{ss} = \frac{f_{dist}}{k_p\,m}$$

Use: f_{dist} enters the force junction with a minus; $E = x_d - x$ flips it. k_v, b, k do not appear. Tolerance E_{max}: $k_p \geq f_{dist}/(mE_{max})$, then $k_v = 2\sqrt{k_p}$.

RECIPE Steady-state error and its cure

1. Error equation with f_{dist}/m on the right. Set $\ddot{E} = \dot{E} = 0$.
2. $E_{ss} = f_{dist}/(k_pm)$; PD cannot reach 0 unless $k_p \to \infty$.
3. Cure: integral action. $f' = \ddot{x}_d + k_v\dot{E} + k_pE + k_i\int E\,dt$.
4. Differentiate: $\dddot{E} + k_v\ddot{E} + k_p\dot{E} + k_iE = \dot{f}_{dist}/m$; steady state $k_iE = 0 \Rightarrow E = 0$.
5. Stable only if $k_vk_p > k_i$ (poly $s^3 + k_vs^2 + k_ps + k_i$; [added]).

WORKED

$f_{dist} = 10$, $m = 2$, $k_p = 25$: $E_{ss} = 0.2$ m; $m = 5$: 0.08 m. $E_{ss} \leq 5$ mm: $k_p \geq 1000$, $k_v = 63.25$. PID k_p=25, k_v=10, k_i=40: roots -7.335, $-1.332 \pm 1.918i$; $\int E \to 10/(2 \cdot 40) = 0.125$. k_i=250: $(s + 10)(s^2 + 25)$, oscillation at 5 rad/s; k_i=300 unstable. [checked]

TRAP

"E_{ss} independent of m", "$E_{ss} = f/k_v$", "raising k_v cuts the steady error", "PD removes it", "integral never hurts".

DC motor and gearbox [p. 46–50]

Reflected inertia [p. 49–50]

$$Q_m = k_4i_a, \quad I_m\ddot{q}_m = k_4i_a - c_m\dot{q}_m, \quad \dot{q}_m = \eta\dot{q}_{axle}$$

$$Q = (I + \eta^2I_m)\ddot{q} + (c + \eta^2c_m)\dot{q}, \qquad Q_m = Q/\eta, \quad i_a = Q_m/k_4$$

Use: $\eta > 1$: motor faster than axle. Not η, not ηI_m. $\eta \cdot \eta$: inertia torque $I_m\eta\ddot{q}$, counted η times at the axle. p.51 is blank: no combined PID+motor formula.

WORKED

$\eta = 50$, $I_m = 2\times10^{-5}$, $I = 0.04$, $c = 0.1$, $c_m = 10^{-4}$, $k_4 = 0.04$; $\ddot{q} = 10$, $\dot{q} = 2$: $I_{eff} = 0.04+0.05 = 0.09$, $c_{eff} = 0.35$, $Q = 0.9 + 0.7 = 1.6$ N m, $Q_m = 0.032$, $i_a = 0.8$ A, motor speed 100 rad/s. Wrong: 1.6 (no $\div\eta$), 0.012 (no η^2). [checked]

Partitioned control, scalar [p. 5–11]

$f = \alpha f' + \beta$ [p. 10]

Isolate acceleration: (coeff)$\ddot{x}$ + (rest) = f. α = coefficient of $\ddot{x}$; β = everything else (all non-linear terms, with sign), using measured $x, \dot{x}$. Then $\ddot{x} = f'$.

$$f' = \ddot{x}_d + k_v\dot{e} + k_p e, \quad e = x_d - x \;\Rightarrow\; \ddot{e} + k_v\dot{e} + k_p e = 0$$

Use: critical $k_v = 2\sqrt{k_p}$. f' is an acceleration, f a force: $\alpha f'$ must have units of f. Cubic spring $m\ddot{x} + b\dot{x} + kx^3 = f$: $\alpha = m, \beta = b\dot{x} + kx^3$ [p. 9–10].

WORKED cubic spring

$m = 2, b = 4, k = 50, k_p = 100, k_v = 20$; $x = 0.10, \dot{x} = 0.30$; $x_d = 0.12, \dot{x}_d = 0.40, \ddot{x}_d = 0.50$: $f' = 0.5 + 20(0.10) + 100(0.02) = 4.5$; $\beta = 1.2 + 0.05 = 1.25$; $f = 2(4.5) + 1.25 = 10.25$ N. Plant $\ddot{x} = (10.25 - 1.25)/2 = 4.5 = f'$ ✓. [checked]

Single link, gravity and friction [p. 12–15]

Point mass at tip, q from horizontal [p. 12–14]

$K = \frac{1}{2}mL^2\dot{q}^2$, $P = mgL\sin q$ (Lagrange):

$$\tau = mL^2\ddot{q} + b_v\dot{q} + b_c\,\mathrm{sgn}(\dot{q}) + mgL\cos q$$

$$\alpha = mL^2, \quad \beta = b_v\dot{q} + b_c\,\mathrm{sgn}(\dot{q}) + mgL\cos q, \quad \tau = \alpha\tau' + \beta$$

Use: $\tau' = \ddot{q}_d + k_v\dot{e} + k_p e$. Check: at rest on the path $\tau = mgL\cos q$ (max horizontal, 0 vertical).

WORKED

$m = 2, L = 0.5, b_v = 0.3, b_c = 0.2$; $q = 30°, \dot{q} = 1, \ddot{q}_d = 2, e = 0.05, \dot{e} = -0.1, k_p = 64, k_v = 16$: $\alpha = 0.5, \tau' = 2 - 1.6 + 3.2 = 3.6, \beta = 0.5 + 8.4957$, $\tau = 10.7957$ N m; plant $\ddot{q} = 1.8/0.5 = 3.6 = \tau'$. [checked]

Quiz: $m = 1.5, L = 0.4, q = 60°, \ddot{q}_d = 0, e = 0.1, k_p = 25$, no friction: $\alpha = 0.24, \tau' = 2.5, \beta = 2.943, \tau = 3.543$ N m. [checked]

TRAP

Degrees in $\cos q$ on a radian calculator; sgn takes the sign of $\dot{q}$ (not q or e); $\alpha = mL$ (kg m, not kg m^2); $\cos q$ dropped from β; $\ddot{q}_d$ put in β. If the link has its own inertia, mL^2 becomes the total inertia about the pivot.

n-joint computed torque [p. 16–21]

Computed torque [p. 18–21]

$$\boldsymbol{\alpha} = \mathbf{M}(\mathbf{q}), \quad \boldsymbol{\beta} = \mathbf{C} + \mathbf{g} + \mathbf{f}$$

$$\mathbf{Q} = \mathbf{M}(\mathbf{q})\left(\ddot{\mathbf{q}}_d + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_p\mathbf{E}\right) + \mathbf{C} + \mathbf{g} + \mathbf{f}$$

$$\Rightarrow\; \ddot{\mathbf{E}} + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_p\mathbf{E} = \mathbf{0}$$

Use: $\mathbf{M}$ SPD so $\ddot{\mathbf{q}} = \mathbf{Q}'$. $\boldsymbol{\alpha}$ is a full matrix (it decouples); $\mathbf{K}$ diagonal ($k_p = \omega_n^2, k_v = 2\omega_n$) gives n separate loops. Model block linearises, servo block stabilises [p. 49]. Exact model, continuous law only.

RECIPE Write the computed-torque law

1. EoM as $\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C} + \mathbf{g} + \mathbf{f} = \mathbf{Q}$; $\boldsymbol{\alpha} = \mathbf{M}$, $\boldsymbol{\beta} = \mathbf{C} + \mathbf{g} + \mathbf{f}$ at the measured state.
2. $\mathbf{Q}' = \ddot{\mathbf{q}}_d + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_p\mathbf{E}$, $\mathbf{E} = \mathbf{q}_d - \mathbf{q}$; $\mathbf{Q} = \mathbf{M}\mathbf{Q}' + \boldsymbol{\beta}$.
3. State the closed loop and "decoupled, exact model only". Check units; at rest on path $\mathbf{Q} = \mathbf{g}$.

WORKED α not diagonal

3-link arm, $q = (0, 0, 90°)$: $\mathbf{M} = \begin{bmatrix} 103 & 31 & 1 \\ 31 & 13 & 1 \\ 1 & 1 & 1 \end{bmatrix}$; $\mathbf{Q}' = (1, 0, 0)$ needs (103, 31, 1) N m: 31 and 1 hold joints 2, 3 still. [checked]

Imprecise model, discretisation [p. 22–28]

Slide 28: hatted law [p. 27–28]

$\boldsymbol{\alpha} = \hat{\mathbf{M}}$, $\boldsymbol{\beta} = \hat{\mathbf{C}} + \hat{\mathbf{g}} + \hat{\mathbf{f}}$; equate to the true plant:

$$\ddot{\mathbf{E}} + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_p\mathbf{E} = \hat{\mathbf{M}}^{-1}\left[(\mathbf{M}-\hat{\mathbf{M}})\ddot{\mathbf{q}} + (\mathbf{C}-\hat{\mathbf{C}}) + (\mathbf{g}-\hat{\mathbf{g}}) + (\mathbf{f}-\hat{\mathbf{f}})\right]$$

Single link, mass only wrong [added]:

$$\ddot{e} + k_v\dot{e} + k_p e = \frac{m - \hat{m}}{\hat{m}}\left[\ddot{q} + \frac{g}{L}\cos q\right], \quad e_{ss} \approx \frac{m - \hat{m}}{\hat{m}}\frac{g}{L}\frac{\cos q}{k_p}$$

Use: $\ddot{\mathbf{q}}$ on the right is the actual one. Error no longer decays to 0; $\approx$ forcing$/k_p$. $\hat{m} < m$: gravity under-compensated, $e > 0$. Dynamics are slow, error×gain is fast: move the model block to the outer loop (evaluated along the plan), servo inside [p. 23–25].

WORKED practical $\hat{m} = 2.8, m = 3, L = 3, k_p = 100$

$(m - \hat{m})/\hat{m} = 1/14$, $g/L = 3.27$: horizontal $e_{ss} = 2.34$ mrad (0.134°); vertical ≈ 0. Forcing at $t = 0$ on the cubic: $\frac{1}{14}(2.3562 + 3.27) = 0.4019$; peak error 3.35 mrad at $t \approx 0.47$ s. Full arm with $\hat{m}_1 = 2.8$: $\hat{M}_{11} = 101.2$, $\hat{g}_1 = 229.554$. diag(M) as $\boldsymbol{\alpha}$ at $t = 0$: row-1 forcing $(31-1)(2.3562)/103 = 0.6863$ rad/s^2 (29%). [checked]

Cartesian control [p. 29–33]

Cartesian law [p. 32–33]

$$\mathcal{F} = \mathbf{M}_x\left(\ddot{\mathbf{x}}_d + \mathbf{K}_v\dot{\mathbf{e}}_x + \mathbf{K}_p\mathbf{e}_x\right) + \mathbf{c}_x + \mathbf{g}_x, \quad \mathbf{Q} = \mathbf{J}^\top\mathcal{F}$$

Feedback $\mathbf{x} = \mathcal{K}(\mathbf{q})$, $\dot{\mathbf{x}} = \mathbf{J}\dot{\mathbf{q}}$. [added] $\mathbf{M}_x = \mathbf{J}^{-\top}\mathbf{M}\mathbf{J}^{-1}$, $\mathbf{g}_x = \mathbf{J}^{-\top}\mathbf{g}$, $\mathbf{c}_x = \mathbf{J}^{-\top}(\mathbf{C} - \mathbf{M}\mathbf{J}^{-1}\dot{\mathbf{J}}\dot{\mathbf{q}})$.

Use: Joint-space route needs $\mathcal{K}^{-1}$, $\mathbf{J}^{-1}$, $\frac{d}{dt}\mathbf{J}^{-1}$ along the whole path (p.31): costly. $\mathbf{M}_x$ blows up near a singularity.

SLIDE CHECK p.31

Slide prints $\dot{J}^{-1}\dot{x}$. Correct: $\frac{d}{dt}(\mathbf{J}^{-1}) = -\mathbf{J}^{-1}\dot{\mathbf{J}}\mathbf{J}^{-1}$ (from $\ddot{\mathbf{q}} = \mathbf{J}^{-1}(\ddot{\mathbf{x}} - \dot{\mathbf{J}}\dot{\mathbf{q}})$). [checked]

WORKED $\mathbf{Q} = \mathbf{J}^\top\mathcal{F}, l_1 = l_2 = 1$

$q = (30°, 60°)$: $\mathbf{J} = \begin{bmatrix} -1.5 & -1 \\ 0.8660 & 0 \end{bmatrix}$, $\mathcal{F} = (3, -4)$: $\mathbf{Q} = (-7.9641, -3.0)$ N m. Check: tool at (0.8660, 1.5), $xF_y - yF_x = -7.9641$; joint 2 ($\mathbf{r} = (0, 1)$): -3. [checked]

Force control [p. 34–44]

Plant and the two feasible laws [p. 36–40]

Wall $f_e = k_e x$: $f = m\ddot{x} + f_{dist} + f_e$, $\ddot{x} = k_e^{-1}\ddot{f}_e$. $\alpha = mk_e^{-1}$ (s^2), $\beta = f_{dist} + f_e$ (infeasible), $E = f_d - f_e$, $X = mk_e^{-1}k_p$ (dimensionless).

$$\text{Opt. 1: } f = \alpha[\ddot{f}_d + k_v\dot{E} + k_p E] + f_e, \quad E_{ss} = \frac{f_{dist}}{mk_e^{-1}k_p} = \frac{f_{dist}}{X}$$

$$\text{Opt. 2: } f = \alpha[\ddot{f}_d + k_v\dot{E} + k_p E] + f_d, \quad E_{ss} = \frac{f_{dist}}{1 + X}$$

Use: Opt. 2 loop: $\ddot{E} + k_v\dot{E} + (k_e/m + k_p)E = f_{dist}k_e/m$. $E_2/E_1 = X/(1+X) < 1$: Opt. 2 always better, most on a stiff wall. Implement [p. 43]: $f = m[k_e^{-1}k_p E - k_v\dot{x}] + f_d$ (do not differentiate noisy f_e).

WORKED $m = 2, f_{dist} = 5, k_p = 100$

$k_e = 1000$: $X = 0.2$, $E_1 = 25$, $E_2 = 4.17$ N. $k_e = 20$: $X = 10, 0.5, 0.455$ N. $E_2 = 1\%$ needs $k_p = 49\,500$. [checked]

Hybrid force/position [p. 45–47]

Selection matrices [p. 47]

$$\mathcal{F} = \mathbf{S}\mathcal{F}_{pos} + \mathbf{S}'\mathcal{F}_{force}, \quad \mathbf{Q} = \mathbf{J}^\top\mathcal{F}$$

Use: diagonal 0/1, complementary $\mathbf{S}' = \mathbf{I} - \mathbf{S}$ (inferred). Surface normal → force ($\mathbf{S}$ entry 0, $\mathbf{S}'$ entry 1); free directions → position. $\mathbf{S}$ goes with the position law.

WORKED wipe a table, z normal

$\mathbf{S} = \mathrm{diag}(1, 1, 0)$, $\mathbf{S}' = \mathrm{diag}(0, 0, 1)$; $\mathcal{F}_{pos} = (12, -3, 9)$, $\mathcal{F}_{force} = (1, 2, 10)$: $\mathcal{F} = (12, -3, 10)$ N. Wall normal x: $\mathbf{S} = \mathrm{diag}(0, 1, 1)$, $\mathbf{S}' = \mathrm{diag}(1, 0, 0)$; the other law's entries on that axis are discarded. [checked]

Which law, when? [p. 49]

scalar non-linear	$f = \alpha f' + \beta$; $\ddot{e} + k_v\dot{e} + k_p e = 0$
n joints	$\mathbf{Q} = \mathbf{M}\mathbf{Q}' + \mathbf{C} + \mathbf{g} + \mathbf{f}$; decoupled
model inexact	hats; RHS $\hat{\mathbf{M}}^{-1}[\dots] \neq 0$
task-space path	$\mathcal{F} = \mathbf{M}_x(\dots) + \mathbf{c}_x + \mathbf{g}_x$, $\mathbf{Q} = \mathbf{J}^\top\mathcal{F}$
contact force	Option 2, $E_{ss} = f_{dist}/(1+X)$
mixed axes	$\mathbf{S}\mathcal{F}_{pos} + \mathbf{S}'\mathcal{F}_{force}$

Not in this deck: Lyapunov proofs, impedance/admittance control.

Practical 3-link arm ($L = 3, 2, 1$ m; $3, 2, 1$ kg)

RECIPE Trajectory, matrices, law

1. Start $(5, 1, 90°)$: wrist $(5, 0)$, $q_0 = (0, 0, 90°)$. Goal $(-3, 3, 180°)$: wrist $(-2, 3)$, $\cos\theta_2 = 0$, $q_f = (90°, 90°, 0)$ (other elbow $(157.38°, -90°, 112.62°)$).
2. Cubic $T = 2$ s (assumed), $q = q_0 + \Delta(3s^2 - 2s^3)$: peak speed $1.5\Delta/T = 1.1781$ rad/s ($\Delta/T = 0.785$ is wrong); peak accel. $6\Delta/T^2 = 2.3562$.
3. Start: $\mathbf{C} = \mathbf{0}$, $\mathbf{g} = (235.44, 58.86, 0)$; $\mathbf{Q}(0) = \mathbf{M}\ddot{\mathbf{q}}_d + \mathbf{g} = (548.8, 160.2, 2.4)$.
4. Link 1 alone: $\alpha = 27$, $\beta = 88.29\cos q$, $\tau = 27(\ddot{q}_d + 20\dot{e} + 100e) + 88.29\cos q$; $\tau(0) = 27(2.3562) + 88.29 = 151.91$ N m. Joint 1 of the full arm instead: $\alpha = 103$, $\beta = 235.44$.
5. Simulator = forward dynamics (RK4), $\ddot{\mathbf{q}} = \mathbf{M}^{-1}(\mathbf{Q} - \mathbf{C} - \mathbf{g} - \mathbf{f})$; $\ddot{q}(0) = 2.3562 = \ddot{q}_d$.

k_p=100, k_v=20: roots $-10, -10$; $e(0) = 0.1$: $e = 0.1(1 + 10t)e^{-10t}$, $e(0.1) = 0.0736$ rad. Slow update: peak 0.26 mrad (10 ms), 1.13 mrad (50 ms). A joint-space cubic does not move the tool straight: at $t = 1$, $q = (45°, 45°, 45°)$, tool (1.414, 4.828), not (1, 2). The start is singular ($\sin\theta_2 = 0$): $\mathbf{J}^{-1}$ fails. Simulator: keep the "true" and "model" parameters separate. [checked]

TRAP claims to reject

"Guarantees $\ddot{E} + K_v\dot{E} + K_p E = 0$ for any robot" (exact model only). "K diagonal so α diagonal". "f' has units of force" (acceleration). Adding $k_p e$ to f without α. $E = q - q_d$. $\mathbf{J}$ for $\mathbf{J}^\top$ or the inverse. Swapped E_1, E_2, missing "1+", or "Option 2 improves on a stiffer wall" (both errors grow; Opt. 2 saturates at f_{dist}).

Frames and chassis velocity [p. 11–17]

Chassis velocity $\zeta = (u, v, \dot\theta)$ in $\{C\}$; $\nu = (\dot x, \dot y, \dot\theta)$ in $\{O\}$

$$\nu = \mathbf{J}(\theta)\zeta = \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} u \\ v \\ \dot\theta \end{bmatrix}, \quad \zeta = \mathbf{J}^{-1}\nu = \mathbf{J}^{\top}\nu$$

Use: u forwards along x_C, v sideways along y_C, θ anticlockwise from x_O. Wheel speeds are commanded: $\zeta = \mathbf{W}\dot\varphi$. Forward kinematics (wheels → motion) = simulation; inverse (motion → wheels) = control.

Wheel types and pure rolling [p. 18–25]

Fixed standard (axle fixed), steerable standard (axle turns about vertical), castor (offset, passive), Swedish/omni (rollers on rim), spherical. **Only the two standard wheels constrain the chassis** (cannot slide sideways). Castor, Swedish, spherical: no constraint. Assume pure rolling, no slip [p. 20].

Omni wheel in its own frame [p. 22–25]

$$\dot\varphi = \frac{v_{drive}}{r} = \begin{bmatrix} \frac{1}{r} & \frac{\tan\gamma}{r} \end{bmatrix}\begin{bmatrix} v_x \\ v_y \end{bmatrix}, \quad v_{drive} = v_x + v_y\tan\gamma$$

Use: x_W = drive direction, y_W = wheel-plane normal; rollers free along angle γ from y_W. Rollers ⊥ drive ⇒ $\gamma = 0$, $\dot\varphi = v_x/r$ (sideways component ignored).

Generalised wheel model [p. 26–36]

Wheel i: position $({}^C x_i, {}^C y_i)$, drive angle β_i (from x_C), roller angle γ_i, radius r_i

$$\dot\varphi_i = \frac{1}{r_i}\begin{bmatrix} 1 & \tan\gamma_i \end{bmatrix}\begin{bmatrix} c\beta_i & s\beta_i \\ -s\beta_i & c\beta_i \end{bmatrix}\begin{bmatrix} 1 & 0 & -{}^C y_i \\ 0 & 1 & {}^C x_i \end{bmatrix}\mathbf{J}^{-1}(\theta)\nu$$

Use: read right to left: inertial → chassis; twist → contact-point velocity $(u - \dot\theta y,\ v + \dot\theta x)$; chassis axes → wheel axes; keep the driven part; divide by r. Start from ζ and drop $\mathbf{J}^{-1}$ if the question gives chassis velocities.

Standard wheel ($\gamma = 0$): rolling row and no-slide row [p. 37–42,68]

$$r\dot\varphi_i = c\beta_i\,(u - \dot\theta\,{}^C y_i) + s\beta_i\,(v + \dot\theta\,{}^C x_i)$$

$$0 = -s\beta_i\,(u - \dot\theta\,{}^C y_i) + c\beta_i\,(v + \dot\theta\,{}^C x_i)$$

Use: the no-slide rows are the non-holonomic constraints. Fixed wheel: β constant; steerable: $\beta = \beta(t)$.

RECIPE Wheel speeds of any wheeled robot (W13 Q10)

1. Draw $\{C\}$. For each wheel write ${}^C x, {}^C y, \beta, \gamma, r$ and which way positive $\dot\varphi$ drives.
2. One row per wheel: $r\dot\varphi_i = \cos\beta_i(u - \dot\theta y_i) + \sin\beta_i(v + \dot\theta x_i)$ ($\gamma = 0$).
3. Standard wheels: add the no-slide row (usually forces $v = 0$).
4. Stack: $\dot\varphi = \mathbf{W}^{-1}\zeta$. Invert (or pseudo-invert) for $\zeta = \mathbf{W}\dot\varphi$; $\det \neq 0 \Rightarrow$ all of $(u, v, \dot\theta)$ reachable (holonomic).
5. $\nu = \mathbf{J}(\theta)\zeta$ for world velocities. **Check:** straight ahead, and spin on the spot.

TRAP

(1) Sign of the C_y term: it is $-\dot\theta\,{}^C y$ in the x-row. (2) $\mathbf{R}(\theta)$ where $\mathbf{R}^{-1}(\theta)$ is needed. (3) $\gamma \neq 0$ for a standard wheel (it is 0, and the wheel also cannot slide sideways). (4) Forgetting the no-slide equation, which is where non-holonomy comes from.

Differential drive and skid-steer [p. 43–52]

Two fixed wheels at $(0, \pm a)$, left $= +a$ [p. 44–47]

$$\dot\varphi_1 = \tfrac{u - a\dot\theta}{r}\ \text{(left)}, \quad \dot\varphi_2 = \tfrac{u + a\dot\theta}{r}\ \text{(right)}, \quad v = 0$$

$$\begin{bmatrix} u \\ v \\ \dot\theta \end{bmatrix} = \begin{bmatrix} \frac{r}{2} & \frac{r}{2} \\ 0 & 0 \\ -\frac{r}{2a} & \frac{r}{2a} \end{bmatrix}\begin{bmatrix} \dot\varphi_1 \\ \dot\varphi_2 \end{bmatrix}$$

$$y_{ICR} = \frac{u}{\dot\theta} = a\,\frac{\dot\varphi_1 + \dot\varphi_2}{\dot\varphi_2 - \dot\varphi_1}\ \text{(on the axle, } x = 0)$$

Use: equal speeds: ICR at infinity (straight). Opposite speeds: ICR at the centre (spin). r cancels in y_{ICR}. Slower wheel is on the ICR side. Turning left slows the left wheel.

WORKED P10.3 and slide-style numbers

$r = 0.05$, $a = 0.15$, $\dot\varphi = (4, 6)$: $u = 0.05(10)/2 = 0.25$ m/s, $\dot\theta = 0.05(2)/0.3 = 0.3333$ rad/s, ICR $y = 0.75$ m left. [checked]
$a = 0.3$, $\dot\varphi = (2, 5)$: $y_{ICR} = 0.3 \cdot 7/3 = 0.7$ m. [checked]
$r = 0.1$, $a = 0.2$, $\dot\varphi = (-5, 5)$: $u = 0$, $\dot\theta = 0.1(10)/0.4 = 2.5$ rad/s, a spin about C.

Skid-steer (4 standard wheels at $(\pm b, \pm a)$ [p. 49–52]): rows $\frac{1}{r}[1, 0, -a]$ (left pair), $\frac{1}{r}[1, 0, +a]$ (right pair); b drops out. Slide's 3×4 inverse $u = \frac{r}{4}\sum\dot\varphi_i$, $\dot\theta = \frac{r}{4a}(-\dot\varphi_1 - \dot\varphi_2 + \dot\varphi_3 + \dot\varphi_4)$ is the least-squares inverse. A real skid-steer must slip to turn; pure rolling ignores that.

Omnidirectional triangle [p. 53–55]

Three Swedish wheels, centre-to-wheel d, drive directions tangential (clockwise), $\gamma = 0$. Wheels at $(0, d)$, $(\frac{\sqrt3}{2}d, -\frac{d}{2})$, $(-\frac{\sqrt3}{2}d, -\frac{d}{2})$ with $\beta = 0°, -120°, +120°$.

Sample Q10 / practical Q1

$$\dot\varphi = \frac{1}{r}\begin{bmatrix} 1 & 0 & -d \\ -\frac{1}{2} & -\frac{\sqrt3}{2} & -d \\ -\frac{1}{2} & \frac{\sqrt3}{2} & -d \end{bmatrix}\zeta$$

$$\zeta = \frac{r}{3}\begin{bmatrix} 2 & -1 & -1 \\ 0 & -\sqrt3 & \sqrt3 \\ -\frac{1}{d} & -\frac{1}{d} & -\frac{1}{d} \end{bmatrix}\dot\varphi$$

Use: row 1: $r\dot\varphi_1 = u - d\dot\theta$. $\det$ of the unscaled matrix $= 3\sqrt3 d/2 \neq 0 \Rightarrow$ invertible, any $(u, v, \dot\theta)$ possible: holonomic. Four-Swedish layout (p. 55) is the same with 45° rollers ($\tan\gamma \neq 0$).

WORKED P10.2: where does it go?

$d = 0.10$, $r = 0.05$, $\dot\varphi = (4, -1, -1)$: $u = \frac{0.05}{3}(8 + 1 + 1) = 0.1667$ m/s, $v = \frac{0.05}{\sqrt3}(-1 + 1) = 0$, $\dot\theta = -\frac{0.05}{0.3}(4 - 1 - 1) = -0.3333$ rad/s. ICR $(0, u/\dot\theta) = (0, -0.5)$: clockwise circle, radius 0.5 m. After 2 s: $\theta = -0.6667$, $x = 0.5\sin 0.6667 = 0.3092$, $y = -0.5(1 - \cos 0.6667) = -0.1071$ m. Euler $dt = 0.01$ s is within 0.55 mm. [checked]

Tricycle, Ackermann, bicycle [p. 56–61,69]

Tricycle: front wheel at $(b, 0)$ driven + steered; rear axle through C so $v = 0$ [p. 56–58]

$$\dot\varphi_1 = \tfrac{1}{r}\sqrt{u^2 + b^2\dot\theta^2}, \qquad \beta_1 = \operatorname{atan2}(b\dot\theta,\ u)$$

Use: front contact velocity is $(u,\ b\dot\theta)$ and equals $r\dot\varphi_1(\cos\beta_1, \sin\beta_1)$. The slide assumes the rear axle passes through C without saying so.

WORKED P10.5

$b = 0.6$, $r = 0.1$, $u = 0.4$, $\dot\theta = 0.5$: contact velocity $(0.4, 0.3)$, $|\cdot| = 0.5$ m/s, $\dot\varphi_1 = 5$ rad/s, $\beta_1 = \operatorname{atan2}(0.3, 0.4) = 36.870°$. [checked]

Bicycle: rear wheel fixed at C (driven), front at $(L, 0)$ steered by β [p. 69]

Rear: $r\dot\varphi_r = u$, no-slide $v = 0$. Front no-slide: $-u\sin\beta + L\dot\theta\cos\beta = 0$

$$\dot\theta = \frac{u\tan\beta}{L}, \qquad R = \frac{L}{\tan\beta}, \qquad \dot x = u\cos\theta,\ \dot y = u\sin\theta$$

Use: $\beta = 0$ (jammed straight): $\dot\theta = 0$, $v = 0$, only forward/back.

WORKED P10.4

$u = 2$, $\beta = 20°$, $L = 1.2$: $\dot\theta = 2\tan 20°/1.2 = 0.6066$ rad/s, $R = 3.297$ m. [checked]

Ackermann steering [p. 61] [added]

Front wheels steered differently so every axle line meets one ICR on the rear-axle line: no scrub. $\cot\beta_{out} - \cot\beta_{in} = \text{track/wheelbase}$.

Example: wheelbase 2.5, track 1.5, $R = 6$ m about the rear-axle midpoint: inner 25.463°, outer 20.323°. [checked]

Manoeuvrability [p. 62–64]

$\delta_M = \delta_m + \delta_s$. Informally: δ_m = directions the chassis can be driven directly; δ_s = independent steering angles. Precise [added]: stack the no-slide rows of all standard wheels into C_1; $\delta_m = 3 - \operatorname{rank} C_1$, $\delta_s = \operatorname{rank} C_{1s}$ (steered wheels' rows).

Robot (slide p. 64)	δ_m	δ_s	δ_M
2 fixed on one axle + castor (diff. drive)	2	0	2
2 fixed + 1 steered standard (tricycle, car)	1	1	2
2 steered standard + 1 Swedish	1	2	3
3 Swedish (omni triangle)	3	0	3

Holonomic vs non-holonomic [p. 65–68]

Definitions

Holonomic constraint: positions only, $f(x, y, \theta, \ldots) = 0$. **Non-holonomic**: involves velocities and cannot be integrated into a position-only equation. Classic: differential-drive no-slide $-\dot x\sin\theta + \dot y\cos\theta = 0$.

It forbids sideways *velocity* at every instant, not any final pose (parallel parking works, just not directly). Holonomic robot: no non-holonomic constraints, $\delta_M = \delta_m = 3$ (omni triangle).

RECIPE "Is the bicycle / car / diff. drive holonomic?" (explain-type)

1. Name the constraint: rear no-slide $-\dot x\sin\theta + \dot y\cos\theta = 0$.
2. Say "velocity, not position; not integrable".
3. Consequence: it can still reach any (x, y, θ) by manoeuvring, but not along any path.
4. Omni triangle: $\det \neq 0$, no no-slide row, so holonomic.

TRAP

Left wheel at $+a$ is $(u - a\dot\theta)/r$, not +; and fixed standard wheels force $v = 0$ (not "any v").

Forward vs inverse dynamics [p. 9–11]

Planar pose $\boldsymbol{\xi} = (x, y, \theta)$ and net load $\mathbf{Q} = (f_x, f_y, \tau_z)$, both in $\{O\}$ unless marked ${}^{C}\mathbf{Q}$. Frames: $\{O\}$ world, $\{C\}$ chassis, $\{G\}$ centre of mass (same orientation as $\{C\}$). Mass m, inertia ${}^{G}I$ about the vertical axis through G; $G = ({}^{C}x_G, {}^{C}y_G)$ in $\{C\}$.
Forward: given $\mathbf{Q}$, find the motion (simulation). **Inverse**: given the motion, find $\mathbf{Q}$ (control, feed-forward). Same equation both ways. Check: straight line $f = m\ddot{x}$.

TRAP
"Inverse dynamics gives the trajectory" is wrong. Keep $\boldsymbol{\xi}$ (global) and $\boldsymbol{\zeta}$ (chassis) separate: mixing them is the commonest error.

Newton–Euler balance on the chassis [p. 14–17]

Applied loads at C, inertial loads at G, all in $\{C\}$ [p. 15,17]

$$ {}^{C}F_x = {}^{C}f_x, \quad {}^{C}F_y = {}^{C}f_y, \quad {}^{C}N_z = {}^{C}n_z + \left({}^{C}y_G\,{}^{C}f_x - {}^{C}x_G\,{}^{C}f_y\right) $$

$$ {}^{C}\mathbf{F}_G = m\,{}^{C}\ddot{\mathbf{P}}_G, \qquad {}^{C}N_z = {}^{G}I\,\ddot{\theta} $$

Use: the applied force acts at C, the inertial force at G: moving one to the other gives the moment-transfer bracket $(y f_x - x f_y)$ (the slides write $\mathbf{P}_{CG}$, later $\mathbf{P}_G$: same vector $C \to G$). $\boldsymbol{\omega} \times {}^{G}\mathbf{I}\boldsymbol{\omega} = 0$ in planar motion. If $C = G$: ${}^{C}n_z = {}^{G}I\ddot{\theta}$.

WORKED
G 0.05 m ahead of C, ${}^{C}f = (10, 4)$ N, ${}^{C}n_z = 0.4$ N m: ${}^{C}N_z = 0.4 + (0 \cdot 10 - 0.05 \cdot 4) = 0.2$ N m. [checked]

Acceleration of G in the rotating frame [p. 19–24]

Four-term sum, then components

$$ {}^{C}\ddot{\mathbf{P}}_G = {}^{C}\ddot{\mathbf{P}}_C + \dot{\boldsymbol{\omega}} \times {}^{C}\mathbf{P}_{CG} + \boldsymbol{\omega} \times {}^{C}\dot{\mathbf{P}}_C + \boldsymbol{\omega} \times \boldsymbol{\omega} \times {}^{C}\mathbf{P}_G $$

$$ {}^{C}\ddot{x}_G = \dot{u} - {}^{C}y_G\ddot{\theta} - v\dot{\theta} - {}^{C}x_G\dot{\theta}^2 $$

$$ {}^{C}\ddot{y}_G = \dot{v} + {}^{C}x_G\ddot{\theta} + \boxed{u}\,\dot{\theta} - {}^{C}y_G\dot{\theta}^2 $$

Use: ${}^{C}\dot{\mathbf{P}}_C = (u, v, 0)$; $\boldsymbol{\omega} = (0, 0, \dot{\theta})$; $(0, 0, k) \times (a, b, 0) = (-kb, ka, 0)$. Unit vectors: $\dot{\hat{x}} = \dot{\theta}\hat{y}$, $\dot{\hat{y}} = -\dot{\theta}\hat{x}$, so $\mathbf{a}_C = (\dot{u} - v\dot{\theta},\ \dot{v} + u\dot{\theta})$.

SLIDE CHECK pp. 24–25
The slides print $+v\dot{\theta}$ in the y-acceleration (and in ${}^{C}F_y$, ${}^{C}f_y$). The correct term is $+u\dot{\theta}$: (1) the slide's own cross product $(0, 0, \dot{\theta}) \times (u, v, 0) = (-v\dot{\theta}, u\dot{\theta}, 0)$; (2) the p.26 matrix already uses u; (3) constant u on a circle needs centripetal $u\dot{\theta}$ in y, the printed form gives 0. The x-equation $(-v\dot{\theta})$ is right. Show the cross-product line so $u\dot{\theta}$ is justified. [checked]

Pattern: x gets $-v\dot{\theta}$, y gets $+u\dot{\theta}$; $C = G$ and no accelerations: $\mathbf{a} = \dot{\theta}(-v, u) \perp$ velocity.

WORKED P11.3
$C = G$, $m = 10$, $u = 2$, $v = 0$, $\dot{\theta} = 0.5$: $f_y = mu\dot{\theta} = 10$ N (circle $R = u/\dot{\theta} = 4$ m, $mu^2/R = 10$ N); printed form gives 0. [checked]

Equations of motion $\mathbf{M}\dot{\boldsymbol{\zeta}} + c(\boldsymbol{\zeta}) = {}^{C}\mathbf{Q}$ [p. 25–27]

Matrix form in $\{C\}$ (substitute f_x, f_y into n_z, then collect)

$$ \begin{bmatrix} m & 0 & -m\,{}^{C}y_G \\ 0 & m & m\,{}^{C}x_G \\ -m\,{}^{C}y_G & m\,{}^{C}x_G & {}^{G}I + m({}^{C}x_G^2 + {}^{C}y_G^2) \end{bmatrix} \begin{bmatrix} \dot{u} \\ \dot{v} \\ \ddot{\theta} \end{bmatrix} + \begin{bmatrix} -m\dot{\theta}(v + {}^{C}x_G\dot{\theta}) \\ m\dot{\theta}(u - {}^{C}y_G\dot{\theta}) \\ m\dot{\theta}({}^{C}x_G\,u + {}^{C}y_G\,v) \end{bmatrix} = \begin{bmatrix} {}^{C}f_x \\ {}^{C}f_y \\ {}^{C}n_z \end{bmatrix} $$

Use: $\mathbf{M}$: constant, symmetric, M_{33} = inertia about C (parallel axis); c: centripetal/Coriolis from the rotating frame, $= 0$ at rest. Accelerations: $\dot{\boldsymbol{\zeta}} = \mathbf{M}^{-1}({}^{C}\mathbf{Q} - c)$. $C = G$: $\mathbf{M} = \mathrm{diag}(m, m, {}^{G}I)$, $c = (-mv\dot{\theta},\ mu\dot{\theta},\ 0)$.

RECIPE Derive the equations of motion (W13 Q4 style)
1. Draw $\{O\}$, $\{C\}$, $\{G\}$; applied loads at C, inertial loads at G, all in $\{C\}$.
2. $F_x = f_x$, $F_y = f_y$; moment about G with the lever arm $-\mathbf{P}_{CG}$.
3. $\ddot{\mathbf{P}}_G$ by the four-term sum (use $u\dot{\theta}$ in y).
4. Substitute f_x, f_y into n_z; read $\mathbf{M}$ from the coefficients of $\dot{u}, \dot{v}, \ddot{\theta}$, c from the rest.
5. **Check:** $\mathbf{M}$ symmetric; $M_{33} = I_G + m(x_G^2 + y_G^2)$; $c = 0$ at rest; $C = G$ gives a diagonal $\mathbf{M}$.

WORKED off-centre COM
$m = 10$, ${}^{G}I = 0.25$, G at $x_G = 0.05$ ahead of C; at rest ${}^{C}f = (10, 0)$, ${}^{C}n_z = 0.4$. $M_{33} = 0.25 + 10(0.05)^2 = 0.275$.

$$ \begin{bmatrix} 10 & 0 & 0 \\ 0 & 10 & 0.5 \\ 0 & 0.5 & 0.275 \end{bmatrix} \ddot{\mathbf{z}} = \begin{bmatrix} 10 \\ 0 \\ 0.4 \end{bmatrix} $$

Row 1: $\dot{u} = 1.0$. Row 2: $\dot{v} = -0.05\ddot{\theta}$. Row 3: $(0.275 - 0.025)\ddot{\theta} = 0.4 \Rightarrow \ddot{\theta} = 1.6$ rad/s^2, $\dot{v} = -0.08$ m/s^2 (C swings sideways about G). Naive $0.4/0.275 = 1.4545$ ignores the coupling and is wrong. [checked]

TRAP
${}^{G}I$ alone in M_{33} when $C \neq G$; sign of the moment-transfer term; $v\dot{\theta}$ for $u\dot{\theta}$; chassis and inertial quantities mixed.

Chassis frame → inertial frame [p. 28–34]

Loads transform with $\mathbf{J}^{\top}$ (power invariance) [p. 30–32]

$$ \dot{\boldsymbol{\xi}} = \mathbf{J}\boldsymbol{\zeta}, \quad \boldsymbol{\zeta} = \mathbf{J}^{-1}\left[\dot{\boldsymbol{\xi}} - \dot{\mathbf{J}}\mathbf{J}^{-1}\boldsymbol{\xi}\right] $$

$$ {}^{O}\mathbf{Q}\cdot\dot{\boldsymbol{\xi}} = {}^{C}\mathbf{Q}\cdot\boldsymbol{\zeta} \;\Rightarrow\; {}^{C}\mathbf{Q} = \mathbf{J}^{\top}\,{}^{O}\mathbf{Q}, \quad {}^{O}\mathbf{Q} = \mathbf{J}\,{}^{C}\mathbf{Q}\ \text{(rotation)} $$

Use: velocities map with $\mathbf{J}$, loads with $\mathbf{J}^{\top}$ (same as $\boldsymbol{\tau} = \mathbf{J}^{\top}\mathbf{F}$ for arms). Say in words: power is frame-independent.

Inertial form [p. 34]

$$ {}^{O}\mathbf{M}\ddot{\boldsymbol{\xi}} + {}^{O}c = {}^{O}\mathbf{Q}, \quad {}^{O}\mathbf{M} = (\mathbf{J}^{\top})^{-1}\mathbf{M}\mathbf{J}^{-1} $$

$$ {}^{O}c = (\mathbf{J}^{\top})^{-1}\left[c - \mathbf{M}\mathbf{J}^{-1}\dot{\mathbf{J}}\mathbf{J}^{-1}\dot{\boldsymbol{\xi}}\right] $$

Use: $C = G$: ${}^{O}\mathbf{M} = \mathrm{diag}(m, m, {}^{G}I)$ and ${}^{O}c = 0$: a free body obeys $m\ddot{x} = f_x$, $m\ddot{y} = f_y$, ${}^{G}I\ddot{\theta} = \tau_z$ in world axes. The velocity terms exist only because the chassis axes rotate.

SLIDE CHECK p. 34, third line
The slide prints ${}^{O}c = (\mathbf{J}^{\top})^{-1}[c - \mathbf{M}\dot{\mathbf{J}}\mathbf{J}^{-1}\dot{\boldsymbol{\xi}}]$, dropping a $\mathbf{J}^{-1}$ in front of $\dot{\mathbf{J}}$ (expand line 2 of the same slide). They agree only at $\theta = 0$. Test $C = G$, $m = 10$, $\theta = 90°$, $u = 1$, $v = 0$, $\dot{\theta} = 2$: correct ${}^{O}c = 0$, printed gives $(-20, 20, 0)$. The p.30 equations are right; "$\mathbf{J}(\eta)$" on p.30 is a typo for $\mathbf{J}(\boldsymbol{\xi})$. [checked]

WORKED wrench rotation
${}^{C}\mathbf{Q} = (10, 0, 0.4)$, $\theta = 30°$: ${}^{O}\mathbf{Q} = (8.660, 5.000, 0.4)$; $\ddot{\boldsymbol{\xi}} = (0.866, 0.5, 1.6)$ (chassis $(1, 0, 1.6)$ with the linear part rotated). [checked]

Forward simulation [p. 35]

RECIPE One step, chassis frame
1. $c(\boldsymbol{\zeta})$ from the current velocity (0 at rest).
2. Solve $\mathbf{M}\dot{\boldsymbol{\zeta}} = {}^{C}\mathbf{Q} - c$ (diagonal $\mathbf{M}$: divide).
3. $\boldsymbol{\zeta} \leftarrow \boldsymbol{\zeta} + \Delta t\,\dot{\boldsymbol{\zeta}}$ (Euler); then $\dot{\boldsymbol{\xi}} = \mathbf{J}(\theta)\boldsymbol{\zeta}$.
4. $\boldsymbol{\xi} \leftarrow \boldsymbol{\xi} + \Delta t\,\dot{\boldsymbol{\xi}}$. **Check:** θ in radians; order is acceleration, velocity, pose.

Example: $\dot{\boldsymbol{\zeta}} = (1, 0, 1.6)$, $\Delta t = 0.1$: $\boldsymbol{\zeta}_1 = (0.1, 0, 0.16)$, pose still the origin. [checked]

Wheel forces to chassis loads: differential drive [p. 36–37]

Two driven wheels at $(0, \pm a)$; F_1 left ($+a$), F_2 right ($-a$); $C = G$ [p. 37]

$$ \begin{bmatrix} {}^{C}f_x \\ {}^{C}f_y \\ {}^{C}n_z \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ -a & a \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}, \qquad F_i = \frac{\tau_i}{r} $$

$$ \dot{u} = \frac{F_1 + F_2}{m}, \quad \ddot{\theta} = \frac{a(F_2 - F_1)}{{}^{G}I}, \quad f_y^{req} = mu\dot{\theta}\ (v = 0) $$

Use: $n_z = x f_y - y f_x$. Equivalent to ${}^{C}\mathbf{Q} = \mathbf{J}_w^{\top}\boldsymbol{\tau}$ with $\mathbf{J}_w = \frac{1}{r}\begin{bmatrix} 1 & 0 & -a \\ 1 & 0 & a \end{bmatrix}$ (the W10 map, transposed). Power: $\boldsymbol{\tau}\cdot\dot{\boldsymbol{\varphi}} = f_x u + n_z\dot{\theta}$. Castor: no drive force, adds nothing.

RECIPE Accelerations from wheel torques
1. Top view: $C, G, \pm a, F_1$ (left, $+y$), F_2 (right, $-y$).
2. $F = \tau/r$ each wheel; $f_x = F_1 + F_2$, $f_y = 0$, $n_z = a(F_2 - F_1)$ (lever arm a, not $2a$).
3. $C = G$, no slip: $\dot{u} = f_x/m$, $\ddot{\theta} = n_z/{}^{G}I$, $v = 0$.
4. **Check sign:** right wheel pushing harder turns the robot left ($\dot{\theta} > 0$). Check power. Circle radius $R = u/\dot{\theta} = {}^{G}I f_x/(m n_z)$.

WORKED numbers are ours; no numbers in the deck
$m = 10$, ${}^{G}I = 0.25$, $r = 0.1$, $a = 0.2$, $\tau_L = 0.4$, $\tau_R = 0.6$ N m from rest: $F_1 = 4$, $F_2 = 6$ N; $f_x = 10$ N, $n_z = 0.2(6 - 4) = 0.4$ N m; $\dot{u} = 1.0$ m/s^2, $\ddot{\theta} = 1.6$ rad/s^2; $R = 0.625$ m (constant circle, speeding up). At $t = 1$ s: $u = 1$, $\dot{\theta} = 1.6$; $\dot{\varphi}_L = (1 - 0.32)/0.1 = 6.8$, $\dot{\varphi}_R = 13.2$ rad/s; $mu\dot{\theta} = 16$ N sideways to hold $v = 0$ (if free: $\dot{v} = -1.6$); power $0.4(6.8) + 0.6(13.2) = 10.64$ W $= 10(1) + 0.4(1.6)$. At $t = 2$ s: $(x, y) = (-0.036, 1.249)$ m, $\theta = 3.2$ rad. [checked]

WORKED P11.1 and quiz
$m = 8$, ${}^{G}I = 0.2$, $r = 0.08$, $a = 0.15$, $\tau = (0.3, 0.5)$: $F = (3.75, 6.25)$, $f_x = 10$, $n_z = 0.375$; $\dot{u} = 1.25$, $\ddot{\theta} = 1.875$, $R = 0.667$ m (no lever arm: 12.5; $2a$: 3.75, both wrong). [checked]

SLIDE CHECK gap, not an error
The deck never states no-slip inside the dynamics: the p.37 matrix lists only forward forces; the lateral force holding $v = 0$ is a constraint reaction. Wheel forces given and $\dot{v}$ asked: no slip $\Rightarrow \dot{v} = 0$ (say so); free body $\Rightarrow \mathbf{M}\dot{\boldsymbol{\zeta}} + c = \mathbf{Q}$ with $f_y = 0$.

TRAP
$2a$ as lever arm (doubles $\ddot{\theta}$); sign swap (harder right push turns right); skipping $F = \tau/r$.

Swedish wheels, summary [p. 38–39] [added]

Slide p.38 is a figure only. Wheel i at (x_i, y_i) pushing along unit (d_x, d_y) gives column $(d_x, d_y, x_i d_y - y_i d_x)^{\top}$ of ${}^{C}\mathbf{Q} = \mathbf{B}\mathbf{F}$ ($\mathbf{B}$ = W10 matrix transposed, up to r); 4 wheels, 3 loads: rank 3, a null space of internal forces. [checked] Summary: $\mathbf{M}\dot{\boldsymbol{\zeta}} + c = {}^{C}\mathbf{Q}$ (body); ${}^{O}\mathbf{M}\ddot{\boldsymbol{\xi}} + {}^{O}c = {}^{O}\mathbf{Q}$ (inertial); ${}^{C}\mathbf{Q} = \mathbf{J}^{\top}\,{}^{O}\mathbf{Q}$. No control content (W8–W9).

Framework and completeness [p. 4–9,25–26,80–81]

Planning = given map, start, goal (and robot kinematics), find a continuous collision-free path, or report failure. First squeeze the robot to a point (C-space), then *combinatorial* (describe C_{free}, build a graph, search: visibility graph, Voronoi, cell decomposition, grid + BFS/DFS/Dijkstra/A*) or *sampling-based* (probe C-space with a collision detector: PRM 1996, RRT 1998).

Complete	finds a path whenever one exists, reports failure otherwise (visibility graph, Voronoi, exact cells)
Resolution complete	does so only at the grid resolution; a gap narrower than a cell is invisible (BFS on a grid, approximate cells)
Probabilistically complete	P(path found) $\to 1$ as samples $\to \infty$; may never terminate if no path exists (PRM, RRT)

TRAP
"PRM / RRT are complete" is false. "BFS is complete" needs "on a finite graph / at the grid resolution". DFS fails on infinite trees. Slide p.81 says "probability of samples approaches 1": it means the probability of *finding a path*.

Configuration space and inflation [p. 11–24]

Free space [p. 15]; inflation by a disc

$$C_{free} = \{\mathbf{q} \in C \mid \mathbf{q} \cap C_{obs} = \varnothing\}, \qquad A_{grown} = A + Pr + \pi r^2$$

Use: A, P = area, perimeter of a convex polygon; r = robot radius (Minkowski sum: each edge moves out by r, each corner becomes an arc of radius r). Corridor width w: free width for the point robot = $w - 2r$; negative = cannot pass. Square robot (half-side r): square margin, 2×2 m obstacle $\to$ 3×3 m at $r = 0.5$ (disc: 8.785 m^2). Slide writes $C_{obs} \in C$; it should be $\subset$.

Rotary joints: $q \equiv q + 2\pi$, so C-space is a torus. One workspace point can map to several configurations (2-link $L_1 = L_2 = 1$: tip $(1, 1)$ at $(0°, 90°)$ and $(90°, -90°)$).

WORKED
Triangle $(5,3), (6,3), (5,1)$, $r = 0.5$: $A = 1$, $P = 1 + \sqrt{5} + 2 = 5.236$, $A_{grown} = 1 + 2.618 + 0.785 = 4.403$ m^2. Corridor 1.0 m, $r = 0.4$: 0.2 m free; $r = 0.55$: $1.0 - 1.1 < 0$, blocked. [checked]

Visibility graph, Voronoi, cells [p. 27–46]

	Visibility graph	Voronoi
Optimises	path length	clearance
Path	straight, via corners	equidistant curves
Risk	hugs obstacles (inflate first)	longer; fails with short-range sensor

Visibility: join S, G and every corner pair whose segment avoids obstacle interiors (touching/sliding on an edge is allowed), weight by length, then Dijkstra or A*. Voronoi: "maximise sensor readings".
Cells: exact (trapezoids; complete), approximate (uniform grid, blocked if touched; resolution complete), adaptive (large in open space, small near obstacles).

WORKED practical Q2 map
$S = (1.5, 1.5)$, $G = (9, 9)$ blocked on the diagonal by rectangle $(6, 6)$–$(8, 8)$. Via corner $(8, 6)$: $\sqrt{6.5^2 + 4.5^2} = 7.906$, $\sqrt{1^2 + 3^2} = 3.162$, total 11.068; to the disc edge (radius 1) 10.068. [checked]

BFS and DFS [p. 47–55]

BFS: FIFO queue, mark seen when added; fewest edges, cost-optimal only if all edges equal; $O(|V| + |E|)$. **DFS**: LIFO stack, dives, no cost guarantee, depends on neighbour order. **State your tie-break (alphabetical) first.** Stop when the goal is popped.

WORKED slide graph p.56, our alphabetical tie-break
Edges A–B 10, B–E 9, A–C 6, C–F 5, F–H 7, H–J 4, J–K 3, A–D 8, D–G 9, F–G 3, G–I 7, I–K 5. BFS visits A B C D E F G H I J K, path A-D-G-I-K (4 edges) cost $8 + 9 + 7 + 5 = 29$. DFS visits A B E C F G D I K, path A-C-F-G-I-K cost $6 + 5 + 3 + 7 + 5 = 26$. Best is 25: neither is optimal. [checked]

Dijkstra [p. 56–58]

RECIPE Dijkstra by hand
1. Table: columns = nodes, rows = past cost g and parent. g(start) = 0, rest unknown.
2. Pick the unexpanded node with the smallest g; move to closed.
3. Relax each neighbour m: if $g(n) + w < g(m)$ set $g(m)$, parent$(m) = n$ (strict). Never relax a closed node.
4. **Stop when the goal is expanded**, not when first discovered (K found at 26 via I, final 25 via J). Trace parents; **check** g(goal) = sum of path edges.

Non-negative edges only. Uninformed (no goal knowledge); Dijkstra = A* with $h = 0$.

WORKED Dijkstra on the slide graph, A to K
Expansion order A (0), C (6), D (8), B (10), F (11), G (14), H (18), E (19), I (21), J (22), K (25): ten nodes before K, 11 in all. G falls 17 (via D) to 14 (via F); K falls 26 (via I) to 25 (via J).

	A	B	C	D	E	F	G	H	I	J	K
g	0	10	6	8	19	11	14	18	21	22	25
par	–	A	A	A	B	C	F	F	G	H	J

Path A-C-F-H-J-K: $6 + 5 + 7 + 4 + 3 = 25$. [checked]

TRAP
Stopping at first discovery (gives 26); relaxing a closed node.

A* and heuristics [p. 59–62]

A*

$$f(n) = g(n) + h(n)$$

g: best cost found so far, start to n. h: optimistic cost-to-go. A* = Dijkstra with priority f; stop when the goal is *expanded*.

admissible: $h(n) \le d(n, \text{goal})\ \forall n$ (non-strict; h(goal) = 0)

consistent: $h(n) \le d(n, m) + h(m)\ \forall$ neighbours

Use: consistent $\Rightarrow$ admissible, not conversely. Admissible keeps A optimal; consistent also makes popped f non-decreasing and lets a plain closed set work. Overestimating h can return a worse path. $h = 0$ is Dijkstra.*

RECIPE Fill an A* table (slide pp. 59–60)
1. Write h beside every node. Admissible? compare each $h(n)$ with the true remaining cost (backward Dijkstra).
2. Consistent? check $h(n) \le w + h(m)$ edge by edge, both directions.
3. Pop the smallest $f = g + h$ (tie: smaller h, then alphabetical; say so). Relax with $g(n) + w$; put h of the *child* in f.
4. Stop when the goal is popped. **Check:** f(goal) = g(goal) = path cost; unexpanded nodes are the saving.

Slide graph, heuristic p.59: h: A 20, B 10, C 17, D 13, E 8, G 8, H 7, I 2, J 2; K = 0 (not printed); F has no red label, we use $h(F) = 9$ (the only number printed beside F). True $d(n, K)$: H 7, J 3, I 5, G 12, F 14, C 19, D 21, A 25, B 35, E 44. All $h \le d$: **admissible** (tight at H). **Not consistent**: H→J: $7 > 4 + 2$; C→F: $17 > 5 + 9$.

WORKED A* table, tie-break lowest f, then h, then alphabetical

#	expand	g	h	open after: node $g+h=f$ (parent)
1	A	0	20	B 10+10=20 (A); D 8+13=21 (A); C 6+17=23 (A)
2	B	10	10	D 21 (A); C 23 (A); E 19+8=27 (B)
3	D	8	13	C 23 (A); G 17+8=25 (D); E 27 (B)
4	C	6	17	F 11+9=20 (C); G 25 (D); E 27 (B)
5	F	11	9	G 14+8=22 (F); H 18+7=25 (F); E 27 (B)
6	G	14	8	I 21+2=23 (G); H 25 (F); E 27 (B)
7	I	21	2	H 25 (F); K 26+0=26 (I); E 27 (B)
8	H	18	7	J 22+2=24 (H); K 26 (I); E 27 (B)
9	J	22	2	K 25+0=25 (J); E 27 (B)
10	K	25	0	E 27 (B)

	A	B	C	D	E	F	G	H	I	J	K
g	0	10	6	8	19	11	14	18	21	22	25
h	20	10	17	13	8	9	8	7	2	2	0
f	20	20	23	21	27	20	22	25	23	24	25
par	–	A	A	A	B	C	F	F	G	H	J

Order A B D C F G I H J K, path A-C-F-H-J-K, cost 25, 10 expansions (Dijkstra 11); E discovered ($f = 27$) never expanded. Popped f: 20, 20, 21, 23, 20, 22, 23, 25, 24, 25, so it falls twice: the signature of an inconsistent h. Perfect $h = d$: expands only A C F H J K. Overestimating $h(F) = 30$: order A B D C G I E K, path A-D-G-I-K, cost 29. [checked]

SLIDE CHECK pp. 56–60
(1) A blue 9 sits above F on the A* slide where the red h labels are, and in the same place on the Dijkstra slides where there is no h: probably a leftover edge label, so the slide gives no $h(F)$. Any integer $h(F)$ from 0 to 13 gives the same order and cost 25. (2) Both tables are blank on the slides; the fills are ours. (3) The slide asks for h admissible *and* consistent, but its own h fails consistency at H→J. (4) K has no printed h; 0 is the only valid value.

TRAP
h of the wrong node; f with the new g of a closed node; "A* always optimal" (needs admissible h); "Manhattan is always admissible" (not with diagonals); any heuristic makes A* faster (not so: $h = 0$ is Dijkstra). Slide demo (one grid): A* 277 operations vs Dijkstra 1378 vs BFS 1413; do not quote as a general ratio.

Grids and the curse of dimensionality [p. 63–65]

N cells per dimension, d dimensions: N^d nodes; neighbours $2d$ (axis) or $3^d - 1$ (with diagonals).

Grid	Nodes	Neighb.	Note
2D 100^2	10^4	4 or 8	
3D 100^3	10^6	6 to 26	
6D 100^6	10^{12}	12 or 728	slide leaves it open; ≈1 TB at 1 B/node

4-connected 100×100: $|E| = 2 \cdot 100 \cdot 99 = 19\,800$, $|V| + |E| = 29\,800$ [checked]. Resolution trade-off: too coarse loses narrow gaps, too fine is slow. Say: node count grows exponentially with dimension, so grids fail and sampling is used. **Grid heuristics.** 4-connected, unit cost: Manhattan $|\Delta r| + |\Delta c|$ is admissible and consistent. 8-connected (diagonal $\sqrt{2}$): Manhattan **overestimates** (says 2 for a step costing 1.414); use Euclidean $\sqrt{\Delta r^2 + \Delta c^2}$. Widget "Manhattan trap": A* with Manhattan returns 10.243 against the optimum 9.657.

Sampling-based planning: PRM and RRT [p. 66–79]

Framework: empty graph; sample a point (sampler); pick near vertices (distance function); connect with a local planner; add the edge if collision-free (collision detector); repeat; query start to goal. **PRM**: learning phase (sample, connect to near roadmap vertices), then query (add S, G; A* or Dijkstra). Simple, works in high dimension; holonomic only, can fail in narrow passages, not for dynamic environments (roadmap built first).

RRT: tree from the start; sample q_{rand}, nearest node, step towards it, add q_{new} if free; stop when a node lands in the goal region. Handles non-holonomic robots by sampling controls (steer, speed) and keeping the result closest to the target [p. 77–79].

RRT extension

$$q_{new} = q_{near} + \delta \frac{q_{rand} - q_{near}}{\|q_{rand} - q_{near}\|}$$

Use: discard q_{rand} inside an obstacle; reject the edge if the segment crosses one. Segment $\overline{ab}$ crosses edge $\overline{cd}$ if c, d are on opposite sides of line ab and a, b on opposite sides of line cd. Point in polygon: ray to $+x$, odd crossings = inside.

WORKED practical Q2 map: $C = [0, 10]^2$, **goal disc radius 1 at** $(9, 9)$

Uniform sample in the goal: $\pi/100 = 0.0314$, about 32 samples on average. RRT step: $S = (1.5, 1.5)$, $q_{rand} = (6, 2)$, $\delta = 1$: $\|\cdot\| = \sqrt{4.5^2 + 0.5^2} = 4.528$, $q_{new} = (2.494, 1.610)$. Hand PRM S-(4,2.5)-(7,4.5)-(9,7)-(9,8.5): legs 2.693 + 3.606 + 3.202 + 1.5 = 11.00, 9.3% longer than the visibility optimum 10.068 (PRM/RRT paths are rarely shortest). $(3, 5.2)$ in triangle $(2, 7), (4, 5), (3, 4)$: one crossing, inside. Nearest to $(6, 4)$ among S,(4,2.5),(7,4.5),(9,7): $(7, 4.5)$, 1.118. $(5.4, 2)$ lies inside triangle $(5, 3), (6, 3), (5, 1)$. [checked]

TRAP

"RRT always finds the shortest path" (no); "PRM stops when it reaches the goal" (that is RRT); RRT/PRM may never terminate if no path exists.

Practical Q1: maze (our grid reading)

36 open cells, 38 edges, unit weights. BFS 8 moves (26 cells popped); DFS 16 moves (19 popped, order up/right/down/left); A* with Manhattan (admissible and consistent here) 8 moves, 16 popped. G is 4 cells away in Manhattan terms but a wall forces an 8-move detour. [checked]

Choosing a planner [p. 80–83]

Method	Guarantee	Weakness
BFS	fewest edges; resolution complete	cost-optimal only if equal edges
DFS	finds some path	long, no optimality
Dijkstra	cheapest, costs ≥ 0	expands everywhere
A*	cheapest if h admissible	bad h wastes effort
Visibility	shortest, complete	hugs corners
Voronoi	max clearance, complete	not shortest
PRM	prob. complete	holonomic, narrow gaps, static
RRT	prob. complete	path not optimal

Optimality of Dijkstra/A* are standard results (the slides give them via the heuristic conditions).

RECIPE "Which planner?" scenario

1. Static, known map? If not, a roadmap built once (PRM) is a poor fit.
2. High DoF (arm, snake): sampling; 2D grid: graph search.
3. Non-holonomic (car): RRT with sampled controls.
4. Clearance matters: Voronoi. Length, polygons: visibility graph.
5. Costs plus a good distance-to-goal guess: A*; costs, no guess: Dijkstra.
6. Name the method *and* the property that justifies it.

Matrix formulas

Inverse and determinant

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\det\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = a(ei-fh) - b(di-fg) + c(dh-eg)$$

$$\mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}}\begin{bmatrix} ei-fh & ch-bi & bf-ce \\ fg-di & ai-cg & cd-af \\ dh-eg & bg-ah & ae-bd \end{bmatrix}$$

$\det = 0 \Rightarrow$ no inverse (singular). $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$; $(\mathbf{AB})^{\top} = \mathbf{B}^{\top}\mathbf{A}^{\top}$.

Trigonometry

Identities

$\sin(a \pm b) = \sin a \cos b \pm \cos a \sin b$
$\cos(a \pm b) = \cos a \cos b \mp \sin a \sin b$
$\tan(a \pm b) = \dfrac{\tan a \pm \tan b}{1 \mp \tan a \tan b}$
$\sin 2a = 2\sin a \cos a$
$\cos 2a = \cos^2 a - \sin^2 a = 2\cos^2 a - 1 = 1 - 2\sin^2 a$
$\tan 2a = \dfrac{2\tan a}{1-\tan^2 a}$, $\sin^2 a + \cos^2 a = 1$
$\cos^2 a = \frac{1}{2}(1+\cos 2a)$, $\sin^2 a = \frac{1}{2}(1-\cos 2a)$
$\cos(\pi - q) = -\cos q$, $\sin(\pi - q) = \sin q$
Use: arm algebra: $c_1c_{12} + s_1s_{12} = c_2$ and $s_{12}c_1 - c_{12}s_1 = s_2$ (with $c_{12} = \cos(q_1+q_2)$).

Single-harmonic form

$$a\cos x + b\sin x = R\cos(x-\varphi) = R\sin(x+\psi)$$

$R = \sqrt{a^2+b^2}$, $\varphi = \operatorname{atan2}(b,a)$, $\psi = \operatorname{atan2}(a,b)$.
Use: solving $a\cos x + b\sin x = c$: $x = \varphi \pm \arccos(c/R)$, needs $|c| \le R$ (reach test).

Triangle rules

Cosine rule: $c^2 = a^2 + b^2 - 2ab\cos C$
Sine rule: $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$
Use: arm triangle: interior elbow angle $= \pi - q_2$, so $r^2 = L_1^2 + L_2^2 + 2L_1L_2\cos q_2$ (the sign flip is the classic trap).

atan2(y,x): quadrant table

x	y	value
> 0	any	$\arctan(y/x)$
< 0	≥ 0	$\arctan(y/x) + \pi$
< 0	< 0	$\arctan(y/x) - \pi$
0	> 0 / < 0	$+\pi/2$ / $-\pi/2$

Range $(-\pi, \pi]$. Never use plain $\arctan(y/x)$ for a position angle.

Rotations and frames

Basic rotations (right-hand rule)

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c & -s \\ 0 & s & c \end{bmatrix},\ R_y(\theta) = \begin{bmatrix} c & 0 & s \\ 0 & 1 & 0 \\ -s & 0 & c \end{bmatrix},\ R_z(\theta) = \begin{bmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$R^{\top} = R^{-1}$, $\det R = +1$, columns orthonormal. $R_z(a)R_z(b) = R_z(a+b)$; different axes do not commute.

Composition rules

${}^A\mathbf{R}_C = {}^A\mathbf{R}_B\,{}^B\mathbf{R}_C$ (inner indices cancel).
Right-multiply: next turn about the *current* (moving) axes.
Left-multiply: next turn about the *fixed* axes.
Same for 4×4 $\mathbf{T}$: order of the matrices = order of the moves, read left to right for current axes.

Homogeneous inverse

$$\begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^{\top} & 1 \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{R}^{\top} & -\mathbf{R}^{\top}\mathbf{t} \\ \mathbf{0}^{\top} & 1 \end{bmatrix}$$

Cross product and skew matrix

$$\mathbf{a}\times\mathbf{b} = [\mathbf{a}]_{\times}\mathbf{b} = \begin{bmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{bmatrix},\ [\mathbf{a}]_{\times} = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}$$

$|\mathbf{a}\times\mathbf{b}| = |\mathbf{a}||\mathbf{b}|\sin\theta$; $\mathbf{a}\times\mathbf{b} = -\mathbf{b}\times\mathbf{a}$; $[\mathbf{a}]_{\times}^{\top} = -[\mathbf{a}]_{\times}$.
Rotating frame: $\dot{\mathbf{R}} = [\boldsymbol{\omega}]_{\times}\mathbf{R}$. Velocity of a point on a body: $\mathbf{v} = \mathbf{v}_0 + \boldsymbol{\omega}\times\mathbf{p}$.

Small angles

$|\theta| \lesssim 0.1$ rad ($\approx 6°$)

$\sin\theta \approx \theta$ (error $< 0.2\%$), $\tan\theta \approx \theta$ ($< 0.4\%$), $\cos\theta \approx 1 - \theta^2/2$ (abs. error $< 10^{-5}$; or just 1).
$(1+x)^n \approx 1 + nx$. $R_z(\delta) \approx \mathbf{I} + [\delta\hat{z}]_{\times}$ (rotations commute to first order).

Unit conversions

$1°$	$= 0.017453$ rad
1 rad	$= 57.2958°$
1 rpm	$= 2\pi/60 = 0.10472$ rad/s
1 rad/s	$= 9.5493$ rpm
1 Hz	$= 2\pi$ rad/s
e.g. 3000 rpm	$= 314.16$ rad/s
$90°$	$= 1.5708$ rad

Ideal gearbox, ratio N (motor : load): $\omega_m = N\omega_l$, $\tau_l = N\tau_m$, inertia seen at the load $N^2 I_m$. [added]

Euler-angle extraction

ZYZ: $\mathbf{R} = R_z(\alpha)R_y(\beta)R_z(\gamma)$

$$\mathbf{R} = \begin{bmatrix} c_\alpha c_\beta c_\gamma - s_\alpha s_\gamma & -c_\alpha c_\beta s_\gamma - s_\alpha c_\gamma & c_\alpha s_\beta \\ s_\alpha c_\beta c_\gamma + c_\alpha s_\gamma & -s_\alpha c_\beta s_\gamma + c_\alpha c_\gamma & s_\alpha s_\beta \\ -s_\beta c_\gamma & s_\beta s_\gamma & c_\beta \end{bmatrix}$$

$\beta = \operatorname{atan2}\big(\sqrt{r_{13}^2 + r_{23}^2},\, r_{33}\big)$, $\alpha = \operatorname{atan2}(r_{23}, r_{13})$, $\gamma = \operatorname{atan2}(r_{32}, -r_{31})$.
Second solution (flip): $-\beta$, $\alpha + \pi$, $\gamma + \pi$ (use $\operatorname{atan2}(-r_{23}, -r_{13})$, $\operatorname{atan2}(-r_{32}, r_{31})$).
Singular if $s_\beta = 0$: only $\alpha + \gamma$ (β=0) or $\alpha - \gamma$ (β=π) is fixed.

ZYX (yaw ψ, pitch θ, roll ϕ): $\mathbf{R} = R_z(\psi)R_y(\theta)R_x(\phi)$

$$\mathbf{R} = \begin{bmatrix} c_\psi c_\theta & c_\psi s_\theta s_\phi - s_\psi c_\phi & c_\psi s_\theta c_\phi + s_\psi s_\phi \\ s_\psi c_\theta & s_\psi s_\theta s_\phi + c_\psi c_\phi & s_\psi s_\theta c_\phi - c_\psi s_\phi \\ -s_\theta & c_\theta s_\phi & c_\theta c_\phi \end{bmatrix}$$

$\theta = \operatorname{atan2}\big(-r_{31}, \sqrt{r_{11}^2 + r_{21}^2}\big)$, $\psi = \operatorname{atan2}(r_{21}, r_{11})$, $\phi = \operatorname{atan2}(r_{32}, r_{33})$.
Second solution: $\pi - \theta$, $\psi + \pi$, $\phi + \pi$. Singular at $\theta = \pm 90°$ ($c_\theta = 0$).

Inertia of standard bodies (about the centroid, principal axes)

Body (mass m)	Inertia
Slender rod, length L, axis $\perp$ rod	$\frac{1}{12}mL^2$ (≈ 0 about its own axis)
Box $a\times b\times c$ (along x,y,z)	$I_{xx} = \frac{1}{12}m(b^2+c^2)$; $I_{yy} = \frac{1}{12}m(a^2+c^2)$; $I_{zz} = \frac{1}{12}m(a^2+b^2)$
Solid cylinder, radius r, length h, axis z	$I_{zz} = \frac{1}{2}mr^2$; $I_{xx} = I_{yy} = \frac{1}{12}m(3r^2+h^2)$
Solid sphere, radius r	$\frac{2}{5}mr^2$ (any axis)
Thin disc, radius r	axis $\perp$ disc $\frac{1}{2}mr^2$; diameter $\frac{1}{4}mr^2$

Parallel-axis theorem

Scalar (axes parallel, distance d): $I = I_C + md^2$.
Tensor ($\mathbf{r}$ = vector from centroid to the new origin):

$$\mathbf{I}_O = \mathbf{I}_C + m\big(\|\mathbf{r}\|^2\mathbf{1} - \mathbf{r}\mathbf{r}^{\top}\big)$$

Components: $I'_{xx} = I_{xx} + m(y^2+z^2)$, $I'_{xy} = I_{xy} - mxy$ (convention $\mathbf{I} = \int(\|\mathbf{r}\|^2\mathbf{1} - \mathbf{r}\mathbf{r}^{\top})dm$).
Different frame: $\mathbf{I}_B = \mathbf{R}\,\mathbf{I}_A\,\mathbf{R}^{\top}$.
Use: link inertia about the joint: centroid value + md^2. Never add inertias about different points.

Polynomials and stability

Quadratic

$ax^2 + bx + c = 0$: $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.
$b^2 - 4ac > 0$: two real; $= 0$: double; < 0: complex pair. Sum $= -b/a$, product $= c/a$.

TABLES

Cubic $q(t)$: boundary-condition matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & T & T^2 & T^3 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 2T & 3T^2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} q_0 \\ q_F \\ \dot{q}_0 \\ \dot{q}_F \end{bmatrix}$$

$a_0 = q_0$, $a_1 = \dot{q}_0$, $\Delta = q_F - q_0$:

$$a_2 = \frac{3\Delta - (2\dot{q}_0 + \dot{q}_F)T}{T^2}, \quad a_3 = \frac{-2\Delta + (\dot{q}_0 + \dot{q}_F)T}{T^3}$$

Rest to rest: $a_2 = 3\Delta/T^2$, $a_3 = -2\Delta/T^3$. Quintic rest to rest: $a_3 = 10\Delta/T^3$, $a_4 = -15\Delta/T^4$, $a_5 = 6\Delta/T^5$. [checked]

Routh–Hurwitz

Quadratic $s^2 + a_1 s + a_0$: stable iff $a_1, a_0 > 0$.
Cubic $s^3 + a_2 s^2 + a_1 s + a_0$: stable iff all $a_i > 0$ **and** $a_2 a_1 > a_0$.
PID: $s^3 + k_v s^2 + k_p s + k_i$: need $k_v k_p > k_i$ (equality gives $\pm j\sqrt{k_p}$, undamped).

Second-order step response,

Poles $-\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$; $\omega_d = \omega_n\sqrt{1 - \zeta^2}$.
Overshoot $(0 < \zeta < 1)$: $M_p = \exp\left(-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}\right)$; peak time π/ω_d.
2 % settling $t_s \approx \frac{4}{\zeta\omega_n}$ (approximate rule).
$\zeta = 0.6$: $M_p = 9.48\,\%$. $\zeta = 1$: critical, $k_v = 2\sqrt{k_p}$. $\zeta > 1$: no overshoot.

(a) Proximal DH link transform multiplied out

${}^{i-1}\mathbf{T}_i = R_x(\alpha)\,t_x(a)\,R_z(\theta)\,t_z(d)$ (write $\alpha = \alpha_{i-1}$, $a = a_{i-1}$). Multiply left to right.

Step 1: $R_x(\alpha)\,t_x(a) = \begin{bmatrix} 1 & 0 & 0 & a \\ 0 & c\alpha & -s\alpha & 0 \\ 0 & s\alpha & c\alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.

Step 2: $R_z(\theta)\,t_z(d) = \begin{bmatrix} c\theta & -s\theta & 0 & 0 \\ s\theta & c\theta & 0 & 0 \\ 0 & 0 & 1 & d \\ 0 & 0 & 0 & 1 \end{bmatrix}$.

Step 3, rotation block $R_x(\alpha)R_z(\theta)$, row by row:
row 1 $= (c\theta,\ -s\theta,\ 0)$;
row 2 $= c\alpha\,(s\theta,\ c\theta,\ 0) - s\alpha\,(0,0,1) = (c\alpha s\theta,\ c\alpha c\theta,\ -s\alpha)$;
row 3 $= s\alpha\,(s\theta,\ c\theta,\ 0) + c\alpha\,(0,0,1) = (s\alpha s\theta,\ s\alpha c\theta,\ c\alpha)$.
Step 4, translation: R_z leaves the z-axis alone, so the two shifts add inside $R_x(\alpha)$: $\mathbf{p} = R_x(\alpha)(a,0,d)^\top = (a,\ -s\alpha\,d,\ c\alpha\,d)$.

$$ {}^{i-1}\mathbf{T}_i = \begin{bmatrix} c\theta & -s\theta & 0 & a \\ s\theta c\alpha & c\theta c\alpha & -s\alpha & -s\alpha\,d \\ s\theta s\alpha & c\theta s\alpha & c\alpha & c\alpha\,d \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Checks: $\alpha = 0$: planar rotation, $\mathbf{p} = (a,0,d)$. $\theta = a = \alpha = 0$: pure $t_z(d)$. [checked]

(b) Planar 2R inverse kinematics, both methods

$x = L_1c_1 + L_2c_{12}$, $y = L_1s_1 + L_2s_{12}$.
Algebraic. Square and add; $c_1c_{12} + s_1s_{12} = c_2$:
$x^2 + y^2 = L_1^2 + L_2^2 + 2L_1L_2\,c_2$
$\Rightarrow c_2 = \dfrac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2}$; reachable iff $|c_2| \le 1$.
$s_2 = \pm\sqrt{1 - c_2^2}$, $q_2 = \operatorname{atan2}(s_2, c_2)$: two branches.
Expand c_{12}, s_{12} and put $k_1 = L_1 + L_2c_2$, $k_2 = L_2s_2$:
$x = k_1c_1 - k_2s_1$, $y = k_1s_1 + k_2c_1$.
This is (k_1, k_2) rotated by q_1, so $\arg(x,y) = q_1 + \mu$ with $\mu = \operatorname{atan2}(k_2, k_1)$:

$$ q_1 = \operatorname{atan2}(y,x) - \operatorname{atan2}(L_2s_2,\ L_1 + L_2c_2) $$

Geometric. Triangle: origin O, elbow E, tip P; sides L_1, L_2, $r = |OP|$.
The *interior* angle at E is $\pi - q_2$.
Cosine rule: $r^2 = L_1^2 + L_2^2 - 2L_1L_2\cos(\pi - q_2) = L_1^2 + L_2^2 + 2L_1L_2\cos q_2$ (same c_2).
Drop a perpendicular from P to line OE: height $L_2\sin(\pi - q_2) = L_2s_2$, base $L_1 + L_2c_2$, so $\tan\mu = \dfrac{L_2s_2}{L_1 + L_2c_2}$ (μ = angle at O between link 1 and OP).
$q_2 > 0$ (tip anticlockwise of link 1): link 1 lies *below* OP, $q_1 = \operatorname{atan2}(y,x) - \mu$.
$q_2 < 0$: $\mu < 0$, link 1 lies above OP; the same formula gives $q_1 = \operatorname{atan2}(y,x) + |\mu|$.
Both branches: (q_1, q_2) and $(q_1', -q_2)$ with $\mu \to -\mu$.
Check: $L = (0.5, 0.4)$, $P = (0.5, 0.3)$: $c_2 = -0.1750$; $q = (-11.52°,\ 100.08°)$ or $(73.45°,\ -100.08°)$; FK reproduces P for both. [checked]

(c) Spherical wrist: decoupling and ZYZ wrist angles

Axes 4, 5, 6 meet at the wrist centre W, so $q_4..q_6$ cannot move W: W depends on $q_1..q_3$ only.
1. $\mathbf{p}_W = \mathbf{p}_E - d_6\,{}^0\mathbf{R}_E\,[0\ 0\ 1]^\top$ (third column of $\mathbf{R}$).
2. Top view: $q_1 = \operatorname{atan2}(y_W, x_W)$; second branch $q_1 + \pi$.
3. Vertical plane: $r_W = \sqrt{x_W^2 + y_W^2}$, height $z_W - H$; upper arm G, forearm F, q_3 relative (a 2R problem, as in (b)):

$$ \cos q_3 = \frac{r_W^2 + (z_W - H)^2 - F^2 - G^2}{2FG}, \quad q_3 = \operatorname{atan2}(\pm\sqrt{1 - c_3^2}, c_3) $$

$$ q_2 = \operatorname{atan2}(z_W - H,\ r_W) - \operatorname{atan2}(Fs_3,\ G + Fc_3) $$

4. ${}^3\mathbf{R}_6 = ({}^0\mathbf{R}_3)^\top\,{}^0\mathbf{R}_6 = R_z(q_4)R_y(q_5)R_z(q_6)$. Multiplying out, the entries needed are
$r_{13} = c_4s_5$, $r_{23} = s_4s_5$, $r_{33} = c_5$,
$r_{31} = -s_5c_6$, $r_{32} = s_5s_6$.
5. $r_{13}^2 + r_{23}^2 = s_5^2$, so for $s_5 \ge 0$:

$$ q_5 = \operatorname{atan2}\big(\sqrt{r_{13}^2 + r_{23}^2},\ r_{33}\big) $$

$$ q_4 = \operatorname{atan2}(r_{23}, r_{13}),\ q_6 = \operatorname{atan2}(r_{32}, -r_{31}) $$

6. Flip: $s_5 < 0$ gives $q_5 \to -q_5$ and, dividing by $-s_5$, $q_4 \to q_4 + \pi$, $q_6 \to q_6 + \pi$. If $s_5 = 0$ only $q_4 + q_6$ is fixed (singular).
Count: shoulder × elbow × wrist flip = 8.
Check: $H, G, F = 0.5, 0.4, 0.3$, $W = (0.4813, 0.4039, 0.5974)$: $q_1 = 40°$, $c_3 = 0.6428$; $q_3 = -50° \Rightarrow q_2 = 30°$; $q_3 = +50° \Rightarrow q_2 = -12.378°$. [checked]
Sign convention: q_3 anticlockwise relative angle, as on the W3 page.

(d) Grübler–Kutzbach from free bodies

N bodies including ground; ground is fixed, so $N - 1$ moving bodies, each free with D DoF ($D = 3$ planar, 6 spatial): total freedom $D(N-1)$.
A joint with F_i DoF allows F_i relative motions, so it removes $D - F_i$:

$$ M = D(N-1) - \sum_{i=1}^{J}(D - F_i) = D(N-1) - DJ + \sum F_i $$

$$ \boxed{M = D(N - J - 1) + \textstyle\sum_i F_i} $$

Joint DoF: R, P = 1; U = 2; S = 3. A prismatic pair counts as 2 bodies.
Extended form. The plain count assumes (i) every constraint independent, (ii) every DoF moves the output. Correct both:

$$ M = D(N - J - 1) + \sum F_i + R_C - R_M $$

$R_C = \sum F_i - \sum_k u_k$ (u_k: distinct platform DoF allowed by loop k, union over loops); R_M: idle local spins.
Worked: planar 3-RPR $3(8 - 9 - 1) + 9 = 3$. Planar four-bar $3(4 - 4 - 1) + 4 = 1$. Common delta $6(11 - 15 - 1) + 39 = 9$, then $9 + 0 - 6 = 3$ ($R_M = 6$ rod spins). 4RRR: $6(10 - 12 - 1) + 12 = -6$; $R_C = 12 - 5 = 7$, so $M = -6 + 7 - 0 = 1$ (platform only translates in z). Gough–Stewart 6-UPS: $6(14 - 18 - 1) + 36 = 6$. [checked]

(e) Four-bar: loop closure and FK by circle intersection

Frames: $A = L_1(c\theta_1, s\theta_1)$, $D = (L_0, 0)$, coupler $AB = L_2 = a + b$ with K on it ($|AK| = a$, $|KB| = b$), rocker $DB = L_3$ at angle φ_1.
Break at K (pose of K from both sides):

$$L_1 c_{\theta_1} + a\cos(\theta_1{+}\varphi_2) = L_0 + L_3 c_{\varphi_1} + b\cos(\varphi_1{+}\varphi_3)$$
$$L_1 s_{\theta_1} + a\sin(\theta_1{+}\varphi_2) = L_3 s_{\varphi_1} + b\sin(\varphi_1{+}\varphi_3)$$
$$\theta_1 + \varphi_2 = \varphi_1 + \varphi_3 + \pi$$

Three equations; unknowns $\varphi_1, \varphi_2, \varphi_3$.
Break at both ends of the coupler: only its length: $|A - B|^2 = L_2^2$ with $B = D + L_3(c_{\varphi_1}, s_{\varphi_1})$:

$$(L_1 c_{\theta_1} - L_0 - L_3 c_{\varphi_1})^2 + (L_1 s_{\theta_1} - L_3 s_{\varphi_1})^2 - L_2^2 = 0$$

FK. θ_1 known: A, D fixed. B is on circle(A, L_2) and circle(D, L_3). With $d = |AD|$, $\hat{u} = (D - A)/d$, $\hat{n} = \hat{u}$ turned +90°, foot distance x and half-chord h from $x^2 + h^2 = L_2^2$, $(d - x)^2 + h^2 = L_3^2$:

$$x = \frac{L_2^2 - L_3^2 + d^2}{2d},\quad h = \sqrt{L_2^2 - x^2},\quad B = A + x\hat{u} \pm h\hat{n}$$

Assembles iff $|L_2 - L_3| \le d \le L_2 + L_3$ (two modes, one at the limit, none otherwise). Then $\varphi_1 = \operatorname{atan2}(B_y, B_x - L_0)$, $\varphi = \operatorname{atan2}(B - A)$, $K = A + a(\cos\varphi, \sin\varphi)$.
IK (pose of K, angle φ): $A = K - a(\cos\varphi, \sin\varphi)$, $\theta_1 = \operatorname{atan2}(A_y, A_x)$; $B = K + b(\cos\varphi, \sin\varphi)$, $\varphi_1 = \operatorname{atan2}(B_y, B_x - L_0)$.
Check ($L_0, L_1, L_2, L_3 = 1, 0.5, 0.45, 0.6$, $a = 0.2$, $\theta_1 = 70°$): $A = (0.1710, 0.4698)$, $d = 0.9529$, $x = 0.3938$, $h = 0.2178$. Lower mode ($-h\hat{n}$): $B = (0.4062, 0.0862)$, $\varphi = -58.486°$, $K = (0.2756, 0.2993)$. Upper mode: $B = (0.6210, 0.4651)$. IK from K, φ returns $\theta_1 = 70°$. [checked]

(f) Cubic trajectories: coefficients

$q = a_0 + a_1 t + a_2 t^2 + a_3 t^3$, $\dot{q} = a_1 + 2a_2 t + 3a_3 t^2$, $\ddot{q} = 2a_2 + 6a_3 t$.
$q(0) = q_0 \Rightarrow a_0 = q_0$; $\dot{q}(0) = \dot{q}_0 \Rightarrow a_1 = \dot{q}_0$.
Let $\Delta = q_F - q_0$, $r = \Delta - \dot{q}_0 T$, $s = \dot{q}_F - \dot{q}_0$. The end conditions are $a_2 T^2 + a_3 T^3 = r$ and $2a_2 T + 3a_3 T^2 = s$.
(2/T) first − second: $-a_3 T^2 = 2r/T - s$, so $a_3 T^2 = s - 2r/T$:

$$a_3 = \frac{-2\Delta + (\dot{q}_0 + \dot{q}_F)T}{T^3},\quad a_2 = \frac{3\Delta - (2\dot{q}_0 + \dot{q}_F)T}{T^2}$$

Rest to rest ($\dot{q}_0 = \dot{q}_F = 0$): $a_2 = \dfrac{3\Delta}{T^2}$, $a_3 = -\dfrac{2\Delta}{T^3}$.
Then $\dot{q} = \dfrac{6\Delta}{T^2}t - \dfrac{6\Delta}{T^3}t^2$, peak at $t = T/2$: $\dot{q}_{max} = \dfrac{3\Delta}{2T} = 1.5\Delta/T$. $\ddot{q}(0) = +6\Delta/T^2$, $\ddot{q}(T) = -6\Delta/T^2$.
Check: 10° → 50° in 2 s: $(10, 0, 30, -10)$, peak 30°/s. 20° → 80° in 3 s, $\dot{q}_F = 10$: $a_2 = 16.667$, $a_3 = -3.333$, $q(3) = 80$, $\dot{q}(3) = 10$. [checked]

(f) Two cubics through a via point, continuous acceleration

Segment 1 on $[0, T_1]$, segment 2 on $[0, T_2]$ (own clock), a_{ij} = coefficient j of segment i: **8 unknowns**.
Conditions (8): (1) $q_1(0) = q_0$; (2) $\dot{q}_1(0) = 0$; (3) $q_1(T_1) = q_v$; (4) $q_2(0) = q_v$; (5) $q_2(T_2) = q_F$; (6) $\dot{q}_2(T_2) = 0$; (7) $\dot{q}_1(T_1) = \dot{q}_2(0)$; (8) $\ddot{q}_1(T_1) = \ddot{q}_2(0)$.
Let $w = \dot{q}_v$ (unknown via speed), $d_1 = q_v - q_0$, $d_2 = q_F - q_v$. Conditions 1–7 are met by the closed cubic with end speeds $(0, w)$ and $(w, 0)$:

$$a_{10} = q_0,\ a_{11} = 0,\ a_{12} = \frac{3d_1 - wT_1}{T_1^2},\ a_{13} = \frac{-2d_1 + wT_1}{T_1^3};$$
$$a_{20} = q_v,\ a_{21} = w,\ a_{22} = \frac{3d_2 - 2wT_2}{T_2^2},\ a_{23} = \frac{-2d_2 + wT_2}{T_2^3}.$$

Condition 8: $\ddot{q}_1(T_1) = 2a_{12} + 6a_{13}T_1 = \dfrac{-6d_1 + 4wT_1}{T_1^2}$, $\ddot{q}_2(0) = 2a_{22} = \dfrac{6d_2 - 4wT_2}{T_2^2}$. Equate and solve for w:

$$w = \frac{3\left[T_1^2(q_F - q_v) + T_2^2(q_v - q_0)\right]}{2T_1T_2(T_1 + T_2)} \xrightarrow{T_1=T_2=T} w = \frac{3(q_F - q_0)}{4T}$$

Equal durations: w does not depend on q_v; shared acceleration = $3(q_0 + q_F - 2q_v)/T^2$ and $a_{21} = 3(q_F - q_0)/(4T)$.
Check ($q_0, q_v, q_F = 10, 40, 20$, $T = 1$): $w = 7.5$; seg 1 = $10 + 82.5t^2 - 52.5t^3$; seg 2 = $40 + 7.5t - 75t^2 + 47.5t^3$; acceleration −150 both sides. With $T_1 = 1$, $T_2 = 2$: $w = 25$, acceleration −80. [checked]

(g) LSPB (trapezoid) blend time

Blend $+a$ for t_B, coast at $v = a\,t_B$, blend $-a$ for t_B; total time t_F, $\Delta = q_F - q_0 > 0$.
Blend 1 moves $\frac{1}{2}at_B^2$. Coast lasts $t_F - 2t_B$: moves $at_B(t_F - 2t_B)$. Blend 2 moves $vt_B - \frac{1}{2}at_B^2 = at_B^2 - \frac{1}{2}at_B^2$.
Sum: $\Delta = \frac{1}{2}at_B^2 + at_Bt_F - 2at_B^2 + at_B^2 - \frac{1}{2}at_B^2 = a\,t_Bt_F - a\,t_B^2$.

$$a\,t_B^2 - a\,t_F\,t_B + \Delta = 0 \Rightarrow t_B = \frac{t_F}{2} \mp \frac{\sqrt{a^2t_F^2 - 4a\Delta}}{2a}$$

Two blends must fit ($t_B \le t_F/2$): take the minus sign.
Real iff $a^2t_F^2 - 4a\Delta \ge 0 \iff a \ge \dfrac{4\Delta}{t_F^2}$ (non-strict). Equality: $t_B = t_F/2$, no coast, triangle with peak $2\Delta/t_F$.
Coast speed $v = at_B = \frac{1}{2}\left[at_F - \sqrt{a^2t_F^2 - 4a\Delta}\right]$, always in $(\Delta/t_F,\ 2\Delta/t_F]$.
Check: 120° in 2 s: $a_{min} = 120°/s^2$ ($t_B = 1$, peak 120°/s). $a = 150$: $t_B = 0.5528$, $v = 82.9°/s$. $a = 300$: $t_B = 0.2254$, $v = 67.6°/s$. [checked]

(h) 2R Jacobian, $\det \mathbf{J}$, and $\boldsymbol{\tau} = \mathbf{J}^\top\mathbf{F}$

$x = L_1c_1 + L_2c_{12}$, $y = L_1s_1 + L_2s_{12}$. Differentiate with $\dot{c}_{12} = -s_{12}(\dot{q}_1 + \dot{q}_2)$:

$$\mathbf{J} = \begin{bmatrix} \partial x/\partial q_1 & \partial x/\partial q_2 \\ \partial y/\partial q_1 & \partial y/\partial q_2 \end{bmatrix} = \begin{bmatrix} -L_1s_1 - L_2s_{12} & -L_2s_{12} \\ L_1c_1 + L_2c_{12} & L_2c_{12} \end{bmatrix}$$

Determinant:
$\det \mathbf{J} = (-L_1s_1 - L_2s_{12})L_2c_{12} + L_2s_{12}(L_1c_1 + L_2c_{12})$
$= L_1L_2(s_{12}c_1 - c_{12}s_1) + L_2^2(s_{12}c_{12} - s_{12}c_{12})$
$= L_1L_2\sin(q_{12} - q_1) = L_1L_2\sin q_2$.
Singular at $q_2 = 0, \pi$ (arm straight or folded): no tool velocity along the arm.
Statics by virtual work. Tiny joint motion $\delta\mathbf{q}$ moves the tool by $\delta\mathbf{x} = \mathbf{J}\delta\mathbf{q}$. Actuators do $\boldsymbol{\tau}^\top\delta\mathbf{q}$; the tool force $\mathbf{F}$ (applied to the environment) does $\mathbf{F}^\top\delta\mathbf{x} = \mathbf{F}^\top\mathbf{J}\delta\mathbf{q}$. Equilibrium: equal for *every* $\delta\mathbf{q}$:

$$\boldsymbol{\tau}^\top = \mathbf{F}^\top\mathbf{J} \Rightarrow \boldsymbol{\tau} = \mathbf{J}^\top\mathbf{F}$$

Same frame for **J** and **F**; opposite sign if **F** is the load on the tool.
Check: $L = (0.5, 0.4)$, $q = (30°, 60°)$, $\mathbf{F} = (10, -20)$ N: $\mathbf{J} = \begin{bmatrix} -0.65 & -0.4 \\ 0.4330 & 0 \end{bmatrix}$, $\boldsymbol{\tau} = (-15.160, -4.000)$ N m (**JF** would wrongly give $(1.5, 4.330)$). [checked]

(i) 2R point-mass Lagrange, every step

Masses m_1 at the tip of L_1, m_2 at the tip of L_2; q_1 from horizontal, q_2 relative; gravity $-y$.
Positions: $x_1 = L_1c_1$, $y_1 = L_1s_1$; $x_2 = L_1c_1 + L_2c_{12}$, $y_2 = L_1s_1 + L_2s_{12}$.
Velocities: $\dot{x}_2 = -L_1s_1\dot{q}_1 - L_2s_{12}(\dot{q}_1{+}\dot{q}_2)$, $\dot{y}_2 = L_1c_1\dot{q}_1 + L_2c_{12}(\dot{q}_1{+}\dot{q}_2)$.
Kinetic: $K_1 = \frac{1}{2}m_1L_1^2\dot{q}_1^2$. $v_2^2 = \dot{x}_2^2 + \dot{y}_2^2$, cross term $2L_1L_2(s_1s_{12} + c_1c_{12}) = 2L_1L_2c_2$:
$v_2^2 = L_1^2\dot{q}_1^2 + L_2^2(\dot{q}_1{+}\dot{q}_2)^2 + 2L_1L_2c_2\,\dot{q}_1(\dot{q}_1{+}\dot{q}_2)$.

$$K = \tfrac{1}{2}\left(M_{11}\dot{q}_1^2 + 2M_{12}\dot{q}_1\dot{q}_2 + M_{22}\dot{q}_2^2\right)$$

$M_{11} = m_1L_1^2 + m_2(L_1^2 + 2L_1L_2c_2 + L_2^2)$, $M_{12} = m_2(L_1L_2c_2 + L_2^2)$, $M_{22} = m_2L_2^2$.
Potential: $P = m_1gL_1s_1 + m_2g(L_1s_1 + L_2s_{12})$. $\mathcal{L} = K - P$.
Let $h = m_2L_1L_2s_2$. Only c_2 varies: $\partial M_{11}/\partial q_2 = -2h$, $\partial M_{12}/\partial q_2 = -h$, so $\dot{M}_{11} = -2h\dot{q}_2$, $\dot{M}_{12} = -h\dot{q}_2$.
Joint 1. $\partial K/\partial q_1 = 0$.
$\partial\mathcal{L}/\partial\dot{q}_1 = M_{11}\dot{q}_1 + M_{12}\dot{q}_2$
$\frac{d}{dt}(\cdot) = M_{11}\ddot{q}_1 + M_{12}\ddot{q}_2 - 2h\dot{q}_1\dot{q}_2 - h\dot{q}_2^2$
$-\partial\mathcal{L}/\partial q_1 = +\partial P/\partial q_1 = g_1$.
Joint 2. $\partial\mathcal{L}/\partial\dot{q}_2 = M_{12}\dot{q}_1 + M_{22}\dot{q}_2$
$\frac{d}{dt}(\cdot) = M_{12}\ddot{q}_1 + M_{22}\ddot{q}_2 - h\dot{q}_1\dot{q}_2$
$\partial K/\partial q_2 = \frac{1}{2}(-2h)\dot{q}_1^2 + (-h)\dot{q}_1\dot{q}_2 = -h\dot{q}_1^2 - h\dot{q}_1\dot{q}_2$
$Q_2 = \frac{d}{dt}(\cdot) - \partial K/\partial q_2 + g_2$: the $-h\dot{q}_1\dot{q}_2$ terms cancel.

$$\mathbf{M} = \begin{bmatrix} M_{11} & M_{12} \\ M_{12} & M_{22} \end{bmatrix},\ \mathbf{C} = \begin{bmatrix} -h(2\dot{q}_1\dot{q}_2 + \dot{q}_2^2) \\ h\dot{q}_1^2 \end{bmatrix}$$

$$\mathbf{g} = \begin{bmatrix} (m_1{+}m_2)gL_1c_1 + m_2gL_2c_{12} \\ m_2gL_2c_{12} \end{bmatrix},\quad \mathbf{Q} = \mathbf{M}\ddot{\mathbf{q}} + \mathbf{C} + \mathbf{g}$$

Check ($m = 2, 1$ kg; $L = 1, 0.8$ m; $q = (30°, 45°)$; $\dot{q} = (1, -0.5)$; $\ddot{q} = (0.5, 2)$): $\mathbf{Q} = (32.7397,\ 4.4797)$ N m. [checked]

(j) Partitioned control: error equation, gains, steady state, PID

Plant $m\ddot{x} + b\dot{x} + kx = f$. Write $f = \alpha f' + \beta$ with $\alpha = m$, $\beta = b\dot{x} + kx$: then $\ddot{x} = f'$ (unit mass).
Servo law with $E = x_d - x$: $f' = \ddot{x}_d + k_v\dot{E} + k_pE$. Substitute $x = x_d - E$ in $\ddot{x} = f'$:
$\ddot{x}_d - \ddot{E} = \ddot{x}_d + k_v\dot{E} + k_pE$

$$\ddot{E} + k_v\dot{E} + k_pE = 0, \qquad s^2 + k_vs + k_p = 0$$

Roots $s = \frac{1}{2}\left(-k_v \pm \sqrt{k_v^2 - 4k_p}\right)$. Equal roots iff $k_v^2 - 4k_p = 0$:

$$k_v^2 = 4k_p \iff k_v = 2\sqrt{k_p}, \quad \omega_n = \sqrt{k_p}, \ \zeta = \frac{k_v}{2\sqrt{k_p}}$$

(The slide's "$k_p^2 = 4k_v$" has the subscripts swapped.) Example $k_p = 25 \Rightarrow k_v = 10$, double pole at -5.

Constant disturbance. Plant sees $f - f_d$: $m\ddot{x} = mf' - f_d \Rightarrow \ddot{x} = f' - f_d/m$. With $E = x_d - x$ the sign flips:

$$\ddot{E} + k_v\dot{E} + k_pE = f_d/m$$

Steady state ($\dot{E} = \ddot{E} = 0$): $k_pE = f_d/m$:

$$E_{ss} = \frac{f_d}{k_p\, m}$$

k_v, b, k drop out. Check: $f_d = 10$, $m = 2$, $k_p = 25$: 0.2 m; $m = 5$: 0.08 m.
PID. $f' = \ddot{x}_d + k_v\dot{E} + k_pE + k_i\int E\,dt$ gives $\ddot{E} + k_v\dot{E} + k_pE + k_i\int E\,dt = f_d/m$. Differentiate once:

$$\dddot{E} + k_v\ddot{E} + k_p\dot{E} + k_iE = \dot{f}_d/m$$

Constant f_d: all derivatives vanish, $k_iE = 0$, so $E_{ss} = 0$ (integral settles at $f_d/(mk_i)$).
Stable iff $k_vk_p > k_i$ (Routh–Hurwitz); at equality $(s + k_v)(s^2 + k_p)$.
Check $k_p = 25, k_v = 10, k_i = 40$: roots -7.335, $-1.332 \pm 1.918j$; $k_i = 250$: $-10, \pm 5j$. Simulation on the real plant ($m = 2, b = 3, k = 8$): PD error 0.2 m; PID $\to 0$, $\int E = 0.125$. [checked]

(k) Generalised wheel model, three steps; omni triangle

Notation: $\boldsymbol{\zeta} = (u, v, \dot{\theta})$ chassis velocity in $\{C\}$; wheel i at $({}^Cx_i, {}^Cy_i)$, drive direction β_i from x_C, roller angle γ_i, radius r.
Step 1 (wheel frame turned by β): ${}^C\mathbf{v}_W = R(\beta)\,{}^W\mathbf{v}$, so ${}^W\mathbf{v} = R(\beta)^\top\,{}^C\mathbf{v}$.
Step 2 (rigid chassis): contact-point velocity = chassis velocity $+\dot{\theta}\hat{z}\times\mathbf{p}$:

$${}^C\mathbf{v}_W = \begin{bmatrix} u - \dot{\theta}\,{}^Cy \\ v + \dot{\theta}\,{}^Cx \end{bmatrix} = \begin{bmatrix} 1 & 0 & -{}^Cy \\ 0 & 1 & {}^Cx \end{bmatrix} \boldsymbol{\zeta}$$

Step 3 (drive component). Roller direction $(-\sin\gamma, \cos\gamma)$ is free, so $v_{drive} = v_x + v_y\tan\gamma$:

$$r\dot{\varphi} = \begin{bmatrix} 1 & \tan\gamma \end{bmatrix} \begin{bmatrix} c\beta & s\beta \\ -s\beta & c\beta \end{bmatrix} \begin{bmatrix} 1 & 0 & -{}^Cy \\ 0 & 1 & {}^Cx \end{bmatrix} \boldsymbol{\zeta}$$

$\gamma = 0$: $r\dot{\varphi} = c\beta\,(u - \dot{\theta}\,{}^Cy) + s\beta\,(v + \dot{\theta}\,{}^Cx)$.
Standard wheel also has no-slide: $0 = -s\beta\,(u - \dot{\theta}\,{}^Cy) + c\beta\,(v + \dot{\theta}\,{}^Cx)$.
Omni triangle (*this site's convention*: drive directions clockwise, $\beta = 0°, -120°, +120°$; wheels at $(0, d)$, $(\frac{\sqrt{3}}{2}d, -\frac{d}{2})$, $(-\frac{\sqrt{3}}{2}d, -\frac{d}{2})$; $\gamma = 0$):
W1: $c\beta = 1, s\beta = 0$: $r\dot{\varphi}_1 = u - d\dot{\theta}$.
W2: $c\beta = -\frac{1}{2}$, $s\beta = -\frac{\sqrt{3}}{2}$: $r\dot{\varphi}_2 = -\frac{1}{2}(u + \frac{d}{2}\dot{\theta}) - \frac{\sqrt{3}}{2}(v + \frac{\sqrt{3}}{2}d\dot{\theta}) = -\frac{1}{2}u - \frac{\sqrt{3}}{2}v - d\dot{\theta}$.
W3: $c\beta = -\frac{1}{2}, s\beta = \frac{\sqrt{3}}{2}$: $r\dot{\varphi}_3 = -\frac{1}{2}u + \frac{\sqrt{3}}{2}v - d\dot{\theta}$.

$$\dot{\boldsymbol{\varphi}} = \frac{1}{r}\begin{bmatrix} 1 & 0 & -d \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} & -d \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} & -d \end{bmatrix} \boldsymbol{\zeta}, \quad \boldsymbol{\zeta} = \frac{r}{3}\begin{bmatrix} 2 & -1 & -1 \\ 0 & -\sqrt{3} & \sqrt{3} \\ -\frac{1}{d} & -\frac{1}{d} & -\frac{1}{d} \end{bmatrix} \dot{\boldsymbol{\varphi}}$$

$\det = \frac{3\sqrt{3}}{2}d \neq 0$: holonomic. Reverse the drive sense and all signs flip.
Check: $r = 0.05$, $d = 0.1$, $\dot{\boldsymbol{\varphi}} = (4, -1, -1)$: $u = 0.1667$, $v = 0$, $\dot{\theta} = -0.3333$; ICR $(-v/\dot{\theta}, u/\dot{\theta}) = (0, -0.5)$ m, clockwise circle of radius 0.5 m. [checked]

(l) Differential drive: inverse model and ICR

Fixed standard wheels, $\beta = 0$, $\gamma = 0$: left at $(0, +a)$, right at $(0, -a)$.
Rolling rows ($r\dot{\varphi} = u - \dot{\theta}\,{}^Cy$):

$$r\dot{\varphi}_1 = u - a\dot{\theta}, \qquad r\dot{\varphi}_2 = u + a\dot{\theta}$$

No-slide row ($\beta = 0$, ${}^Cx = 0$): $v + \dot{\theta}\cdot 0 = 0 \Rightarrow v = 0$.
Add and subtract:

$$u = \frac{r}{2}(\dot{\varphi}_1 + \dot{\varphi}_2), \qquad \dot{\theta} = \frac{r}{2a}(\dot{\varphi}_2 - \dot{\varphi}_1)$$

ICR: the chassis point with zero velocity, (x, y) with $u - \dot{\theta}y = 0$ and $v + \dot{\theta}x = 0$:

$$(x, y) = \left(-\frac{v}{\dot{\theta}}, \frac{u}{\dot{\theta}}\right) = \left(0,\ a\,\frac{\dot{\varphi}_1 + \dot{\varphi}_2}{\dot{\varphi}_2 - \dot{\varphi}_1}\right)$$

r cancels. Equal speeds: ICR at infinity (straight). Opposite speeds: ICR at the centre (spin). $y > 0$ means left of the robot.
Inertial: $\dot{x} = u\cos\theta$, $\dot{y} = u\sin\theta$, $\dot{\theta}$ as above.
Check: $r = 0.05$, $a = 0.15$, $\dot{\varphi} = (4, 6)$: $u = 0.25$ m/s, $\dot{\theta} = 0.3333$ rad/s, ICR $y = 0.75$ m left. $a = 0.3$, $\dot{\varphi} = (2, 5)$: $y = 0.3 \cdot 7/3 = 0.7$ m. $r = 0.1$, $a = 0.2$, $\dot{\varphi} = (-5, 5)$: $u = 0$, $\dot{\theta} = 2.5$ rad/s. [checked]